

TECHNICAL UNIVERSITY OF CRETE
SCHOOL OF ELECTRICAL AND COMPUTER ENGINEERING
TELECOMMUNICATIONS DIVISION

Online Convex Optimization based tuning of Mixture of Experts for Sports Outcome Prediction

DIPLOMA THESIS

Stefanos Mousadis

THESIS COMMITTEE

Prof. Thrasyvoulos Spyropoulos, *Supervisor*
Prof. Michail G. Lagoudakis
Prof. Vasileios Samoladas

Chania, 2026



ΠΟΛΥΤΕΧΝΕΙΟ ΚΡΗΤΗΣ
ΣΧΟΛΗ ΗΛΕΚΤΡΟΛΟΓΩΝ ΜΗΧΑΝΙΚΩΝ ΚΑΙ ΜΗΧΑΝΙΚΩΝ ΥΠΟΛΟΓΙΣΤΩΝ
ΤΟΜΕΑΣ ΤΗΛΕΠΙΚΟΙΝΩΝΙΩΝ

Κυρτή Βελτιστοποίηση σε Πραγματικό Χρόνο Βασισμένη στην Παραμετροποίηση Μείγματος Ειδικών για την Πρόβλεψη Αποτελεσμάτων Αθλητικών Αγώνων

ΔΙΠΛΩΜΑΤΙΚΗ ΕΡΓΑΣΙΑ

Στέφανος Μουσάδης

ΕΞΕΤΑΣΤΙΚΗ ΕΠΙΤΡΟΠΗ

Καθ. Θρασύβουλος Σπυρόπουλος, *Επιβλέπων*
Καθ. Μιχαήλ Γ. Λαγουδάκης
Καθ. Βασίλειος Σαμολαδάς

Χανιά, 2026

Abstract

This thesis predicts sports match outcomes (home / draw / away) as a problem of **online learning with expert advice**, rather than as the training of a statistical model. Each betting operator is treated as an expert quoting odds on the three outcomes, from which probabilities are derived. The task is to find a weight vector such that the mixture of experts outperforms each of them individually, learning not retrospectively over the past but match by match.

Three stages of **Online Convex Optimization** (OCO) are implemented. First, three algorithms are built (Hedge, Online Gradient Descent, Follow-The-Regularized-Leader), which handle partially covering operators through the **Sleeping Experts** reduction. A second OCO stage then calibrates the mixture via online temperature scaling. Finally the bet size is computed with the Kelly criterion, whose multiplier is itself learned online with Online Newton Step. Every comparison is tested with a moving block bootstrap. The dataset spans 22 leagues, 10 seasons and 76,584 matches.

The mixture significantly outperforms every individual bookmaker except the sharpest, with which it draws level in the configuration ultimately adopted. Under a deployment test with a genuine temporal cut (fitted through the end of 2024, evaluated on 2025–26) it significantly outperforms every opening price and draws level with the bookmakers' closing prices. The one source of odds that significantly outperforms it is Betfair Exchange, which is not a bookmaker but a betting exchange: a platform on which bettors bet against one another, so that its odds are set by the bettors themselves and carry no profit margin added by a company.

The three broader findings are that learning pays only where experts are heterogeneous in quality; that better prediction does not translate directly into profit, since the quality gain falls short of the market's margin; and that part of the advantage over the uniform average is due not to learning but to the geometry of the Sleeping Experts reduction.

Keywords: online convex optimization, prediction with expert advice, sleeping experts, regret, probability calibration, betting markets

Περίληψη

Η παρούσα εργασία προβλέπει τα αποτελέσματα αθλητικών αγώνων (1/X/2) ως πρόβλημα **online μάθησης με συμβουλή ειδικών**, και όχι ως εκπαίδευση στατιστικού μοντέλου. Κάθε στοιχηματικός πάροχος θεωρείται ένας expert που δίνει αποδόσεις για τα τρία ενδεχόμενα, από τις οποίες προκύπτουν πιθανότητες. Ζητούμενο είναι η εύρεση ενός διανύσματος βαρών έτσι ώστε το μείγμα των experts να τους ξεπερνά ατομικά, με μάθηση όχι αναδρομικά πάνω στο παρελθόν, αλλά αγώνα προς αγώνα.

Υλοποιούνται τρία στάδια Κυρτής Βελτιστοποίησης σε Πραγματικό Χρόνο (**Online Convex Optimization**, OCO). Αρχικά κατασκευάζονται τρεις αλγόριθμοι (Hedge, Online Gradient Descent, Follow-The-Regularized-Leader), που με την αναγωγή σε **Sleeping Experts** διαχειρίζονται τους παρόχους μερικής κάλυψης. Έπειτα, με δεύτερο στάδιο OCO, γίνεται βαθμονόμηση μέσω online temperature scaling. Τέλος υπολογίζεται το μέγεθος του στοιχήματος με το κριτήριο Kelly, όπου ο πολλαπλασιαστής του μαθαίνεται online με Online Newton Step. Κάθε σύγκριση ελέγχεται με moving block bootstrap. Το σύνολο δεδομένων καλύπτει 22 πρωταθλήματα, 10 σεζόν και 76.584 αγώνες.

Το μείγμα ξεπερνά στατιστικά σημαντικά κάθε μεμονωμένο bookmaker πλην του σκληρότερου, με τον οποίο ισοπαλεί στη διαμόρφωση που επιλέγεται τελικά. Σε έλεγχο ανάπτυξης με πραγματική χρονική τομή (εκπαίδευση έως το τέλος του 2024, μέτρηση στο 2025–26) κερδίζει σημαντικά κάθε τιμή ανοίγματος και ισοπαλεί με τις τιμές κλεισίματος των bookmakers. Η μόνη πηγή αποδόσεων που το ξεπερνά σημαντικά είναι το Betfair Exchange, το οποίο δεν είναι στοιχηματική εταιρεία αλλά ανταλλακτήριο στοιχημάτων: οι παίκτες στοιχηματίζουν ο ένας απέναντι στον άλλον, οπότε τις αποδόσεις τις διαμορφώνουν οι ίδιοι και δεν περιέχουν περιθώριο κέρδους κάποιας εταιρείας.

Τα τρία γενικότερα ευρήματα είναι ότι η μάθηση αποδίδει μόνο όταν οι experts έχουν ανομοιομορφία στην ποιότητα, ότι η καλύτερη πρόβλεψη δεν μεταφράζεται άμεσα σε κέρδος καθώς η διαφορά ποιότητας υπολείπεται του περιθωρίου της αγοράς, και ότι ένα μέρος της υπεροχής έναντι της ομοιόμορφης μέσης τιμής δεν οφείλεται στη μάθηση αλλά στη γεωμετρία της αναγωγής Sleeping Experts.

Λέξεις-κλειδιά: online κυρτή βελτιστοποίηση, πρόβλεψη με συμβουλή ειδικών, sleeping experts, regret, βαθμονόμηση πιθανοτήτων, αγορές στοιχηματισμού

Contents

Abstract **1**

Περίληψη **2**

1 Introduction **6**
1.1 The problem . . . 6
1.2 Why this is not trivial . . . 6
1.3 Contributions . . . 7
1.4 Thesis outline . . . 7

2 Related work **8**
2.1 Online learning and regret theory . . . 8
2.2 Sports outcome prediction . . . 9
2.3 Efficiency of betting markets . . . 9
2.4 Favourite–longshot bias and odds conversion . . . 10
2.5 Probability calibration . . . 10
2.6 Forecast combination . . . 10
2.7 Position of the present work . . . 11

3 Theoretical background **12**
3.1 The online learning framework . . . 12
3.1.1 Online learning and regret . . . 12
3.1.2 Prediction with expert advice . . . 14
3.2 From odds to probabilities . . . 15
3.2.1 De-vigging . . . 15
3.2.2 Microstructure of the betting market . . . 16
3.3 The algorithms . . . 17
3.3.1 Hedge . . . 17
3.3.2 Online Gradient Descent . . . 18
3.3.3 Follow-The-Regularized-Leader . . . 20
3.3.4 Exp-concavity and the Aggregating Algorithm . . . 21
3.3.5 The Sleeping Experts reduction . . . 22
3.3.6 Fixed-Share . . . 23
3.3.7 Markov chains: three different uses . . . 23
3.3.8 Bandit online convex optimisation . . . 25
3.4 Scoring and statistical testing . . . 26
3.4.1 Proper scoring rules and calibration . . . 26
3.4.2 Statistical significance: the moving block bootstrap . . . 27

3.5 Bet sizing . . . 28
3.5.1 The Kelly criterion . . . 28
3.5.2 Stake size as a third OCO stage . . . 29
3.5.3 A Bayesian version of Kelly and Markov chain Monte Carlo sampling 30

4 Problem formulation and experimental design 32
4.1 The data . . . 32
4.1.1 Source and scope . . . 32
4.1.2 Cleaning . . . 33
4.1.3 Opening and closing odds . . . 33
4.2 Experimental design . . . 34
4.2.1 The expert tensor . . . 34
4.2.2 The three OCO stages . . . 34
4.2.3 Hyperparameter selection protocol . . . 35
4.2.4 Tuning the step . . . 35
4.2.5 Comparing series with different coverage . . . 35
4.2.6 Causality controls in value betting . . . 36
4.2.7 Deployment test . . . 36

5 Results 37
5.1 Tuning the step . . . 37
5.2 Ranking and the choice of panel . . . 40
5.2.1 Ranking on the full panel . . . 40
5.2.2 Which panel . . . 42
5.2.3 Where exactly closing prices are superior . . . 43
5.2.4 The configuration that is selected . . . 45
5.2.5 Without Pinnacle . . . 46
5.3 Robustness over time . . . 48
5.3.1 Recent window . . . 48
5.3.2 Deployment test . . . 50
5.4 Value betting and the learned stake size . . . 50
5.5 How much information does learning need? . . . 56
5.5.1 The bandit experiment . . . 56
5.5.2 Markovian switching of market regimes . . . 59
5.6 Negative results . . . 60
5.7 Discussion . . . 62
5.7.1 What the thesis showed . . . 62
5.7.2 When learning pays off: heterogeneity as a precondition . . . 62
5.7.3 The character of the method . . . 63

6 Conclusions and future work 64
6.1 Conclusions . . . 64
6.2 The odds as a social relation . . . 65
6.3 Limitations . . . 67
6.4 Future work . . . 68

A Full result tables **69**
A.1 Ranking and statistical significance 69
A.2 Panel and de-vig method 71
A.3 Configurations 73
A.4 Without Pinnacle 73
A.5 Recent window and deployment test 74
A.6 Value betting 77
A.7 Extensions and negative results 80

References **89**

Chapter 1

Introduction

1.1 The problem

Predicting the outcome of a football match is usually treated as a supervised learning problem: features of teams and matches are collected, a model is trained, and it is evaluated on data it has not seen. This approach runs into a practical barrier. Betting companies already solve the same problem, with incomparably more data, with teams of analysts, and with a direct financial incentive to solve it correctly. A model trained within the scope of a thesis will struggle to compete with them on their own ground.

This thesis changes the question. Rather than building yet another forecaster, it treats *each betting company as an expert* that publishes, ahead of every match, its own probabilities (p_H, p_D, p_A) over the three outcomes: p_H that the home team wins, p_D that the match is drawn and p_A that the away team wins, with $p_H + p_D + p_A = 1$. The task then becomes: how are these opinions to be combined, online and without knowledge of the future, so that the mixture is a better forecaster than any one of them taken separately?

The framework that answers this is *prediction with expert advice* and, more generally, *Online Convex Optimization* (OCO). Its principal advantage here is that it yields regret guarantees **without any statistical assumption** about the distribution of the data: the outcomes are not required to come from a fixed distribution, nor the experts to be independent. That suits a market which changes over time, where companies appear and disappear and where prices are set strategically.

1.2 Why this is not trivial

Applying the framework to real data meets three difficulties, and these shape the structure of the thesis.

Partial coverage. The classical algorithms assume that every expert issues a prediction in every round. In the data used here, per-bookmaker coverage ranges from 6.9% to 99.8% of matches, and several companies appear or disappear midway through the decade. This calls for the *Sleeping Experts* reduction, which gives each expert a guarantee measured only over the rounds in which it was present.

Odds are not probabilities. Published odds embed the bookmaker's profit margin: their reciprocals sum to more than one. Converting them into probabilities (de-vigging) is a modelling choice that can influence the conclusions, and it is therefore checked with two independent methods.

The best log-loss is not the objective. A mixture may predict better than every bookmaker and still lose money, if its advantage is smaller than the market's margin. The thesis tests this gap explicitly, through a betting simulation based on the Kelly criterion.

1.3 Contributions

The thesis makes the following contributions:

- A complete implementation of three OCO algorithms under the Sleeping Experts reduction, over a dataset of 22 leagues, 10 seasons and 76,584 matches, with 26 bookmakers as experts.
- Two further OCO stages layered on the mixture: calibration via online temperature scaling, and bet sizing via a learned Kelly multiplier, so that the entire chain from odds to bet is online and causal. It is shown that *what* is learned in the third stage is decisive: learning the stake directly fails systematically, whereas learning the multiplier on top of the closed-form rule does not.
- Systematic statistical testing of every comparison with a moving block bootstrap, rather than comparison of point estimates.
- A deployment test with a genuine temporal cut, answering the question of what a model fitted through 2024 would do today.
- A series of negative findings that bound the scope of the method and are, in our view, as useful as the positive ones.

1.4 Thesis outline

The rest of the thesis is structured as follows. Chapter 2 reviews the related literature. Chapter 3 develops the theoretical background. Chapter 4 describes the data, its preprocessing and the experimental design. Chapter 5 gives the results and discusses them. Chapter 6 summarises, states the limits of the method and proposes future work. Appendix A gives the full result tables.

Chapter 2

Related work

This thesis sits at the confluence of three research traditions that developed largely independently: the theory of online learning, sports outcome prediction, and the economic study of betting markets. This chapter sketches each and locates the present contribution.

2.1 Online learning and regret theory

In *prediction with expert advice* a learner must issue a forecast at every round, and before doing so it sees the forecasts of a fixed set of N *experts*; in this thesis each expert is a betting company and its forecast is the probabilities implied by its odds. The learner's performance is measured by its *regret*: how much larger its total loss over T rounds is than that of the best single expert, or the best fixed combination of experts, chosen in hindsight. A regret that grows more slowly than T means that, averaged per round, the learner does as well as that best choice. The formal definitions are in Chapter 3.

The idea of prediction with expert advice predates modern machine learning. In its computational form it appears with the *weighted majority* algorithm of Littlestone & Warmuth (1994) [31], which showed that one can guarantee performance almost as good as that of the best expert in a set, with no assumption whatsoever about how the data are generated. The generalisation to continuous weights and convex losses led to *Hedge* (Freund & Schapire, 1997) [12], which keeps one weight per expert and after every round multiplies it by a factor that shrinks with that expert's loss, and more broadly to the family of *multiplicative updates*.

In parallel came the *aggregating algorithm* of Vovk (1990) [43], which exploits a stronger property of certain losses, *exp-concavity*. Exp-concavity is a curvature condition, stronger than convexity, which the logarithmic loss satisfies (it is defined in §3.3.4). Under it the regret can be bounded by $\ln N$, where N is the number of experts, that is *independently* of the *horizon* T, the total number of rounds, instead of growing as $O(\sqrt{T \log N})$. That result is theoretically stronger, and this thesis provides an empirical case in which it nevertheless fails, for reasons connected to the non-stationarity of the expert set.

The unification of the above under the common heading of *online convex optimization* is due in large part to Zinkevich (2003) [45], who showed that simple projected gradient descent attains $O(\sqrt{T})$ regret for any sequence of convex functions. Modern synthetic treatments of the field are given by Cesa-Bianchi & Lugosi (2006) [7] and Hazan (2016) [23].

Two extensions bear directly on our application. The first is the *sleeping experts* or *specialists* reduction (Freund, Schapire, Singer & Warmuth, 1997; Blum & Mansour, 2007) [13,

3], which handles experts that do not participate in every round. The second is *Fixed-Share* (Herbster & Warmuth, 1998) [24], which changes the comparator from the best fixed vector to the best sequence of vectors with a bounded number of switches, and therefore addresses precisely the problem of non-stationarity.

Finally, adaptive learning rates of the AdaGrad type (Duchi, Hazan & Singer, 2011) [11] propose per-coordinate scaling of the step according to accumulated gradient. The idea is appealing in a setting where experts differ greatly in how often they participate, and it is examined explicitly in §5.6.

2.2 Sports outcome prediction

The literature on football outcome prediction follows, roughly, three directions.

The first is *parametric statistical models*, starting from the observation that goals are approximated reasonably well by a Poisson distribution. Such models estimate attacking and defensive strength per team and produce a score distribution, from which the home/draw/away probabilities follow. Their main advantage is interpretability; their main drawback is that they require sufficient history per team and struggle with transfers, managerial changes and promotion or relegation.

The second is *machine learning over features*, where variables of form, ranking, head-to-head statistics and, in more recent work, player tracking data are assembled. Such models often perform well against parametric alternatives, but their comparison against the market is rarer and usually less rigorous.

The third, and the most relevant here, uses *the odds themselves as the prediction*. This approach recognises that the market price incorporates all available information, including items that never appear in public statistics. Kaunitz et al. (2017) [26] is the clearest empirical precedent: rather than building a competing forecast, they treat the consensus of the published odds as the probability estimate and bet where an individual bookmaker departs from it, a strategy profitable over a decade of simulation and over five months of real-money betting. The present work belongs to this category, with two differences. It does not use a fixed consensus but learns online how to weight the companies, with a regret guarantee; and the weighting is driven by a loss that each company pays on its own quoted matches, so a company covering a fraction of the fixtures can still contribute.

2.3 Efficiency of betting markets

Betting markets are a privileged field for the study of market efficiency, for one specific reason: unlike financial markets, every bet has *a fixed expiry date and an objectively verifiable outcome*. The fundamental value does not have to be estimated; it is revealed.

Levitt (2004) [30] documents that betting markets are organised differently from financial ones: the bookmaker does not merely act as an intermediary balancing the two sides, but takes an active position, exploiting systematic biases of bettors. That observation matters for interpreting the differences between companies: they reflect not only differing predictive ability but also differing business strategy.

Market microstructure theory supplies the framework for understanding how information becomes embedded in price. The models of Kyle (1985) [29] and Glosten & Milgrom

(1985) [16] show that, in the presence of informed traders, price movement is the mechanism through which private information becomes public, and that the bid–ask spread is the market maker's protection against them. That framework bears directly on the difference between opening and closing prices (§4.1.3).

2.4 Favourite–longshot bias and odds conversion

The most systematically documented anomaly of betting markets is the *favourite–longshot bias*: low-probability outcomes are systematically priced more expensively than their true frequency warrants. The phenomenon has been confirmed across multiple sports and decades (Thaler & Ziemba, 1988) [42].

Shin (1992, 1993) [38, 39] offered an explanation that treats it not as a psychological bias of bettors but as a *rational response* by the bookmaker to the existence of informed traders. The model generates the shape of the bias endogenously and, most importantly for us, provides an inversion procedure that recovers fair probabilities together with an estimate of the proportion z of informed traders. Štrumbelj (2014) [41] systematically compares methods for converting odds into probabilities and finds that Shin's method outperforms proportional normalisation in the quality of the resulting probabilities.

2.5 Probability calibration

The distinction between the *reliability* and the *resolution* of a probabilistic forecast goes back to Murphy (1973) [34], who showed that every proper scoring rule decomposes into these components. The observation has an immediate practical consequence: a forecaster may discriminate excellently between likely and unlikely outcomes and still report wrong probabilities, an error correctable with no additional information whatsoever.

In the modern machine learning literature the issue resurfaced with Guo et al. (2017) [18], who showed that deep neural networks are systematically overconfident and that a one-parameter correction (*temperature scaling*) suffices to calibrate them. The present work uses the same family of correction, with the difference that the parameter is not fitted on a validation set but learned *online*, as a second OCO stage on top of the first.

2.6 Forecast combination

The idea that combining many forecasts beats selecting one has a long history in econometrics and meteorology. A persistent empirical finding, known as the *forecast combination puzzle*, is that the simple average often proves more robust than weighted combinations whose weights are estimated from data: the gain from better weighting is cancelled by the estimation noise in the weights themselves.

The combination of online aggregation with the sleeping-experts reduction is established practice outside this domain. Pfitzner et al. (2025) [35] apply exactly that pairing to operational temperature forecasting, waking experts that are poor on average but good in identifiable circumstances, and obtain second-order regret bounds for the result. The reduction this thesis relies on is therefore not novel in itself; what is new is the observation

that in a betting panel the sleeping pattern is not a nuisance to be tolerated but carries information, since which companies quote a given fixture is itself a signal, and that the same reduction induces a movement of the weight vector that must be controlled for before any gain is read as learning (§5.5.1).

That finding bears directly on our results. This thesis shows that learned combination beats the simple average when the experts differ substantively in quality, and falls short of it when they do not. The puzzle is therefore not a general law but a statement about homogeneous panels: the condition it depends on is the spread in expert quality, and that condition is measurable in advance.

2.7 Position of the present work

Taken together, the above delineates the gap this thesis addresses. Online learning theory offers algorithms with guarantees but is more rarely applied to betting markets with genuine partial coverage. The sports prediction literature often uses odds as a benchmark, but less often as *a set of experts to be combined*. And the economic literature on markets explains why companies differ, but does not exploit those differences algorithmically.

Online convex optimisation has been brought to betting markets, but from the opposite side of the counter. Zhu et al. (2024) [44] place the *bookmaker* in the role of the learner, updating posted odds as bets arrive, and prove $O(\sqrt{T})$ regret for the resulting pricing algorithms. Their central result is worth stating here because it is the mirror image of one of ours: they show that maximising the operator's profit and eliciting information from bettors are fundamentally incompatible objectives. Read from the bettor's side, that is the same separation this thesis measures between predictive and economic performance (§5.4), and it supplies a reason to expect it rather than merely an observation that it occurred.

Our contribution is the systematic synthesis of the three: online algorithms with guarantees, applied to the full set of available companies with correct handling of partial coverage, with statistical testing of every comparison and with an explicit distinction between predictive and economic performance.

Chapter 3

Theoretical background

This chapter presents the algorithms the thesis uses, the guarantees that come with them and the quantities by which their output is evaluated, and explains what each one contributes. Proofs are standard and are left to the cited sources.

3.1 The online learning framework

3.1.1 Online learning and regret

In the online learning setting (Cesa-Bianchi & Lugosi, 2006; Shalev-Shwartz, 2012; Hazan, 2016) [7, 37, 23] there is no separation into a training phase and an application phase. The algorithm operates in rounds: at round t it issues a prediction based only on what it has seen up to round $t-1$, the true outcome is then revealed, it incurs a loss, and it updates its state. Every prediction is therefore out of sample by construction, with no need to split the data.

The measure of success is not the absolute loss but the *regret*: how much worse the algorithm did than the best *fixed* choice in hindsight.

Definition 3.1 (Regret). Let $\mathcal{K}$ be a convex set, $f_1, \dots, f_T$ a sequence of convex loss functions, and $x_1, \dots, x_T \in \mathcal{K}$ the algorithm's choices, where x_t depends only on $f_1, \dots, f_{t-1}$. The regret is defined as

$$\text{Regret}_T \;=\; \sum_{t=1}^{T} f_t(x_t) \;-\; \min_{x \in \mathcal{K}} \sum_{t=1}^{T} f_t(x).$$

An algorithm is considered successful when its regret grows sublinearly in T, that is $\text{Regret}_T / T \to 0$: in the long run its average loss approaches that of the best fixed choice. What matters for our application is that this guarantee holds **for every** sequence, with no assumption of independence or of a stationary distribution.

This point deserves to be stated precisely, because it is the reason the OCO framework is chosen here over the statistical one. In statistical learning theory the object of interest is a bound on the *risk* $\mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$, the expected loss ℓ of a predictor h that maps an input x to a forecast of the outcome y, which presupposes that a fixed distribution $\mathcal{D}$ exists and that samples are drawn independently from it. Neither assumption is harmless here: odds are set by a market that changes over the decade, and consecutive matches share matchday, league and season. Regret requires neither. The price is that the guarantee

becomes *relative*: it does not say that we shall predict well, only that we shall not predict substantially worse than the best fixed combination of the experts. If all the experts are bad, the guarantee holds and the result remains bad.

The protocol. Formally, online convex optimization is the following game between the algorithm and a (potentially adversarial) environment:

for $t = 1, 2, \ldots, T$:
the algorithm chooses $x_t \in \mathcal{K}$;
the environment reveals a convex $f_t : \mathcal{K} \to \mathbb{R}$;
the algorithm incurs loss $f_t(x_t)$ and observes f_t (or part of it).

The phrase "or part of it" is essential and returns in §3.3.8: the amount of information returned to the algorithm after each round defines different problems with different optimal bounds. In the full information model the whole of f_t is revealed, and hence its gradient at every point.

Are the bounds tight? The regret bounds proved for the algorithms below are *upper* bounds: each says that the regret is at most so large. An upper bound on its own does not say whether an algorithm is good or merely whether its analysis is loose. That is settled by a *lower* bound, which says that no algorithm whatsoever can do better, and Theorem 3.1 is such a bound. For the experts problem the answer is known, and it rules out any improvement in order of magnitude.

Theorem 3.1 (Lower bound for prediction with expert advice). *For every prediction algorithm with N experts and every sufficiently large T there exists a sequence of losses $\ell_t \in [0,1]^N$ such that*

$$\text{Regret}_T \;\geq\; (1 - o(1))\,\sqrt{\tfrac{1}{2}\,T \ln N}.$$

Hedge's bound (Theorem 3.6) is $M\sqrt{T \ln N/2}$, which therefore matches the lower bound up to the factor M expressing the scale of the losses. The algorithm is thus *optimal in order of magnitude* and there is no room for improvement in the worst case. This matters for interpreting the empirical results of §5.1: when it is found there that the theoretically optimal step is two orders of magnitude smaller than the empirically best one, that does *not* mean the theory is wrong. It means the data are not the worst case, and that the distance between the worst case and the particular sequence we happen to have is large.

The static comparator. A second limitation is worth noting, one that proves essential at several points of this thesis: the comparator is *static*. The guarantee concerns comparison with the best fixed weight vector, chosen in hindsight but remaining a single vector for the entire period. When the object itself changes (when, for example, an expert stops participating) an algorithm that converges quickly to one answer is punished for precisely that. The generalisation that lifts the restriction is defined as follows.

Definition 3.2 (Dynamic regret and tracking regret). For a comparator sequence $u_1, \dots, u_T \in \mathcal{K}$ the *dynamic regret* is defined as

$$\mathrm{Regret}_T^d(u_{1:T}) \;=\; \sum_{t=1}^{T} f_t(x_t) \;-\; \sum_{t=1}^{T} f_t(u_t).$$

When the sequence is restricted to those changing value at most m times, that is $|\{t : u_{t+1} \neq u_t\}| \leq m$, the quantity is called the *tracking regret* of order m.

Static regret is the special case $m = 0$. Without a restriction on m no non-trivial bound is possible: with u_t free to be the hindsight-best expert at every round, the dynamic regret can grow linearly. Fixed-Share, in §3.3.6, is precisely the algorithm that trades a worse constant in the static regret for a non-trivial bound at $m > 0$.

3.1.2 Prediction with expert advice

In the special case of interest here, $\mathcal{K}$ is the probability simplex

$$\Delta_N \;=\; \Big\{ w \in \mathbb{R}^N : w_i \geq 0,\ \textstyle\sum_{i=1}^{N} w_i = 1 \Big\},$$

where N is the number of experts. At round t each expert i publishes a prediction $p_{t,i} \in \Delta_3$ over the three outcomes, the algorithm chooses weights $w_t \in \Delta_N$, and predicts the mixture

$$\hat{p}_t \;=\; \sum_{i=1}^{N} w_{t,i}\, p_{t,i}.$$

The outcome $y_t \in \{H, D, A\}$ is revealed and the algorithm incurs the logarithmic loss $\ell(\hat{p}_t, y_t) = -\log \hat{p}_t[y_t]$.

The logarithmic loss is a *strictly proper scoring rule*: its expectation is uniquely minimised when the forecaster reports its true probabilities. That makes it appropriate here, in contrast to accuracy, which is not a proper rule and ignores entirely how confident a prediction was.

Theorem 3.2 (Propriety of the logarithmic loss). *Let p be the true outcome distribution and q the reported one. Then*

$$\mathbb{E}_{y\sim p}\big[-\log q_y\big] \;=\; H(p) \;+\; \mathrm{KL}(p \,\|\, q) \;\geq\; H(p),$$

with equality if and only if $q = p$.

Here $H(p) = -\sum_y p_y \log p_y$ is the Shannon entropy of p, the loss that remains even with perfect knowledge of the distribution, and $\mathrm{KL}(p \,\|\, q) = \sum_y p_y \log(p_y/q_y)$ is the Kullback–Leibler divergence of q from p, the extra loss paid for reporting q instead (Cover & Thomas, 2006) [9].

The corollary is that a forecaster has no incentive whatsoever to report probabilities other than those it genuinely holds. This property fails for accuracy: a forecaster interested only in the proportion of correct picks may report 100% confidence on every match at no penalty, provided it preserves the same ordering.

Linear versus logarithmic pooling

The definition $\hat{p}_t = \sum_i w_{t,i} p_{t,i}$ is a choice, not the only possible one. It is called *linear pooling* and is the natural choice here, because it is what makes the problem convex in w and therefore solvable with the tools of online convex optimization. The alternative is *logarithmic pooling*, the normalised geometric mean

$$\hat{p}_k^{\log} \propto \prod_{i=1}^{N} p_{t,i}[k]^{\,w_{t,i}}.$$

The two behave qualitatively differently, and that difference predicts one of the empirical findings of this thesis.

Proposition 3.3 (Linear pooling is under-confident). *Let $\hat{p} = \sum_i w_i p_i$ with $w \in \Delta_N$. Then:*

(a) $\max_k \hat{p}_k \leq \max_i \max_k p_{i,k}$*, with equality only if all experts with $w_i > 0$ attain their maximum at the same k with the same value;*

(b) $H(\hat{p}) \geq \sum_i w_i H(p_i)$*, where H is the Shannon entropy, with equality if and only if all experts with $w_i > 0$ agree exactly.*

In other words, the mixture is always *less* confident than the average confidence of its members, and the confidence it loses is exactly their disagreement. This is not a defect of the implementation but a property of the operator: linear pooling systematically pushes probabilities toward the centre of the simplex. Logarithmic pooling does not have this property (it can produce a distribution sharper than any of its members, provided the members agree), and for that reason it is said to be *externally Bayesian*.

The consequence is predictable in advance and is confirmed empirically: stratifying the predictions into ten probability bands, an individual bookmaker is better calibrated than the mixture in *all ten*, even when the mixture has the lower overall log-loss. This is no contradiction. The log-loss decomposes (§3.4.1) into a resolution term and a reliability term; pooling improves the former and worsens the latter.

3.2 From odds to probabilities

3.2.1 De-vigging

Before any algorithm there is a modelling step whose result depends on an assumption, and which can therefore affect every number that follows. Published odds are not probabilities: their reciprocals sum to more than one. If o_i is the decimal odds of outcome i and $\pi_i = 1/o_i$, then

$$B = \sum_i \pi_i > 1,$$

and the excess $B - 1$ is the *overround*, the bookmaker's profit margin. Converting the π_i into probabilities requires an assumption about *how* that margin is distributed across the three outcomes.

Proportional normalisation. The simplest choice assumes the margin is distributed proportionally:

$$p_i \;=\; \frac{\pi_i}{B}.$$

It is transparent and even-handed across outcomes, but it embeds a strong implicit assumption: that the bookmaker charges the same percentage margin on a favourite and on a longshot. The empirical literature does not bear this out. The well-known *favourite–longshot bias* (the systematic overpricing of unlikely outcomes in published odds; Thaler & Ziemba, 1988 [42]) means precisely that the margin is heavier on longshots.

Shin's method. Shin (1992, 1993) [38, 39] derives the distribution of the margin from a microeconomic argument rather than assuming it. The model posits that a proportion z of bettors hold private information and bet with certainty on the true outcome, while the remaining $1 - z$ bet according to public probabilities. The bookmaker, knowing that such bettors exist but not who they are, prices so as to protect itself. That protection is more expensive where private information pays off most, namely on low-probability outcomes, and so the model generates the favourite–longshot bias *endogenously*.

Inverting the model, the fair probabilities are given by

$$p_i(z) \;=\; \frac{\sqrt{z^2 + 4(1-z)\,\pi_i^2/B} \;-\; z}{2(1-z)},$$

with z determined by the requirement $\sum_i p_i(z) = 1$. The equation has no analytic solution; in the implementation z is found per match by bisection over the interval $[0,\, 0.4]$. The interval is a generous bracket rather than a modelling choice: z is a proportion of bettors, and on these data it comes out at 3.1% on average, with a 99th percentile of 6.3%. The upper end is reached by 495 of the $835{,}550$ quotes (0.06%), all but one of them opening prices of Betfair Exchange with overrounds of 68% and above, against a median of 6% for the data as a whole; no z in the interval satisfies the condition there, and the cap acts as a bound.

Why both are retained. The de-vig choice is an assumption, not a datum, and could in principle determine a conclusion. For that reason the entire dataset is built twice, once with each method, and every central result is checked under both. It is of further interest that the estimated parameter z is itself a quantity with an interpretation, which we exploit later as an indicator of information arrival.

3.2.2 Microstructure of the betting market

The de-vigging of §3.2.1 treats the overround as a technical obstacle to be removed. It is worth saying *why* it exists, because the answer explains two of the central findings of this thesis: the opening–closing difference, and why one particular firm is not beaten.

The market maker's problem. The bookmaker publishes prices before learning the outcome and accepts bets from two kinds of counterparty: the *uninformed*, who bet for reasons unrelated to information, and the *informed*, who bet only when the price is wrong in their favour. The second category creates *adverse selection*: the mere fact that someone accepts your price is bad news about your price.

The models of Glosten & Milgrom (1985) [16] and Kyle (1985) [29] show that the rational response is not to refuse to trade but to price with a *margin*: the maker charges enough that the profits from the uninformed cover the losses to the informed. The overround is exactly that margin, and its size is an increasing function of the estimated proportion of informed traders. In Kyle's model the price moves linearly with the *order flow*, the net volume of orders arriving on each side (here, the amount bet on each outcome), with a coefficient λ measuring market *depth*; each transaction reveals part of the private information and the price converges toward fundamental value as expiry approaches.

Remark 3.4 (Why the closing price is the better forecast). The consequence is immediate and testable: if price movement is the mechanism by which private information becomes public, then the closing price incorporates whatever the market learned between opening and kick-off, and must be systematically more accurate. More strongly: its advantage must be *larger where the price moved more*, since that is precisely where more information arrived. This is a quantitative prediction and not a mere qualitative expectation, and it is tested in §5.2.3.

The connection to Shin. Shin's model, which §3.2.1 uses as a de-vigging method, is the same idea in a form one can invert: the parameter z *is* the estimated proportion of informed traders, and the favourite–longshot bias arises endogenously because protection costs more where private information pays off more, namely on unlikely outcomes. The quantity z is therefore two things at once: a nuisance parameter for converting odds into probabilities, and *a measurable quantity with an economic interpretation.*

Two things the framework does not license. First, odds alone do not distinguish *private* from *public* information: a large movement may be due to a leaked line-up or to an announced injury. The claim concerns the *arrival* of information, not its nature, and any talk of "insider detection" would be an overstatement. Second, the model explains why a margin exists but does not predict its *size* for a particular firm; that depends on commercial strategy, cost of capital and risk tolerance, that is on factors outside our data (§6.2).

3.3 The algorithms

Three algorithms are used to learn the weights of the experts: *Hedge*, *Online Gradient Descent* (OGD) and *Follow-The-Regularized-Leader* (FTRL). They are presented in that order, followed by the *Aggregating Algorithm*, which exploits a stronger property of the loss, and by the reduction that lets all of them handle experts that do not quote every match.

3.3.1 Hedge

The Hedge, or multiplicative weights, algorithm (Freund & Schapire, 1997) [12] maintains one weight per expert and decays it exponentially in that expert's loss:

$$w_{t+1,i} \;\propto\; w_{t,i}\, e^{-\eta_t\, \ell_{t,i}}, \qquad \ell_{t,i} = -\log p_{t,i}[y_t].$$

The quantity $\eta_t > 0$ is the *step size*, also called the learning rate: it sets how far one round's loss is allowed to move the weights, and the whole of §5.1 is about choosing it.

With a suitable η_t the algorithm attains regret $O(\sqrt{T \log N})$. The dependence on N is only logarithmic, which is what makes large expert sets tractable.

Lemma 3.5 (Hoeffding, 1963 [25]). *Let X be a random variable taking values in an interval of width M. Then for every $\lambda \in \mathbb{R}$*

$$\ln \mathbb{E}\big[e^{\lambda X}\big] \;\le\; \lambda\, \mathbb{E}[X] \;+\; \frac{\lambda^2 M^2}{8}.$$

Theorem 3.6 (Hedge regret bound). *With a fixed $\eta > 0$ and losses $\ell_{t,i} \in [0, M]$, Hedge satisfies*

$$\sum_{t=1}^{T} \langle w_t, \ell_t \rangle \;-\; \min_i \sum_{t=1}^{T} \ell_{t,i} \;\le\; \frac{\ln N}{\eta} \;+\; \frac{\eta\, M^2 T}{8}.$$

With $\eta = \sqrt{8 \ln N / (M^2 T)}$ the bound becomes $M\sqrt{T \ln N / 2}$.

Note that the optimal η depends on the horizon T, which in an online setting is not known in advance. There are two established remedies. The first is the *doubling trick*: the algorithm runs with a fixed η tuned for horizon 2^j, and each time the elapsed time exceeds 2^j it restarts with $j+1$. The total regret is then the sum of a geometric series, hence of the same order with a multiplicative overhead of $(\sqrt{2}-1)^{-1} \approx 3.41$. The second, used here, is the decaying schedule $\eta_t \propto 1/\sqrt{t}$, which attains the same order without restarts.

Proposition 3.7 (Decaying schedule). *With $\eta_t = c/\sqrt{t}$ the second term of the bound becomes $\frac{cM^2}{8} \sum_{t=1}^{T} t^{-1/2} \le \frac{cM^2}{4}\sqrt{T}$, while the first becomes $\frac{\ln N}{c}\sqrt{T}$. The total bound remains $O(M\sqrt{T \ln N})$ for every T simultaneously, without the horizon being known.*

This choice has consequences examined empirically in §5.1, where the constant c dictated by theory is found to be, for these data, seriously wrong, too large for Hedge, and too small by two orders of magnitude for OGD.

3.3.2 Online Gradient Descent

OGD (Zinkevich, 2003) [45] takes a step in the direction opposite to the gradient and projects back onto the feasible set:

$$w_{t+1} \;=\; \Pi_{\Delta_N}\big(w_t - \eta_t \nabla f_t(w_t)\big),$$

where Π_{Δ_N} is the Euclidean projection onto the simplex, computable in $O(N \log N)$ (Duchi et al., 2008) [10]. For convex functions with bounded gradient the regret is $O(\sqrt{T})$.

Two ingredients of this scheme are used throughout the thesis and deserve explicit justification: the linearisation of the loss and the computation of the projection.

Why linearisation is permitted. The implementation does not use the gradient of the true loss $-\log\langle w, p_t\rangle$ but the vector of per-expert losses ℓ_t, that is the gradient of the *linear* function $\tilde{f}_t(w) = \langle w, \ell_t \rangle$. This substitution is not an approximation to be checked after the fact; it is safe by construction for any convex loss.

Lemma 3.8 (Linearisation does not loosen the bound). *Let f_t be convex and $g_t \in \partial f_t(x_t)$ any subgradient at the played point. Then for every $u \in \mathcal{K}$*

$$\sum_{t=1}^{T} f_t(x_t) - \sum_{t=1}^{T} f_t(u) \;\le\; \sum_{t=1}^{T} \langle g_t,\, x_t - u \rangle,$$

that is, the regret with respect to the true losses is bounded by the regret with respect to their linearisations. Any bound proved for linear losses therefore holds automatically for the original convex ones.

The direction of the inequality is the one we need: the linear problem is *harder*, so solving it well implies that the original was solved well too. The price is that linearisation discards the curvature, and with it the opportunity for the far stronger $O(\log T)$ bound permitted by the exp-concavity of the logarithmic loss (§3.3.4). This is a deliberate trade-off and not an oversight: §5.1 shows that the algorithm which does exploit the curvature performs worse on these data, for reasons that concern not the quality of the bound but the static comparator it refers to.

The projection onto the simplex. The projection is the only non-trivial operation in the step and it is executed tens of thousands of times, so its closed form matters.

Theorem 3.9 (Euclidean projection onto the simplex). *Let $v \in \mathbb{R}^n$ and $\Pi_{\Delta_n}(v) = \arg\min_{w \in \Delta_n} \|w - v\|_2^2$. Then*

$$\Pi_{\Delta_n}(v)_i \;=\; \max\{v_i - \theta,\, 0\},$$

where θ is the unique number satisfying $\sum_i \max\{v_i - \theta, 0\} = 1$. If u denotes the vector v with its entries sorted in decreasing order and $\rho = \max\{j : u_j - \frac{1}{j}(\sum_{r \le j} u_r - 1) > 0\}$, then $\theta = \frac{1}{\rho}\big(\sum_{r \le \rho} u_r - 1\big)$.

Remark 3.10 (The projection is not passive). The projection is not merely a feasibility mechanism. Because it zeroes every coordinate below the threshold θ, it is *sparsifying*: it imposes exactly zero weight on experts, not merely small weight. Under the Sleeping Experts reduction of §3.3.5 it is reapplied every time the awake set changes, and the empirical investigation of §5.3.2 shows that weight movement in the late stages of the period is due mainly to it and not to the gradient step: a finding that overturns the intuitive expectation that a decaying η_t "freezes" the algorithm.

Theorem 3.11 (OGD regret bound). *Let $\mathcal{K}$ be convex with diameter $D = \max_{u,v \in \mathcal{K}} \|u - v\|$ and bounded gradients, $\|g_t\| \le G$. With a fixed η,*

$$\mathrm{Regret}_T \;\le\; \frac{D^2}{2\eta} \;+\; \frac{\eta G^2 T}{2},$$

and with $\eta = D/(G\sqrt{T})$ it follows that $\mathrm{Regret}_T \le DG\sqrt{T}$.

In our case $\mathcal{K} = \Delta_N$, so $D = \sqrt{2}$, and G is bounded by the M of §5.1. This structure (that η multiplies the gradient of *one* round) explains why OGD responds to step scaling in the opposite way to Hedge.

Remark 3.12 (The structure of η and the direction of tuning). The reason Hedge and OGD respond in *opposite* directions to scaling of the step is structural and visible in the formulas. In Hedge, η sits in an *exponent* over a single round's loss: a smaller η means slower exponential decay, hence *longer memory*. In OGD, η *multiplies* a single round's gradient: a larger η simply means a larger step in the same direction. The two parameters, despite sharing a name, do not control the same quantity.

3.3.3 Follow-The-Regularized-Leader

FTRL (Shalev-Shwartz, 2012; Hazan, 2016) [37, 23] selects at each round the point minimising the entire accumulated loss, plus a regularisation term R preventing excessively abrupt movement:

$$w_{t+1} \;=\; \arg\min_{w\in\Delta_N} \Big\{ \eta_t \sum_{s=1}^{t} \langle \ell_s, w\rangle + R(w) \Big\}.$$

With $R(w) = \frac{1}{2}\|w - w_0\|^2$, where w_0 is the uniform starting vector, the solution is again given by projection onto the simplex. The difference from OGD is that FTRL is *"lazy"*: it rebuilds the weights from the whole history each time, instead of stepping from the previous position.

Why the regularisation term is needed. The obvious strategy without R (pick whatever was optimal so far, known as Follow-The-Leader or FTL) fails spectacularly. With two experts and losses alternating as $(\frac{1}{2}, 0), (0, 1), (1, 0), (0, 1), \ldots$, FTL picks at every round exactly the expert about to lose the next round and incurs linear regret, whereas the best fixed expert loses about $T/2$. The term R is what stabilises successive choices; the lemma below shows that stability is precisely what is required.

Lemma 3.13 (Be-The-Leader). *Let $w_t^\star = \arg\min_w\{\sum_{s=1}^{t}\langle \ell_s, w\rangle + R(w)/\eta\}$ be the FTRL choice. If the algorithm were allowed to play $w_t^\star$* after *seeing ℓ_t, then for every u*

$$\sum_{t=1}^{T} \langle \ell_t, w_t^\star\rangle \;\le\; \sum_{t=1}^{T} \langle \ell_t, u\rangle + \frac{R(u) - R(w_0^\star)}{\eta}.$$

Theorem 3.14 (FTRL regret bound). *With R strongly convex with respect to $\|\cdot\|_2$ with constant 1, losses satisfying $\|\ell_t\| \le G$, and R bounded on $\mathcal{K}$ by $D_R = \max_u R(u) - \min_u R(u)$, FTRL satisfies*

$$\mathrm{Regret}_T \;\le\; \frac{D_R}{\eta} \;+\; \eta\, G^2 T,$$

and with $\eta = \sqrt{D_R/(G^2T)}$ it follows that $\mathrm{Regret}_T \le 2G\sqrt{D_R\, T}$.

Remark 3.15 (OGD and FTRL as "greedy" and "lazy" projection). The two algorithms are not unrelated: with quadratic R, FTRL can be written as $w_{t+1} = \Pi_{\mathcal{K}}(w_0 - \eta \sum_{s\le t} \ell_s)$, that is a projection of the accumulated sum, whereas OGD is $w_{t+1} = \Pi_{\mathcal{K}}(w_t - \eta\ell_t)$, a projection after every step. The difference is *where* the projection intervenes: FTRL keeps the unprojected sum and projects at the end ("lazy"), OGD projects and continues from the projected point ("greedy"). Without constraints the two coincide; with constraints they

diverge, and precisely this divergence explains why they react in opposite ways to step scaling: in FTRL, η multiplies the entire accumulated history, so increasing it drives the algorithm towards unstable Follow-The-Leader, whereas in OGD it multiplies the gradient of a single round.

3.3.4 Exp-concavity and the Aggregating Algorithm

The bounds of Theorems 3.6, 3.11 and 3.14 grow as $O(\sqrt{T})$ and hold for any convex loss. For certain losses a substantially stronger result is attainable.

Definition 3.3 (Exp-concavity). A function f is called α-exp-concave if $e^{-\alpha f}$ is concave.

The logarithmic loss of the mixture is 1-exp-concave: if $f(w) = -\log\langle w, p\rangle$, then $e^{-f(w)} = \langle w, p\rangle$, which is *linear* in w and hence concave. This is a stronger property than convexity and a weaker one than strong convexity; its importance is that it permits a logarithmic, or even constant, bound instead of $O(\sqrt{T})$.

The Aggregating Algorithm. Vovk's (1990) [43] algorithm is not a variant of Hedge but a different construction. It maintains weights updated *Bayesianly* with the likelihood as the weight,

$$w_{t+1,i} \;\propto\; w_{t,i}\, e^{-\ell_{t,i}} \;=\; w_{t,i}\, p_{t,i}[y_t],$$

that is exactly Bayes' rule with a uniform prior, and predicts $\hat{p}_t = \sum_i w_{t,i}\, p_{t,i}$. For the logarithmic loss this scheme is identical to Hedge at $\eta = 1$, which is why the investigation of §5.1 examines it as the special case "constant $\eta = 1$".

Theorem 3.16 (Aggregating Algorithm bound). *For an α-exp-concave loss and N experts, the Aggregating Algorithm satisfies*

$$\mathrm{Regret}_T \;\le\; \frac{\ln N}{\alpha},$$

that is a bound independent of the horizon T.

For $N = 26$ experts this means regret at most $\ln 26 \approx 3.26$ *in total*, against hundreds for the $O(\sqrt{T})$ bounds at $T \approx 76{,}000$. Note that the result does not contradict the lower bound of Theorem 3.1: that one concerns *linear* losses, which are not exp-concave, and the difference between the two cases is exactly the curvature that the linearisation of Lemma 3.8 throws away.

This result is theoretically far stronger. It is therefore worth stating explicitly that on our data the corresponding algorithm performs *worse than the uniform average*. The explanation is not that the bound is wrong but that it concerns a *static* comparator: it guarantees that we will not do much worse than the best *fixed* expert. When the best expert stops participating in the middle of the period, the guarantee remains formally true but practically vacuous, and an algorithm that rapidly concentrates its mass on a few experts is penalised for precisely that.

3.3.5 The Sleeping Experts reduction

The algorithms above assume that every expert participates in every round. In our data this does not hold: many bookmakers cover only a small fraction of the matches. We use the classical *Sleeping Experts* or *Specialists* reduction (Freund, Schapire, Singer & Warmuth, 1997; Blum & Mansour, 2007) [13, 3]. Let $A(t) \subseteq \{1, \dots, N\}$ be the set of "awake" experts at round t. Then:

- each expert maintains its own persistent state (a weight for Hedge, a position for OGD, an accumulated loss for FTRL);
- the prediction is formed only from the awake ones, with their weights renormalised so as to constitute a distribution over $A(t)$ and nothing else;
- after the round only the state of the awake experts is updated; the sleeping ones remain *frozen* exactly as they were;
- a round in which nobody is awake is skipped entirely.

The reduction gives each expert i a regret guarantee measured *only over the rounds in which it was awake*, instead of requiring participation in all of them.

Theorem 3.17 (Per-expert regret under partial participation). *Let $T_i = \{t : i \in A(t)\}$ be the rounds in which expert i is awake and $n_i = |T_i|$. Hedge with the above reduction satisfies, for every i simultaneously,*

$$\sum_{t \in T_i} \Big(\langle \bar{w}_t, \ell_t \rangle - \ell_{t,i} \Big) \;\le\; \frac{\ln N}{\eta} \;+\; \frac{\eta M^2 n_i}{8},$$

where $\bar{w}_t$ are the weights renormalised over $A(t)$. With η tuned to n_i the bound becomes $M\sqrt{n_i \ln N/2}$.

The substantive consequence is that a bookmaker covering 7% of the matches does not "dilute" the guarantee of the rest: each expert is judged on its own domain. This is exactly what a panel with coverage ranging from 6.9% to 99.8% requires.

Remark 3.18 (A temptation the bound creates and the data reject). Theorem 3.17 has n_i where the original had T, so it suggests of its own accord that each expert's step should decay as $1/\sqrt{n_i}$ (on its own clock) rather than as $1/\sqrt{t}$ on the global one. The suggestion is theoretically sound and was tried; it was rejected empirically (§5.6). The reason is instructive: an expert with 7% coverage acquires a step roughly 14 times larger than one with full coverage at the same point in time, so the mixture chases the rare experts. The bound is not wrong; it is simply that the quantity it guarantees is not the one we are interested in minimising.

Two details prove critical in the implementation. First, the persistent weight vector must never be renormalised globally, only the copy used for the prediction. The reason lies in how Theorem 3.17 is established: the lower bound relies on the weight $w_{t,i}$ of a sleeping expert remaining *literally* unchanged, and a global renormalisation changes it. Second, the last row of the weight matrix is not "the final weight of each expert", since it is zero for anyone who did not participate in the last round; the interpretable quantity is the weight each one held the last time it was awake.

3.3.6 Fixed-Share

The Sleeping Experts reduction creates a side problem: an expert that sleeps for a long stretch keeps its weight frozen, and that weight is not eroded by the losses of the others. Rare experts can thus end up with disproportionate influence. The Fixed-Share algorithm (Herbster & Warmuth, 1998) [24] introduces a small leak α towards the uniform distribution after every update:

$$w_{t+1,i} \;=\; (1-\alpha)\,\tilde{w}_{t,i} \;+\; \frac{\alpha}{N}\sum_{j}\tilde{w}_{t,j}, \qquad \tilde{w}_{t,i} \;=\; w_{t,i}\,e^{-\eta\ell_{t,i}},$$

where the second form preserves the total mass exactly. Theoretically this changes the comparator from the best fixed vector to the best *sequence* of vectors with a bounded number of switches, that is to the tracking regret of Definition 3.2.

Theorem 3.19 (Fixed-Share tracking regret). *Let the comparison sequence select one expert at each round and switch at most m times. Fixed-Share with suitable η and $\alpha = m/(T-1)$ satisfies*

$$\mathrm{Regret}_T^{d} \;\le\; \frac{1}{\eta}\Big[\ln N + m\ln(N-1) + (T-1)\,\mathcal{H}(\alpha)\Big] + \frac{\eta M^2 T}{8},$$

where $\mathcal{H}(\alpha) = -\alpha\ln\alpha - (1-\alpha)\ln(1-\alpha)$ is the binary entropy. Optimising, the bound is $O\big(M\sqrt{T\,(m\ln N + m\ln\frac{T}{m})}\big)$.

The bound quantifies a trade-off: for $m = 0$ the term $(T-1)\mathcal{H}(\alpha)$ is pure overhead relative to plain Hedge, while for $m > 0$ it is the price that buys a guarantee Hedge does not offer at all. The empirical result of §5.6 is exactly what the trade-off predicts when the underlying sequence does not switch enough: every $\alpha > 0$ worsens the log-loss. More interestingly, the optimal value of α turns out to be 0 *once* the step has been tuned, because the two quantities control the same thing (how fast the algorithm forgets) and the step is the more efficient of the two instruments.

3.3.7 Markov chains: three different uses

The term "Markov chain OGD" is used in the literature for at least three different things, and this thesis tries all three. We separate them explicitly, because the negative results of one say nothing about the others.

(a) A chain on the weight vector. This generalises Fixed-Share along the lines of the mixing schemes of Bousquet & Warmuth (2002) [4]. After the gradient step and the projection, mass is moved by a row-stochastic matrix Q, that is a non-negative matrix whose every row sums to one (Q rather than M, which already denotes the bound on a single round's loss):

$$v_t = \Pi_\Delta(w_t), \qquad \hat{w} = v_t - \eta_t\ell_t, \qquad w_{t+1} = Q^\top\hat{w}.$$

The Fixed-Share of §3.3.6 is the special case $Q = (1-\alpha)I + \frac{\alpha}{N}\mathbf{1}\mathbf{1}^\top$, that is a uniform leak. A general Q is strictly more expressive, because it can say *where* the lost mass goes: e.g. to the opening/closing "twin" of the same house, or to peers in the same market phase. Row-stochasticity guarantees conservation of total mass, which is exactly what makes $\alpha = 0$ leave the weights exactly, not approximately, unchanged.

(b) A chain on a latent state (hidden Markov model). A hidden Markov model (Rabiner, 1989) [36] describes a system that moves between unobserved states according to a Markov chain, while only the effects of those states are observed. Here the hidden variable is not the weights but a state $k \in \{1, \dots, K\}$, and each state carries its own weight vector. The belief b_t over the states is updated by the *forward recursion*:

$$b_{t+1}(k) \;\propto\; b_t(k)\, L_t(k), \qquad L_t(k) = \langle w_t^{(k)}, p_t[y_t] \rangle,$$

followed by the transition $b \;\leftarrow\; (1-\tau)\, b + \tau/K$. The prediction is the mixture $\sum_k b_t(k)\, \langle w_t^{(k)}, p_t \rangle$ and each state is updated scaled by its *responsibility* $b_t(k)$.

Remark 3.20 (Why symmetry is a trap). If all states start close to one another and receive the *same* loss vector, they follow identical trajectories; then the $L_t(k)$ are equal for every k, the belief stays uniform, and the scheme degenerates into an OGD with step divided by K. The symmetric solution is a fixed point of the dynamics. An explicit differentiation mechanism is needed, sharpening of the belief so that it reacts to small per-round differences, forgetting so that it tracks the present instead of the sum of history, and a per-state clock so that none of them under-steps. §5.5.2 shows that without these the model fails in exactly this way.

(c) Markov Chain Gradient Descent: dependent sampling. This is what Sun, Sun & Yin (2018) [40] study, and it has nothing to do with weight transitions. The classical analysis of *stochastic gradient descent* (SGD), which updates on one randomly drawn sample at a time, assumes independent samples; when the samples are consecutive points of a Markov trajectory, after t rounds the algorithm has seen *less* than t independent units of information. If the dependence horizon is τ rounds, the effective number of samples is $\sim t/\tau$ and the schedule becomes

$$\eta_t \;=\; \frac{D}{M\sqrt{t/\tau + 1}} \;\approx\; \sqrt{\tau}\; \eta_t^{\text{iid}},$$

where η_t^{iid} is the schedule that would apply if the samples were independent and identically distributed (iid); that is, a dependent sequence *justifies* a step inflated by $\sqrt{\tau}$.

Remark 3.21 (An independent theoretical explanation of OGD's step multiplier). The empirical tuning of §5.1 finds that OGD performs best with a step 100 times larger than the textbook schedule (a multiplier of $\times 100$). The argument above predicts part of that inflation from theory alone, and it needs two numbers. The first is the dependence horizon τ. For it we take the block length of the moving block bootstrap (§3.4.2), which estimates the same quantity for an entirely different purpose: $b \approx n^{1/3}$, and with $n = 76{,}584$ rounds this gives $b \approx 42$, so the inflation is $\sqrt{42} \approx 6.5$. The second is the loss bound M in the schedule. Theory sets $M = \ln(1/\varepsilon) \approx 13.82$, the largest loss a probability clipped at $\varepsilon = 10^{-6}$ can produce (§4.2.4), whereas the largest loss actually observed in the data is 3.76; since the step is inversely proportional to M, using the observed bound inflates it by a further $13.82/3.76 \approx 3.7$. Together the two give $6.5 \times 3.7 \approx 24$. This does not reach the empirical $\times 100$, but it is a partial explanation obtained *without any grid search*, and for that reason worth reporting, even when the algorithm itself improves nothing.

3.3.8 Bandit online convex optimisation

Everything above assumes *full information*: after round t the algorithm learns the entire vector ℓ_t, that is how well *every* expert did, not only itself. In our application this is feasible because we score all 26 bookmakers after the fact. It is not, however, to be taken for granted. A forecaster that records only its own performance (and this is the usual situation in deployment) observes a single number per round:

$$f_t(w_t) \;=\; -\log\langle w_t, p_t[y_t]\rangle.$$

This model is called *bandit*, and it defines a genuinely harder problem. The difficulty is specific: to move in the direction that decreases a function one needs its gradient, and a single scalar value does not determine it. The solution of Flaxman, Kalai & McMahan (2005) [14] is to *construct* a stochastic estimate of the gradient from function values alone.

Lemma 3.22 (Gradient estimation from a single value). *Let $f : \mathbb{R}^d \to \mathbb{R}$ and $\hat f_\delta(x) = \mathbb{E}_{v\sim\mathbb{B}^d}[f(x+\delta v)]$ be its smoothing with v uniform on the unit* ball. *If u is distributed uniformly on the unit* sphere $\mathbb{S}^{d-1}$*, then*

$$\mathbb{E}_u\Big[\frac{d}{\delta}\, f(x+\delta u)\, u\Big] \;=\; \nabla \hat f_\delta(x).$$

The estimate is therefore *unbiased* for the gradient of the smoothed function, not of the original one, and the smoothing introduces an $O(\delta)$ error. At the same time its variance scales as $O(1/\delta^2)$: small δ gives small bias and enormous noise, large δ the converse. Optimising the trade-off gives the central result.

Theorem 3.23 (Flaxman, Kalai & McMahan, 2005). *For convex losses bounded in absolute value, Bandit Gradient Descent with $\delta \sim T^{-1/4}$ and suitable η achieves*

$$\mathbb{E}\big[\mathrm{Regret}_T\big] \;=\; O\big(T^{3/4}\big).$$

The comparison with the $O(\sqrt{T}) = O(T^{1/2})$ of full information is the quantitative price of ignorance. The bound improves to $O(\sqrt{T})$ if *two* queries to the same function are permitted per round (Agarwal, Dekel & Xiao, 2010) [1], with the estimate

$$\hat g_t \;=\; \frac{d}{2\delta}\big(f_t(x_t+\delta u) - f_t(x_t-\delta u)\big)\, u,$$

which is a finite difference along a random direction. The reason for the improvement is immediately visible: the subtraction cancels the constant term $f_t(x_t)$, which in the one-point estimate is multiplied by d/δ and dominates the variance. As $\delta \to 0$ the two-point estimate converges to the exact directional derivative, that is, the two-query model is not really bandit for smooth functions: an observation with a direct consequence for the measured results.

The geometry of the simplex. Applying this to our $\mathcal{K} = \Delta_N$ requires three adaptations that are not decorative:

- the simplex is $(N-1)$-dimensional (the constraint $\sum_i w_i = 1$ removes one of the N degrees of freedom), so the perturbation must lie in the *tangent space* $\{u : \sum_i u_i = 0\}$ and the factor in Lemma 3.22 is $d = N-1$, not N. A perturbation along a random direction of $\mathbb{R}^N$ would take the point off the plane $\sum_i w_i = 1$;

- the point about which we perturb must stay away from the boundary, otherwise the perturbation is infeasible. We use the *shrunk* simplex $\Delta^{\xi} = \{w \in \Delta_N : w_i \geq \xi/N\}$, where $\xi \in (0, 1)$ is a small shrinkage parameter that fixes the minimum weight, and the choice $\delta = \xi/N$ guarantees feasibility for free: since $|u_i| \leq 1$, every coordinate of $w \pm \delta u$ remains non-negative, while the sum stays exactly 1;
- the shrinkage introduces its own bias, since the optimum may lie on the boundary, and in our data it typically does, since the projection of Theorem 3.9 zeroes coordinates.

Under the Sleeping Experts reduction all of the above is restricted to $A(t)$, so d and δ vary from round to round with the number of awake experts. A round with a single awake expert has $d = 0$: there is no direction to perturb along and nothing is learned.

What "the loss" of a bandit algorithm measures. A bookkeeping detail that changes the meaning of the numbers. The algorithm publishes the perturbed point $w_t + \delta u_t$, not the centre w_t, so the loss it actually pays includes the cost of exploration; this is also the quantity Theorem 3.23 refers to. The centre w_t is, however, the prediction we would deploy, and it is the only quantity directly comparable with OGD across the whole battery of metrics. Both are reported (§5.5.1): the first is the honest bandit bookkeeping, the second answers the question "what did we actually learn".

3.4 Scoring and statistical testing

3.4.1 Proper scoring rules and calibration

Beyond log-loss we use the Brier score, $\frac{1}{T}\sum_t \|\hat{p}_t - e_{y_t}\|^2$, where e_{y_t} is the vector with 1 in the position of the outcome y_t that occurred and 0 elsewhere, so that the score is the mean squared distance of the forecast from what happened. It is also strictly proper. The general theory (Gneiting & Raftery, 2007) [17] shows that every proper scoring rule corresponds to a convex function G on the simplex via Savage's representation,

$$S(q, y) \;=\; G(q) \;+\; \langle \nabla G(q),\, e_y - q \rangle,$$

where the expected score $\mathbb{E}_{y\sim p}[S(q, y)]$ is minimised at $q = p$ precisely because G is convex. Log-loss arises for G the negative entropy and Brier for $G(q) = \|q\|^2$. The two are therefore not independent choices but two points of the same family: a fact that explains why, as observed empirically, they agree on 18 verdicts out of 18 (the comparisons of each algorithm against the plain average on every panel, §5.2.2 and Tables A.6–A.7), whereas accuracy, which does not belong to the family, reproduces none of them.

Murphy's decomposition. If predictions are grouped into classes sharing a common value, every proper rule decomposes into three terms:

$$\underbrace{\overline{S}}_{\text{score}} \;=\; \underbrace{\mathbb{E}\big[d(\hat{p}, \bar{y}_{\hat{p}})\big]}_{\text{unreliability}} \;-\; \underbrace{\mathbb{E}\big[d(\bar{y}_{\hat{p}}, \bar{y})\big]}_{\text{resolution}} \;+\; \underbrace{d(\bar{y}, \cdot)}_{\text{uncertainty}},$$

where $\bar{y}_{\hat{p}}$ is the actual frequency in the class with prediction $\hat{p}$, $\bar{y}$ the overall base rate and d the divergence induced by G. The third term is a property of the data and nobody

can improve it. The second requires genuine information. The first, however (unreliability) is correctable *without any additional information*, by monotonically rearranging the probabilities already in hand. This is exactly where the calibration stage intervenes, and Proposition 3.3 has already predicted that the aggregation operator systematically inflates precisely that term.

Temperature scaling. The correction we use raises the probabilities to an exponent $s = 1/T_{\text{temp}}$ and renormalises,

$$\tilde{p}_k \;=\; \frac{\hat{p}_k^{\,s}}{\sum_j \hat{p}_j^{\,s}}.$$

Values $s < 1$ "soften" the distribution, values $s > 1$ sharpen it. This is a one-parameter family which preserves the ranking of the outcomes and therefore cannot change any argmax decision: a property that explains why calibration improves log-loss substantially while leaving accuracy unchanged.

Proposition 3.24 (Convexity in the exponent). *Let $a_k = \log \hat{p}_k$ and let y be the realised outcome. The loss*

$$\phi(s) \;=\; -\log \tilde{p}_y \;=\; -s\, a_y \;+\; \log \sum_j e^{s\, a_j}$$

is convex in s, with $\phi'(s) = \mathbb{E}_{k\sim\tilde{p}^{(s)}}[a_k] - a_y$ and $\phi''(s) = \mathrm{Var}_{k\sim\tilde{p}^{(s)}}(a_k) \geq 0$.

Convexity is what makes s learnable by projected OGD, that is a *second OCO stage* on top of the output of the first, which remains fully causal: the temperature at round t depends only on outcomes up to round $t-1$. The form of the derivative has a further direct interpretation: the step is positive when the realised outcome had a log-probability below the average, i.e. the algorithm sharpens the distribution when it is surprised and softens it when it is not.

3.4.2 Statistical significance: the moving block bootstrap

Comparing two forecasters requires a test on the difference of their per-round losses. A simple t-test would be inappropriate, since consecutive losses are not independent: matches of the same round and the same season share difficulty. The consequence of ignoring this is specific and one-sided: for a positively correlated sequence,

$$\mathrm{Var}\Big(\tfrac{1}{n}\sum_t d_t\Big) \;=\; \frac{\sigma^2}{n}\Big(1 + 2\sum_{k\geq 1}\big(1 - \tfrac{k}{n}\big)\rho_k\Big) \;>\; \frac{\sigma^2}{n},$$

where d_t is the difference between the two forecasters' losses at round t, σ^2 its variance and ρ_k the correlation between differences k rounds apart. Since the ρ_k are positive, the independence formula *underestimates* the variance, confidence intervals come out far too narrow and results look significant when they are not.

We use the *moving block bootstrap* (Künsch, 1989) [28], which resamples with replacement *blocks* of consecutive observations of length b, thereby preserving the local correlation: each block carries its internal dependence structure with it, and only correlations at lags greater than b are destroyed. The 95% confidence interval is obtained from the quantiles of the bootstrap distribution.

Remark 3.25 (Why $b \approx n^{1/3}$). The choice of length is a bias–variance trade-off. Small b destroys dependence that is really there, leaving an $O(1/b)$ bias in the estimate of the large-sample variance; large b leaves few effectively independent blocks, namely n/b, and raises the variance of the estimate itself by $O(b/n)$. Minimising the mean squared error $O(b^{-2}) + O(b/n)$ gives $b \propto n^{1/3}$ (Hall, Horowitz & Jing, 1995) [19]. For $n \approx 76{,}000$ this gives $b \approx 42$, about one and a half match rounds: an order of magnitude consistent with the natural dependence horizon of the data. The same quantity reappears independently as a dependence horizon in the step-size study, selected there for an entirely different reason.

Sign convention. We define $d_t = \ell_t(A) - \ell_t(B)$, so a **negative** mean difference means the first is *better*. For log-loss, lower is better; for the logarithmic growth rate of capital the opposite holds.

3.5 Bet sizing

3.5.1 The Kelly criterion

Given an estimated probability p and decimal odds o, staking a fraction f of capital multiplies the capital by $1 + f(o-1)$ if the outcome occurs and by $1 - f$ if it does not, so the expected logarithmic growth rate is

$$g(f) \;=\; p \ln\big(1 + f(o-1)\big) \;+\; (1-p)\ln(1-f).$$

The fraction that maximises it is the one of Kelly (1956) [27],

$$f^* \;=\; \frac{p\,(o-1) - (1-p)}{o-1} \;=\; p - \frac{1-p}{o-1}.$$

The criterion is not an arbitrary preference for the logarithm. Breiman (1961) [5] showed that the strategy maximising the expected logarithmic rate is *asymptotically optimal* in two strong senses: it maximises the growth rate of capital almost surely, and it minimises the expected time for the capital to reach any prescribed large level. Logarithmic utility therefore follows from the repeated structure of the problem, not from an assumption about the bettor's preferences.

Why fractional Kelly. Staking the whole fraction f^* is called *full Kelly*; staking a fixed share of it, λf^* with $0 < \lambda < 1$, is called *fractional Kelly*. Full Kelly is notorious for its volatility and for its sensitivity to errors in estimating p. The trade-off is quantified by the second-order expansion of g around zero. Writing $b = o - 1$ and $e = pb - (1-p)$ for the expected return per unit,

$$g(f) \;\approx\; e\,f \;-\; \tfrac{1}{2}\sigma^2 f^2, \qquad \sigma^2 = \mathbb{E}[b^2] \text{ approximately},$$

a concave parabola with maximum at $f^* = e/\sigma^2$ and maximal value $g(f^*) = e^2/(2\sigma^2)$.

Proposition 3.26 (The cost of fractional Kelly). *In the quadratic approximation, betting a fraction λf^*, with the* Kelly multiplier $\lambda \in (0, 1]$, *retains a proportion $\lambda(2-\lambda)$ of the maximal growth rate, while the variance of the capital is reduced by a factor λ^2. (The symbol λ is used here and for the rest of the thesis in this sense only; it is unrelated to Kyle's depth coefficient of §3.2.2, which is written with the same letter by convention and appears nowhere else.)*

For $\lambda = \frac{1}{4}$, the value used here, one retains $\frac{1}{4} \cdot \frac{7}{4} = \frac{7}{16} \approx 44\%$ of the growth rate with $\frac{1}{16}$ of the variance: a very favourable trade. The argument is stronger than it looks, because p here is an *estimate*: if the true probability is lower than the estimated one, full Kelly overbets, and the damage from overshooting is asymmetrically larger than the gain from the corresponding undershoot, since g falls to $-\infty$ as $f \to 1$.

Betting when there are three possible outcomes. With three mutually exclusive outcomes the exact solution requires simultaneous optimisation of the vector of stakes. The implementation simplifies: at each round the outcome with the largest positive expected value is selected and only that one is backed, using the two-outcome formula. The simplification is conservative (a portfolio of simultaneous bets on mutually exclusive outcomes can only increase the rate, since it reduces the variance), and it matches how one bets in practice. We bet only when $p\,o > 1$, which requires our estimate to exceed the market's implied probability *after* the margin: the threshold is not "we are better than the bookmaker" but "we are better by more than its margin", and the difference between the two is exactly why improved log-loss does not imply profitability.

3.5.2 Stake size as a third OCO stage

The coefficient $\frac{1}{4}$ of Proposition 3.26 is a rule of thumb. Its value does not come from the data, and there is no reason for it to be the same against a bookmaker that prices tightly and one that does not. It is therefore a natural candidate to be learned online, as the weights and the temperature already were.

What exactly is learned, and why it matters. The obvious formulation (learn the fraction f of capital directly) is wrong, and the reason is the *comparator*. A regret bound for such a learner is a promise against the best *constant* f in hindsight. But the closed-form rule does not even belong to that class: it produces a different f every round, computed from $\hat{p}_t$ and the odds, both of which are known *before* the bet is placed. Even zero regret against the best constant stake does not reach a rule that varies from round to round. Moreover such a learner throws away $\hat{p}_t$, exactly the quantity the two preceding stages exist to produce, and relearns it from one bit per bet.

The correct formulation keeps the closed form and learns only the *multiplier*:

$$f_t \;=\; \lambda_t \cdot f_t^*, \qquad f_t^* = f^*(\hat{p}_t, o_t), \qquad \lambda_t \in [0, \lambda_{\max}].$$

The round's information stays inside f_t^* and only the shrinkage is learned. The comparator "best constant λ" now becomes a meaningful object: it is precisely the quantity that the coefficient $\frac{1}{4}$ pins to a constant.

Proposition 3.27 (The loss in the multiplier). *With $w \in \{0, 1\}$ the win indicator, the loss of a single round*

$$\ell_t(\lambda) \;=\; -\log\left(1 - \lambda f_t^* + \lambda f_t^* o_t w\right)$$

is convex in λ, with derivative

$$\ell_t'(\lambda) \;=\; \begin{cases} -\,f_t^*(o_t - 1)/g_t, & w = 1, \\ \;\;f_t^*/g_t, & w = 0, \end{cases} \qquad g_t = 1 - \lambda f_t^* + \lambda f_t^* o_t w.$$

It is moreover 1*-exp-concave on any interval where* g_t *stays bounded away from zero, that is where* $\lambda_{\max} \max_t f_t^* < 1$.

The constraint $\lambda_{\max} \max_t f_t^* < 1$ is not a technical detail but a *solvency* condition: above it a single loss wipes out the capital and the loss ceases to be bounded. The admissible interval is therefore determined by the data and not by choice.

Which algorithm. Exp-concavity is the critical point. Projected OGD gives $O(\sqrt{T})$ here, whereas the Online Newton Step of Hazan, Agarwal & Kale (2007) [22] exploits the curvature and gives $O(\log T)$. In one dimension the scheme is unusually short,

$$A_t \;=\; A_{t-1} + \ell_t'(\lambda_t)^2, \qquad \lambda_{t+1} \;=\; \Pi_{[0,\lambda_{\max}]}\Big(\lambda_t - \tfrac{1}{\gamma}\,\frac{\ell_t'(\lambda_t)}{A_t}\Big),$$

where the projection in the norm defined by A_t coincides with simple clipping, since in one dimension that norm is a positive rescaling of the Euclidean one.

Where it starts. The initial value $\lambda_0 = 1$, that is full Kelly, is not a neutral choice but the theoretically correct one: if $\hat{p}$ were exact, full Kelly would be optimal, and any shrinkage below one is a response to *estimation error*. Starting from there, the algorithm is asked to discover from the data how wrong its own estimate is: a quantity no a-priori constant knows.

Why Universal Portfolios does not carry over. The pure Online Portfolio Selection formulation (Cover, 1991) [8] compares against the best constant rebalanced portfolio over a *fixed* set of assets. Here the "assets" are the H/D/A of a different match every round, so a fixed weight vector corresponds to nothing interpretable. The reduction to a scalar multiplier is the way to retain a meaningful comparator in a problem where the set of available bets changes completely from round to round.

3.5.3 A Bayesian version of Kelly and Markov chain Monte Carlo sampling

Both the closed form and the learned multiplier treat $\hat{p}$ as *known*. It is not: it is an estimate, and how uncertain it is varies from match to match: when the awake experts agree the estimate is reliable, and when they disagree it is not. The natural formulation replaces the point maximum with the *posterior expected* one:

$$f^{\mathrm{B}} \;=\; \arg\max_f \; \mathbb{E}_{\theta\sim\pi(\cdot\mid\mathcal{D})}\big[\, g(f,\theta)\,\big],$$

$$g(f,\theta) \;=\; \theta_k \log\big(1 + f(o_k - 1)\big) \;+\; (1-\theta_k)\log(1-f).$$

Here $\theta = (\theta_H, \theta_D, \theta_A)$ is the unknown true outcome distribution of the match, $\pi(\cdot \mid \mathcal{D})$ is the posterior distribution of θ given the data $\mathcal{D}$ (the experts' forecasts for that match), k is the outcome being backed, and $g(f,\theta)$ is the expected logarithmic growth of capital when a fraction f is staked on it at odds o_k, the quantity the Kelly criterion maximises. Maximising its posterior expectation, rather than its value at a point estimate, is the Bayesian Kelly criterion of Browne & Whitt (1996) [6].

Proposition 3.28 (The Bayesian version is always more conservative). *For every non-degenerate posterior π with mean $\bar{\theta}$ we have $f^{\mathrm{B}} \leq f^*(\bar{\theta})$, where f^* is the closed form at the point $\bar{\theta}$. The inequality is strict when π has positive variance.*

The corollary is conceptually attractive: the *shrinkage emerges* from the uncertainty instead of being imposed by an arbitrary constant $\frac{1}{4}$, and its magnitude varies from round to round according to how much the experts agree.

The model. The prior is taken to be $\theta \sim \mathrm{Dirichlet}(1,1,1)$, i.e. uniform on Δ_3. Each awake expert i contributes its prediction p_i as evidence, modelled as $p_i \sim \mathrm{Dirichlet}(\kappa\,\theta)$ and weighted by the learned weight w_i. The parameter κ controls how tightly the experts are assumed to cluster around the truth: large κ means a concentrated posterior and returns the method to the closed form, small κ means a wide posterior and strong shrinkage.

Why Markov chain Monte Carlo (MCMC) is needed. The resulting posterior is not conjugate (θ appears inside the Dirichlet parameters of the evidence), so there is no closed form either for it or for the integral $\mathbb{E}_\pi[g(f,\theta)]$. We use random-walk *Metropolis–Hastings* sampling (Hastings, 1970) [21]: from the current state θ a proposal θ' is made and accepted with probability

$$a(\theta \to \theta') \;=\; \min\!\left\{1,\; \frac{\pi(\theta' \mid \mathcal{D})\, q(\theta \mid \theta')}{\pi(\theta \mid \mathcal{D})\, q(\theta' \mid \theta)}\right\}.$$

With a symmetric proposal ($q(\theta' \mid \theta) = q(\theta \mid \theta')$) the ratio simplifies to the ratio of posterior densities, so the *normalising constant is not needed*, which is exactly the method's reason for being here. This rule satisfies *detailed balance* $\pi(\theta)P(\theta \to \theta') = \pi(\theta')P(\theta' \to \theta)$, so π is the stationary distribution of the chain; with ergodicity, time averages converge to expectations under π.

Remark 3.29 (Parallel chains instead of one long one). The implementation runs N *parallel* chains for a few iterations instead of one long one. The two options are equivalent with respect to the stationary distribution, but not with respect to cost: parallel chains turn a scalar Python loop into vectorised operations of the NumPy library (Harris et al., 2020) [20], which is what makes the per-bet cost manageable over tens of thousands of bets. The price is that no chain has time to forget its initialisation completely, so the approximation rests on the burn-in being sufficient: an assumption not formally checked here and reported as a limitation.

Theory therefore clearly favours this formulation over the fixed $\frac{1}{4}$. §5.6 reports that on these data the measurement does not, which is itself part of the thesis's argument: the theoretical superiority of a formulation guarantees nothing when its assumptions (here, that the Dirichlet model really describes how bookmakers' disagreements are generated) do not hold.

Chapter 4

Problem formulation and experimental design

This chapter states what the data are and how every number in Chapter 5 was produced. The guiding principle throughout is that every quantity reported must have been computed without access to information that would not have been available at the moment of prediction, and must be compared against quantities measured on exactly the same data.

4.1 The data

4.1.1 Source and scope

Table 4.1 summarises the result. The data come from `football-data.co.uk` [15], section "Main Leagues", which publishes for each match the result and the odds of a number of betting companies. The "Extra Leagues" section has a different format and usually does not contain odds, so it is out of scope.

Leagues	22
Seasons	10 (2016/17 – 2025/26)
Matches with $\geq$1 usable price	76,584
Bookmakers as experts	26 (13 opening / 13 closing)
Coverage range per bookmaker	6.9% – 99.8%

Table 4.1: The dataset after preprocessing, that is after the cleaning steps of §4.1.2: aggregate columns removed, rebranded companies merged into one identity, and matches without a single usable price dropped.

The 22 leagues are distributed over England ($\times$5), Scotland ($\times$4), Germany / Italy / Spain / France ($\times$2 each) and the Netherlands / Belgium / Portugal / Turkey / Greece ($\times$1). The combination of top and lower divisions is deliberate: lower divisions attract smaller betting volume and therefore less attention from the companies' analysts, so the heterogeneity among the experts is greater there.

4.1.2 Cleaning

Aggregate columns. Beyond the individual companies, football-data.co.uk also publishes columns with the maximum and the average of the market (`Max`, `Avg`), as well as the equivalents computed by Betbrain, an odds-comparison service, for the older seasons [15]. These are excluded: they are not independent experts but functions of the rest, and their presence would violate the framework's assumption that the experts give separate opinions.

Company rebranding. The columns `VC` (VC Bet) and `BV` (BetVictor) correspond to the same company before and after a rebrand, and their presence does not overlap in time. They are merged into a single continuous identity, so that the expert is not "reborn" halfway through the decade losing its learned history. The merge is inferred from the non-overlapping presence, not from an official source, and is noted as a limitation.

Companies with partial temporal presence. Several columns do not cover the whole decade: some start late and others stop early. None is excluded for that reason: the Sleeping Experts reduction (§3.3.5) exists precisely to handle them, and each expert's guarantee is measured only over the rounds in which it was active. Temporal presence does, however, have a consequence for *coverage thresholds*: a threshold computed over the decade says very little about a company's presence in a narrower window, which is why every restricted-window analysis recomputes its own threshold inside the window it examines (§5.3.1).

4.1.3 Opening and closing odds

For each company the source publishes two prices: the *opening* one, when the market opens, and the *closing* one, shortly before kick-off. The naming convention is that the suffix `C` denotes the closing price of the identically named company: `B365` and `B365C`, `PS` and `PSC`, and so on. The panel of 26 therefore splits cleanly into 13 opening and 13 closing. Table 4.2 lists the thirteen companies with their codes.

Code	Company	opening	closing (C)
B365	Bet365	99.8%	69.6%
BW	Bet&Win	95.0%	65.0%
PS	Pinnacle	93.1%	93.8%
VC_BV	VC Bet / BetVictor	89.3%	59.3%
WH	William Hill	85.5%	55.3%
IW	Interwetten	74.7%	44.4%
LB	Ladbrokes	27.4%	7.3%
BFE	Betfair Exchange	19.3%	19.4%
BF	Betfair	10.0%	10.0%
1XB	1XBet	9.8%	10.0%
BFD	Betfred	9.9%	9.8%
BMGM	BetMGM	9.5%	9.8%
CL	Coral	7.0%	6.9%

Table 4.2: The thirteen companies of the panel, with the codes of the source. Each company enters twice: with its opening prices under the code shown and with its closing prices under the same code followed by `C`. The last two columns give the share of the 76,584 matches for which each series quoted a price. `VC_BV` merges VC Bet and BetVictor, the same company before and after a change of name (§4.1.2), and Betfair Exchange is a betting exchange rather than a bookmaker.

The distinction is not a technical detail but a substantive variable. The closing price has absorbed whatever information reached the market in the interim (team-sheet announcements, injuries, moves by large bettors) and is for that reason considered in the literature the most efficient available signal. The distinction also imposes a causality constraint: when we bet into an opening price, no closing price exists yet, not even that of the same company.

4.2 Experimental design

4.2.1 The expert tensor

The basic structure is a tensor $P \in \mathbb{R}^{T\times N\times 3}$ with T rounds (matches in chronological order), N experts and three outcomes, together with a binary matrix $A \in \{0,1\}^{T\times N}$ indicating which expert quoted which match. The chronological ordering is essential: the entire framework presupposes that round t precedes $t+1$.

4.2.2 The three OCO stages

The chain from raw odds to the bet consists of three successive stages, all online and causal.

1. **Weighting the experts.** The three algorithms of Chapter 3 with the Sleeping Experts reduction, plus the uniform average as a baseline. They produce a mixture $\hat{p}_t$ per round.

2. **Calibration.** Online temperature scaling on top of $\hat{p}_t$ (§3.4.1), with the temperature learned by projected OGD.

3. **Stake size**. The Kelly criterion on the calibrated probabilities, with the *multiplier* learned online instead of being pinned to a constant (§3.5.2). The fixed coefficient $\frac{1}{4}$ is retained as a reference point for comparison.

All three stages are the same recipe applied to a different object: convex loss, causal update, projection onto the feasible set. The only difference between them is which quantity is treated as unknown (the weight vector, the temperature, the multiplier) and which loss class each one belongs to, which also determines the algorithm appropriate for each.

The uniform average is not decorative. It is the critical baseline: if an algorithm does not beat it, then learned weighting offers nothing over simply taking the market average, and the complexity is not justified.

4.2.3 Hyperparameter selection protocol

The framework does not require a train/test split, since every prediction is already out of sample. It does, however, require care about whatever is *selected* by looking at the data. Two such quantities exist: the step multiplier of each algorithm and the learning rate of the calibrator. Both are selected by the same protocol:

- a chronological prefix of 75% of each series' rounds as *train*;
- a grid sweep over candidate values, evaluated exclusively on the prefix;
- a single final causal pass over the whole history with the selected value frozen.

Wherever a *configuration choice* is made on the basis of reported numbers (for example the choice between panels) a chronological 25% tail that the choice never saw is additionally reported, so that it is visible whether the choice was fitting noise.

4.2.4 Tuning the step

The theoretically optimal step of all three algorithms contains a bound M on the maximum loss of a single round. Because probabilities are clipped at $\varepsilon = 10^{-6}$ so that the log-loss stays finite, this gives

$$M \;=\; \ln(1/\varepsilon) \;=\; \ln 10^{6} \;=\; 6 \ln 10 \;\approx\; 13.82.$$

This bound presupposes a bookmaker pricing the outcome that actually occurred at one in a million. The largest loss actually observed in the data is 3.76, that is the worst-case step is an order of magnitude more conservative than the data warrant. The multipliers are therefore tuned empirically, with the protocol of §4.2.3 and over the same grid for all three algorithms, so that the comparison between them stays fair.

4.2.5 Comparing series with different coverage

Partial coverage creates a trap that determines a large part of the design. A log-loss computed over one set of matches is not comparable with a log-loss computed over another: the matches differ in difficulty, and a series covering fewer of them may look better simply because it happened to cover the easier ones.

Wherever series with different coverage are compared, therefore:

- **fitting** uses each series' full available support;
- **scoring** is restricted to the intersection of the rounds in which *all* compared series are defined;
- **the paired tests** run on that same set.

This principle has a cost. The more series enter a comparison, the smaller the intersection, and in extreme cases requiring everyone to participate drives the statistical power to zero. Where that happens, the low-coverage series is reported separately, in a one-to-one comparison over its own rounds, rather than being imposed on the common table.

4.2.6 Causality controls in value betting

The betting simulation uses a *leave-one-bookmaker-out* design: the bookmaker we bet against is removed from the panel before fitting, so that the mixture does not "predict" using the very price it is asked to beat. This is not enough. When the target is an opening price, the *whole* closing side of the panel must be removed, because no closing price exists yet at that moment, not even that of the same company. A panel that leaves them in bets knowing where the market will close, and ends up producing final bankrolls of order 10^6–10^9, that is a pure artefact of construction.

The training panel therefore matches the phase of the target: a closing panel for a closing target, an opening panel for an opening target.

4.2.7 Deployment test

Beyond the above, the thesis includes a test that answers a different question from regret: *what would a model trained up to yesterday do today?* The cut is a real date rather than a percentage of the series, fitting is done on the rounds before it, and measurement on those after. The opponents are selected by their coverage *inside the test window*, so that the comparison concerns the market that exists then and not the one that existed ten years ago.

Two readings are reported, because they answer different questions: one in which the mixture keeps learning inside the test window, which is the realistic deployment scenario, and one in which the weights are frozen at the cut, which is the strict reading of "trained once".

Chapter 5

Results

5.1 Tuning the step

The step multipliers are tuned first, since they determine the algorithms with which all the remaining results are produced. The multipliers of all three algorithms needed changing, and (this is the non-obvious part) **in opposite directions**. Table 5.1 lists the multipliers that are in force everywhere else in this chapter.

	Mult.	**train**	**full**	**tail**	**train min.**	**tail min.**
OGD	$\times 100$	0.99739	0.99745	0.99765	$\times 110$	$\times 100$
Hedge	$\times 0.25$	0.99828	0.99842	0.99883	$\times 0.25$	$\times 0.05$
FTRL	$\times 0.05$	0.99828	0.99843	0.99889	$\times 0.05$	$\times 0.005$

Table 5.1: Step multipliers in force. **Mult.** is the factor applied to the textbook step schedule. **train**, **full** and **tail** are the log-loss obtained with it on the chronological 75% prefix, on the whole sample and on the held-out 25% tail. **train min.** and **tail min.** are the multipliers that would minimise the log-loss on the prefix and on the tail respectively. Selection is made exclusively on the chronological prefix, so the decisive column is **train min.**: for all three algorithms it coincides with the multiplier actually used, the one exception being OGD's $\times 110$, which gives a tail identical to $\times 100$ (0.99765). The **tail min.** column is reported for comparison and not as a criterion; the difference between the two is discussed below.

Table A.28 in the appendix lists every setting of the sweep. As Figure 5.1 shows, OGD wants a step a hundred times larger than the theoretical one, with a genuine interior optimum: a plateau over the range $\times 75$ to $\times 110$, with $\times 200$, $\times 500$ and $\times 1000$ clearly worse. Hedge and FTRL want a *smaller* step, by factors of 4 and 20 respectively.

A caveat that turned out to be unfounded. The *tail* minimum for Hedge and FTRL lies below the values used, at $\times 0.05$ and $\times 0.005$ respectively, and for a time this thesis carried the caveat that those two algorithms were reported with a sub-optimal step. The caveat was wrong, and the reason is worth stating: **selecting on the tail is selecting on the test set**, which is exactly what the protocol of §4.2.3 exists to prevent. Because the log-loss is a plain mean over rounds, the prefix figure is recoverable exactly from the full sample and

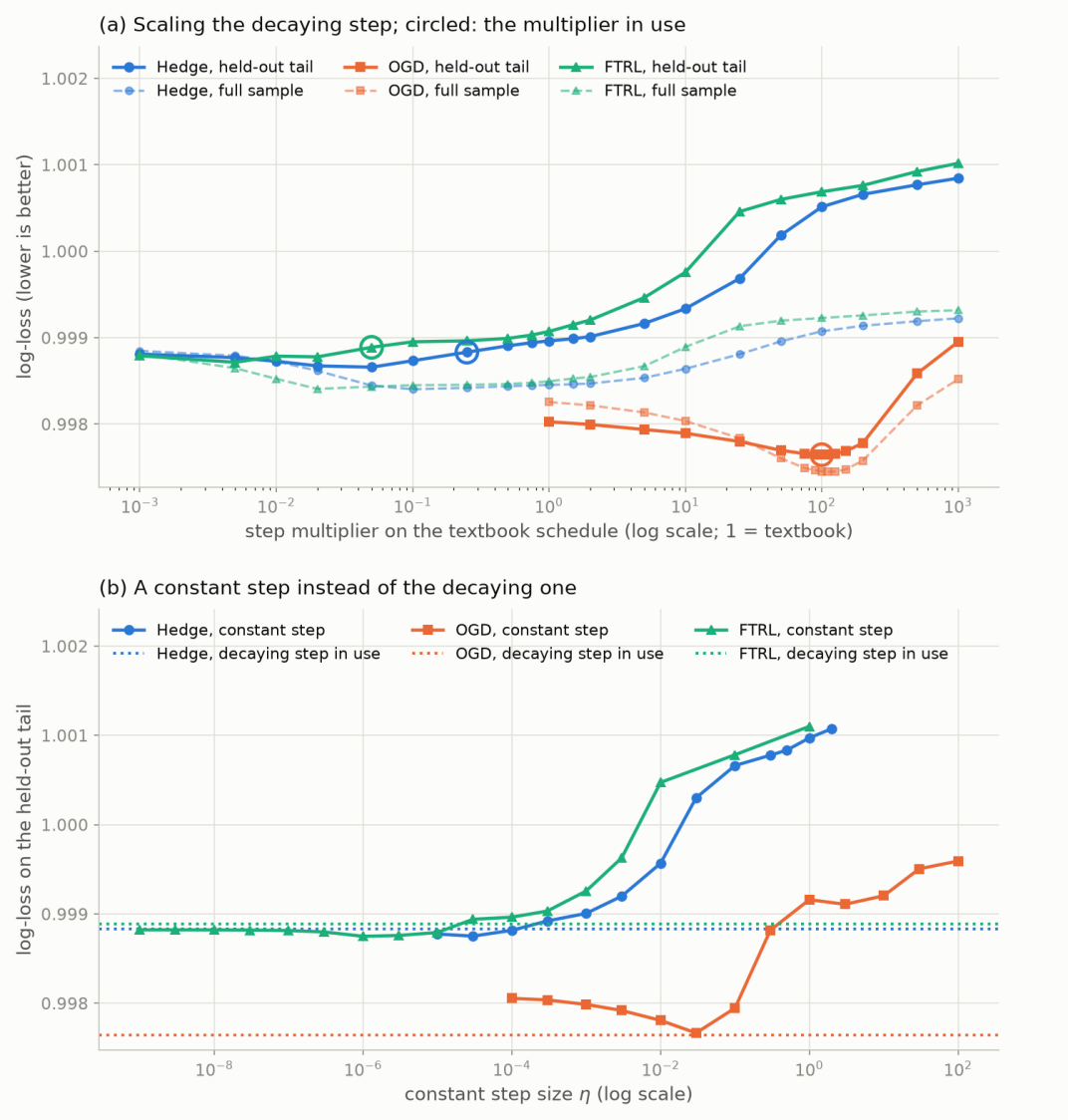


Figure 5.1: Sweep of the step size for all three algorithms; in both panels lower log-loss is better, and blue is Hedge, orange OGD and green FTRL. (a) The multiplier applied to the textbook decaying schedule, on a logarithmic horizontal axis with 1 the textbook value. Solid lines give the log-loss on the held-out tail and dashed lines on the full sample; the circled point on each solid line is the multiplier in use. OGD has a genuine interior optimum. The tail curves of Hedge and FTRL keep falling to the left, but the selection is made on the training prefix, not on them (see the caveat in the text of this section). (b) A constant step η in place of the decaying schedule, on a logarithmic horizontal axis. Each solid line is the held-out log-loss of one algorithm; the dotted line of the same colour is that algorithm's decaying step in use.

the tail, as $(\text{full} \cdot T - \text{tail} \cdot n_{\text{tail}})/n_{\text{train}}$, and under the protocol the answer reverses: Hedge selects $\times 0.25$ and FTRL $\times 0.05$, that is *exactly* the multipliers in force.

The numbers also show why. For Hedge, $\times 0.05$ gives a train figure of 0.99838 against 0.99828 for $\times 0.25$; for FTRL, $\times 0.005$ gives 0.99862 against 0.99828 for $\times 0.05$. In both cases the ordering *reverses* between prefix and tail, which is the signature of fitting the tail rather than evidence of better tuning. OGD remains optimal at $\times 100$, with $\times 110$ giving an identical tail: a tie inside the plateau already described.

A constant step: the one schedule idea that had not been tested properly. The decaying $1/\sqrt{t}$ is derived for a *static* comparator, whereas the panel here is not static, so a step that never stops moving is the theoretically indicated alternative. It had been tried only for Hedge and only over $[0.1, 2]$, while Hedge's decaying schedule passes $\eta \approx 0.033$ at the first round and $\eta \approx 1.7 \cdot 10^{-4}$ by the middle of the period, i.e. the entire interesting region lay *below* the old grid $[0.1, 2]$. The sweep was extended over five decades and to all three algorithms, with selection on the prefix and a paired test on the tail. Table 5.2 gives the comparison.

	constant η	**tail**	**decaying**	p	
Hedge	$3 \cdot 10^{-4}$	0.99892	0.99883	< 0.001	decaying
OGD	0.1	0.99795	0.99765	0.021	decaying
FTRL	10^{-5}	0.99879	0.99889	0.001	**constant**

Table 5.2: Constant against decaying step. **constant** η is the constant step selected on the chronological prefix and **tail** the log-loss it gives on the held-out tail; **decaying** is the tail log-loss of the decaying schedule in force (the **tail** column of Table 5.1). p is the moving block bootstrap p-value of the paired difference between the two on the tail, and the last column names the better of the two, in bold where it is the constant step.

The result is negative for two of them and **positive for one**. For Hedge and OGD the constant step loses significantly, by $+0.000089$ and $+0.000299$ respectively. For FTRL, however, it wins significantly, by -0.000096 ($p = 0.001$), and the critical point is that the two schemes have an *identical* score on the prefix (0.99828 for both). With nothing to separate the two on the training data, the selection cannot have favoured the constant step in order to flatter the tail, and the advantage emerges only out of sample. The explanation is the same asymmetry that explains the opposite tuning directions: FTRL's η multiplies the entire accumulated loss, not one round's gradient, so it is the one of the three for which "non-decaying" means something qualitatively different.

The sweep needed extending here too, for the same reason as before: FTRL's minimum initially sat on the grid's lower endpoint. Extended down to 10^{-9} it became genuinely interior, and the extension simultaneously confirmed the theoretical limit: as $\eta \to 0$ FTRL degenerates to the uniform average, and indeed at $\eta = 10^{-9}$ it gives 0.99882, the uniform average's value on the tail.

A methodological error worth reporting. The original sweep had the grid $1, 2, 5, \ldots, 1000$ and concluded that Hedge and FTRL were "already optimal at $\times 1$". But 1 was the

lower endpoint of the grid, not an optimum, and nobody had looked to the left of it. Extending downwards revealed genuine interior minima, with gains of $+0.00013$ and $+0.00018$ respectively on the tail. The conclusion generalises: when a sweep's answer sits at the endpoint of the grid, the grid must be extended before the answer is believed.

The opposite direction has a structural explanation. OGD's η multiplies the gradient of *one* round, so increasing it means moving further. Hedge's η sits in an exponent over one round's loss, so decreasing it *lengthens the memory*. FTRL's η multiplies the entire accumulated loss, so increasing it collapses it towards follow-the-leader.

5.2 Ranking and the choice of panel

5.2.1 Ranking on the full panel

		phase	**log-loss** (raw / cal.)	**Brier** (raw / cal.)
PSC (Pinnacle)	bookmaker	closing	0.99714 / 0.99685	0.59592 / 0.59577
OGD	algorithm	—	0.99745 / 0.99693	0.59612 / 0.59586
Hedge	algorithm	—	0.99842 / 0.99773	0.59674 / 0.59637
FTRL	algorithm	—	0.99843 / 0.99774	0.59674 / 0.59638
Uniform	baseline	—	0.99886 / 0.99813	0.59703 / 0.59665
B365C	bookmaker	closing	0.99953 / 0.99910	0.59732 / 0.59711
BWC	bookmaker	closing	0.99998 / 0.99919	0.59769 / 0.59728
VC_BV	bookmaker	opening	1.00075 / 1.00026	0.59833 / 0.59806
WH	bookmaker	opening	1.00119 / 1.00048	0.59860 / 0.59818
PS (Pinnacle)	bookmaker	opening	1.00083 / 1.00055	0.59838 / 0.59824
IW	bookmaker	opening	1.00186 / 1.00061	0.59899 / 0.59826
BW	bookmaker	opening	1.00140 / 1.00065	0.59873 / 0.59833
B365	bookmaker	opening	1.00113 / 1.00073	0.59852 / 0.59832

Table 5.3: Ranking by calibrated log-loss, bookmakers with coverage $\geq$60%. **phase** says whether a bookmaker's prices are opening or closing ones; the algorithms and the uniform average combine both. Each cell gives the raw score and, after the slash, the score after temperature calibration (§3.4.1); for both log-loss and Brier lower is better. Each series is scored over the matches it quoted, so differences between rows are tested in paired form in the text rather than read off the table.

Table 5.3 gives the ranking and Figure 5.2 plots it; Table A.1 in the appendix adds each series' coverage, number of rounds and accuracy. The paired tests give:

- All three algorithms significantly beat the uniform average: OGD -0.00141, Hedge -0.00044, FTRL -0.00043, all $p < 0.001$. OGD's margin is about threefold.
- OGD significantly beats Hedge (-0.00097), FTRL (-0.00098) and B365C (-0.00063).
- Calibration significantly improves all four methods, by $+0.00053$ to $+0.00073$.
- PSC remains significantly ahead of OGD in this configuration: $+0.00043$ calibrated, $p < 0.001$, over the 71,818 common matches.

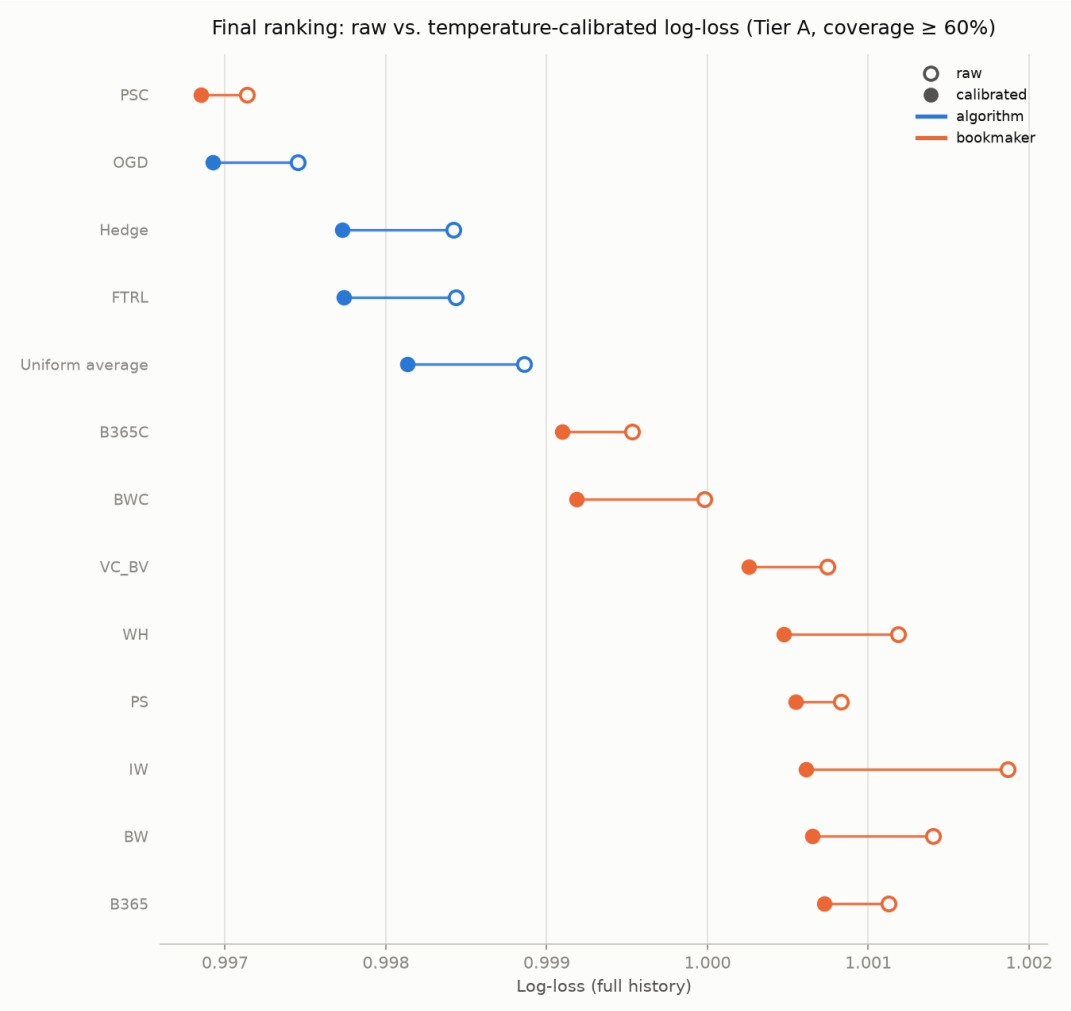


Figure 5.2: Final ranking of the bookmakers and the algorithms over the full decade (coverage $\geq 60\%$). Horizontal axis log-loss, lower better; one row per series, ordered best at the top. Each row is a short segment whose hollow circle is the raw log-loss and whose filled circle is the calibrated one, so the segment's length is what calibration bought. Orange marks a bookmaker and blue an algorithm. A dot plot is used rather than a bar chart because log-loss values cluster around 1.0 with no meaningful zero, and bars on a truncated axis would visually exaggerate minute differences.

A necessary caveat. Three of the twelve closing columns clear the 60% coverage floor (PSC at 93.8%, B365C at 69.6%, BWC at 65.0%); the rest start around 2019/20 and do not reach it over the decade. The ranking therefore compares the mixture against three closing prices and six opening ones, which is enough for the comparison that matters but not the whole closing market. The fuller treatment is in §5.3.1 and §5.3.2, where the window is short enough for most closing columns to qualify.

A second caveat applies to the table as a whole and is more serious. Each series is scored over the rounds it actually quotes, and those differ: 76,584 for the algorithms, 71,818 for PSC, 53,267 for B365C, 49,755 for BWC. The columns are therefore not strictly comparable with one another, which is why every claim below rests on a *paired* test over the rounds the two compared series share, never on the difference of two table entries (§4.2.5).

Note that Interwetten occupies the *last* place in the raw log-loss ranking (1.00186) despite having the seventh best coverage. Its addition therefore makes the statement "we beat everyone except PSC" somewhat cheaper, and for that reason the substantive comparison remains the one against PSC and B365C.

Note also that log-loss and Brier give almost identical orderings, the only exception being the last two places. Accuracy gives a different ordering, but none of its differences is statistically significant, as expected, since it is not a proper scoring rule.

5.2.2 Which panel

	opening (13)		**all** (26)		**closing** (13)	
	basic	Shin	basic	Shin	basic	Shin
OGD	1.00076	1.00024	0.99745	0.99709	0.99722	0.99689
Hedge	1.00088	1.00033	0.99842	0.99781	0.99762	0.99714
FTRL	1.00088	1.00033	0.99843	0.99782	0.99761	0.99714
Uniform	1.00080	1.00022	0.99886	0.99824	0.99748	0.99703

Table 5.4: Raw log-loss of each method fitted on each panel, over the whole decade; lower is better. The number in parentheses is the number of bookmaker series in the panel, and *basic* and *Shin* are the two de-vig methods of §3.2.1 (proportional normalisation and Shin's method). The three panels cover between 76,521 and 76,584 matches, and the differences are tested in paired form in the text.

Table 5.4 reports the grid and, as Figure 5.3 makes visible, the ordering closing $<$ all $<$ opening holds in every cell, and in all 24 comparisons (4 algorithms $\times$ 3 comparisons $\times$ 2 normalisations) the difference is statistically significant. That closing prices beat opening ones is expected. What is unexpected is that the closing-only panel also beats the *full* one: the opening bookmakers do not act as useful diversification but as pure dead weight in the mixture.

Comparing each algorithm with the plain average *inside* the same panel explains the mechanism and is, in our view, the most generally useful finding of the thesis:

- On the **full** panel all three algorithms significantly beat the average.
- On the **restricted** panels Hedge and FTRL *significantly lose* to the average, in 8 tests out of 8. OGD wins on closing and ties on opening.

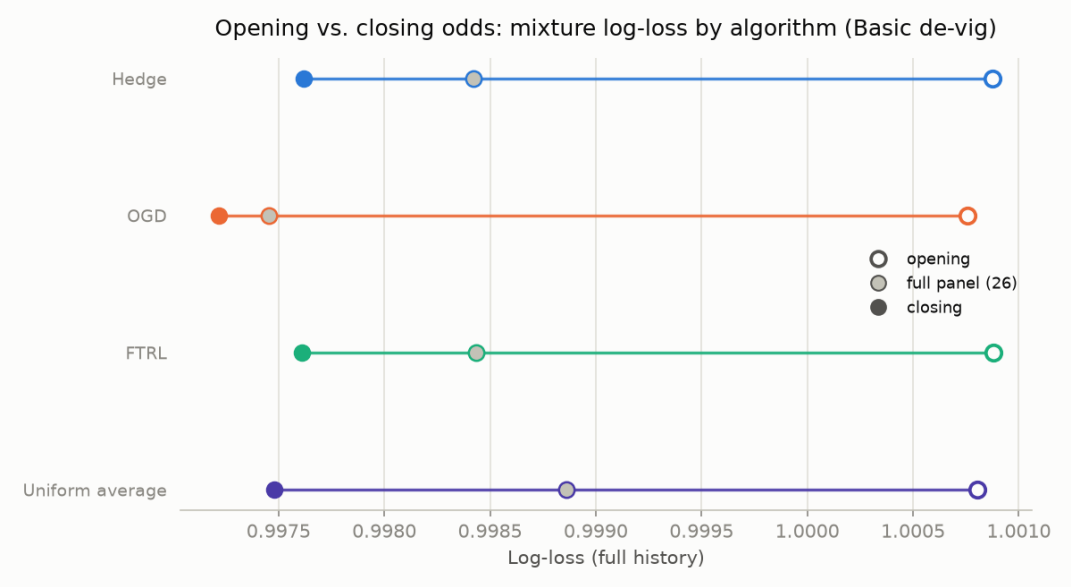


Figure 5.3: Log-loss of each algorithm on the three panels. Horizontal axis log-loss over the full history, lower better; one row per algorithm, each in its own colour. The three marks on a row are the same algorithm fitted on the opening-only panel (hollow), on the full 26-expert panel (grey) and on the closing-only panel (filled), so the spread within a row is the effect of the panel and not of the method. The ordering closing $<$ all $<$ opening holds in every row.

The full panel contains 13 good closing and 13 systematically worse opening experts, that is a clear quality difference available to be learned. In a homogeneous panel that difference is absent, and for two of the three algorithms the adaptation costs more than it returns.

5.2.3 Where exactly closing prices are superior

§5.2.2 shows that the closing panel wins by ~ 0.0035 in log-loss. That is an average, and Remark 3.4 makes a sharper prediction: if the superiority is due to information arrival, it must be concentrated where the price *actually moved*. To test this, each match is characterised by the total variation distance between the initial and the final probability distribution of each house, averaged over the 13 available opening/closing pairs, and the matches are split into quintiles. Table 5.5 reports the result and Figure 5.4 plots it beside the same exercise run on Shin's z.

Movement quintile	Q1	Q2	Q3	Q4	Q5
Mean movement	0.0082	0.0144	0.0215	0.0324	0.0598
log-loss opening	0.99161	1.00581	1.00068	1.00666	1.00790
log-loss closing	0.99118	1.00509	0.99868	1.00214	0.99632
closing − opening	−0.00044	−0.00072	−0.00200	−0.00452	**−0.01158**
p	0.052	0.050	< 0.001	< 0.001	< 0.001

Table 5.5: Closing advantage by quintile of line movement, 10,649–10,650 matches per quintile: Q1 holds the matches whose prices moved least and Q5 those whose prices moved most. *Mean movement* is the mean total variation distance between the opening and the closing probabilities in the quintile. The two log-loss rows are the raw log-loss of OGD fitted on the opening-only and on the closing-only panel, over the matches of the quintile; *closing − opening* is their difference (negative means the closing panel is better) and p the moving block bootstrap p-value of that paired difference.

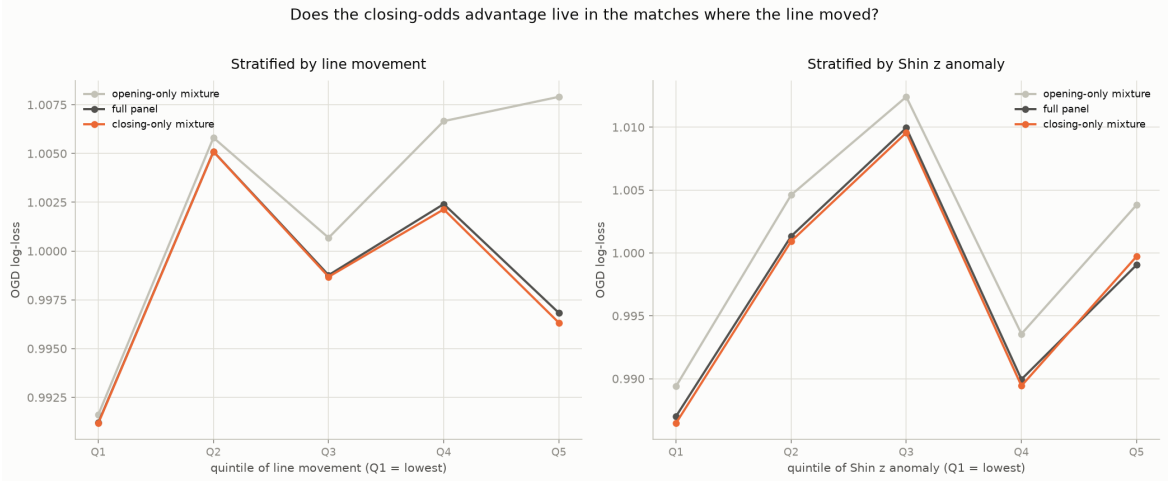


Figure 5.4: Left, the stratification by line movement; right, by anomaly of Shin's z. Horizontal axis the quintile, Q1 quietest and Q5 noisiest; vertical axis the log-loss of OGD, lower better. The three curves are the mixture fitted on the opening-only panel (grey), on the full panel (black) and on the closing-only panel (orange). The vertical gap between the grey and orange curves is the closing advantage, and in the left panel it is that gap which widens from left to right.

The prediction is confirmed, with a 26-fold gradient. The closing advantage runs from -0.00044 in the quiet fifth of the matches to -0.01158 in the noisy one. In Q1 it is in fact **not even statistically significant** ($p = 0.052$): in the fifth of matches where the line barely moved, there is no measurable closing advantage. The "superiority of closing prices" is therefore not a general property of final prices but a phenomenon of the top quintile.

A necessary caveat. Movement is measured from the *same* closing prices that the closing panel uses, so the null result in Q1 is partly mechanical: if the two prices nearly coincide, the two mixtures cannot differ much. The non-trivial part is the *direction* at the other extreme. A large movement could perfectly well go the wrong way; that the later price is

systematically the better one by 0.0116 is exactly the market-efficiency claim, localised where it actually operates. A useful control: the *difficulty* of the matches is not monotone across the quintiles (the full panel's log-loss runs 0.99120, 1.00510, 0.99877, 1.00241, 0.99683), whereas the advantage is, so the gradient is not an artefact of the form "harder matches, larger differences".

Shin's z is a much weaker indicator. Stratifying instead by the anomaly of z (the quantity that §3.2.2 identifies with the estimated proportion of informed traders) the gradient is only -0.00295 to -0.00498, that is a factor of 1.7 against 26, with all quintiles significant. So z does detect something, but the actual price movement is an incomparably cleaner indicator that information has arrived, reasonably so, since z is estimated from a single moment of the market whereas movement observes the difference of two.

5.2.4 The configuration that is selected

Three improvements have been measured separately (the corrected step, the closing-only panel, Shin) and are combined here. Every configuration is scored on the same 71,818 matches, the intersection of all of them with PSC, following the principle of §4.2.5. Table 5.6 stacks them.

Configuration	full	train	25% tail
PSC alone	0.99685	0.99746	0.99502
closing-only + Shin	0.99692	0.99754	0.99507
closing-only, proportional	0.99693	0.99756	0.99506
closing-only + Shin, no single-season	0.99692	0.99754	0.99509
full panel, proportional	0.99728	0.99801	0.99508
full panel, Shin	0.99731	0.99804	0.99512

Table 5.6: Calibrated log-loss of OGD by configuration, the same 71,818 matches throughout; lower is better. **full** is over all of them, **train** over their chronological 75% prefix and **25% tail** over the remaining matches, which no selection has looked at. *PSC alone* is the benchmark bookmaker itself, calibrated the same way; *closing-only* means the mixture uses only the closing panel, *Shin* and *proportional* name the de-vig method, and *no single-season* drops the bookmakers active in only one season.

- **The panel does all the work**. Closing-only + Shin beats the full panel by -0.00036 ($p < 0.001$), and closing-only with proportional normalisation by -0.00034 ($p < 0.001$).
- **Shin on its own offers nothing measurable** over the full panel ($+0.00004$, $p = 0.181$), consistent with Shin and temperature scaling correcting the same favourite-longshot bias, so that the second absorbs the first.
- **Against PSC**: the full panel significantly loses ($+0.00043$, $p < 0.001$), while closing-only + Shin *ties* ($+0.00007$, $p = 0.234$) and closing-only with proportional normalisation likewise ($+0.00009$, $p = 0.143$).

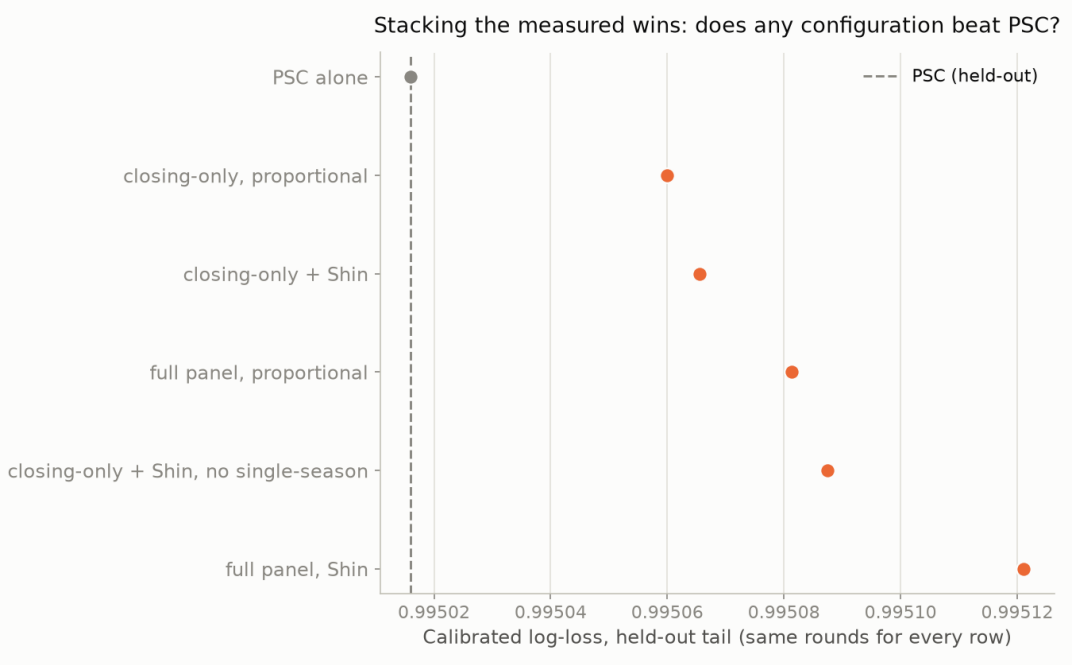


Figure 5.5: The configurations on the held-out tail, with PSC as the reference line. Horizontal axis the calibrated log-loss on the held-out 25%, lower better; one dot per configuration. The dashed vertical line is PSC scored on the same rounds, so a dot to its left beats Pinnacle and a dot to its right does not. All of them cluster within 0.00005.

The 25% tail does not produce a winner: all six rows gather within less than 0.00005 (Table 5.6, Figure 5.5). It thus confirms the tie, and it would not be honest to present it as a win. Note also that in every closing configuration PSC is *inside* the mixture: "we reach PSC" reads as "the mixture reaches its own best member".

5.2.5 Without Pinnacle

The obvious counter-argument is that the mixture merely reproduces Pinnacle, which is inside it anyway. Removing both of its columns and refitting from scratch on three panels gives Table 5.7:

Panel	OGD rank	log-loss	First	Paired
full (24)	#3 of 9	0.99782	FTRL 0.99771	5 wins, 0 losses
closing-only (12)	#1 of 9	0.99919	OGD (itself)	3 wins, 2 ties
opening-only (12)	#5 of 9	0.99999	VC_BV 0.99987	3 wins, **1 loss**, 1 tie

Table 5.7: Each panel is refitted from scratch without the two Pinnacle columns (the number of experts left is in parentheses), and every series is scored on the matches common to all of them. **OGD rank** is OGD's position among the nine series compared, the four methods and five bookmakers, and **log-loss** is its calibrated log-loss. **First** is the series ranked first, with its log-loss, and **Paired** summarises OGD's paired tests against the five bookmakers.

On the **full** panel the result holds comfortably (Figure 5.6): all four methods sit above

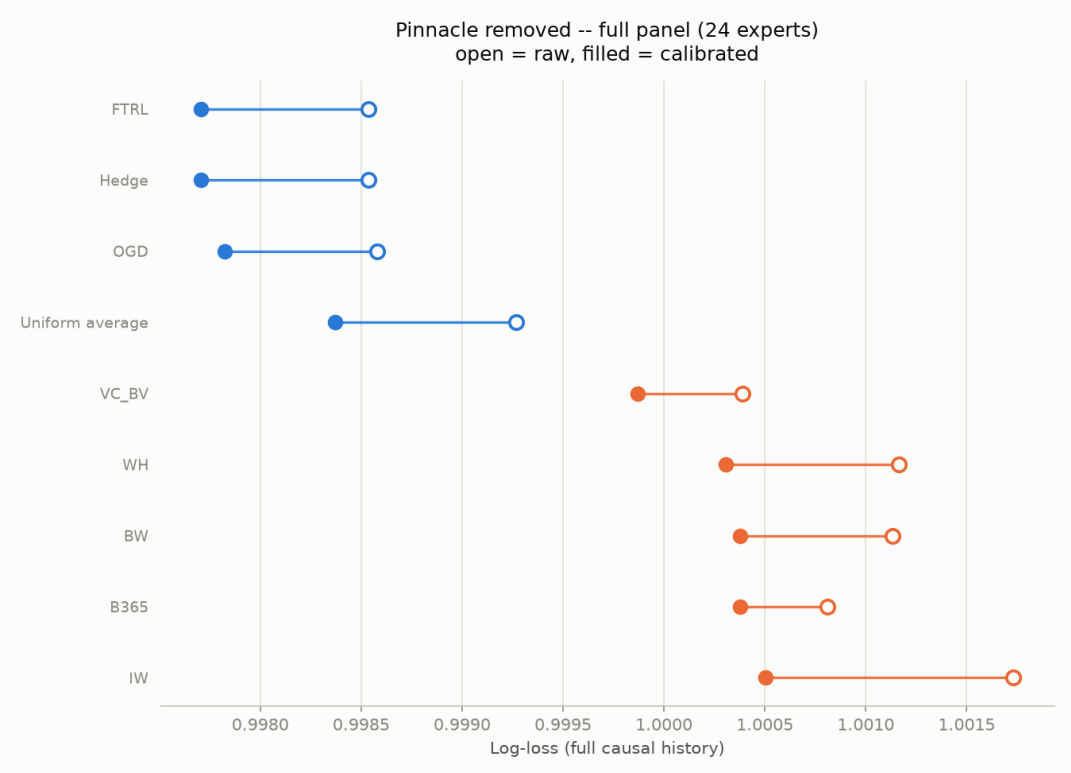


Figure 5.6: Log-loss of the algorithms and the bookmakers on the full panel with both Pinnacle columns removed. Horizontal axis log-loss, lower better; one row per series, drawn as a short segment whose hollow end is the raw value and whose filled end is the calibrated one. All four methods (in blue) sit above all remaining bookmakers (in orange).

every bookmaker, and OGD significantly beats all *five*, with margins from -0.00181 (VC_BV) to -0.00315 (IW) and $p < 0.001$ throughout. All of them, however, are opening prices.

On the two **homogeneous** panels the result changes. On closing, OGD beats three (WHC -0.00088, IWC -0.00077, BWC -0.00051) and ties with B365C ($p = 0.505$) and VC_BVC ($p = 0.284$); it does rank first, but with the plain average second at a distance of 0.00002, i.e. practically indistinguishable. On opening it again beats three, ties with B365 ($p = 0.203$), but **significantly loses to VC_BV** ($+0.00030$, $p = 0.004$): the only case in the whole thesis where a *bookmaker* other than Pinnacle significantly beats us, and there the plain average moves ahead of OGD. The statement is deliberately restricted to bookmakers: in the deployment test (§5.3.2) Betfair Exchange also significantly beats us, but that is not a bookmaker but an exchange.

The pattern is stable enough not to be attributed to chance, and it repeats the finding of §5.2.2: once Pinnacle leaves, the remaining members of the same phase resemble one another so closely that no quality difference is left to learn. The configuration that is selected is therefore the best partly *because* it contains the market's best odds-setter.

5.3 Robustness over time

5.3.1 Recent window

The coverage threshold is defined over the decade, which favours whoever was quoting long ago. Restricting to the last five seasons (38,749 matches) brings VC_BVC and WHC above it as well, so the mixture is measured against five closing prices instead of three and the comparison becomes correspondingly more demanding (Table 5.8).

#	Series	phase	log-loss
1	PSC	closing	0.99439
2	**OGD**	—	**0.99495**
3	VC_BVC	closing	0.99498
4	WHC	closing	0.99502
5	BWC	closing	0.99504
6	B365C	closing	0.99543
7	Uniform	—	0.99591
8	Hedge	—	0.99604
9	FTRL	—	0.99607
10–14	WH, PS, VC_BV, BW, B365	opening	0.99771 and below

Table 5.8: Last five seasons (38,749 matches), full panel: calibrated log-loss of each series over the matches it quoted in the window, lower better. **phase** is opening or closing for a bookmaker and — for the methods; the last row collects the five opening bookmakers.

OGD sits at #2 (Figure 5.7), ahead of every closing bookmaker other than Pinnacle, and the gap to PSC shrinks to $+0.00036$ ($p = 0.014$) from the decade's $+0.00043$. Two findings, however, *reverse* and are reported explicitly: Hedge and FTRL lose to the plain average on *all* panels here, not only the restricted ones ($+0.00012$ and $+0.00015$, $p = 0.006$ and $p = 0.002$), and log-loss and Brier cease to agree on the ordering.

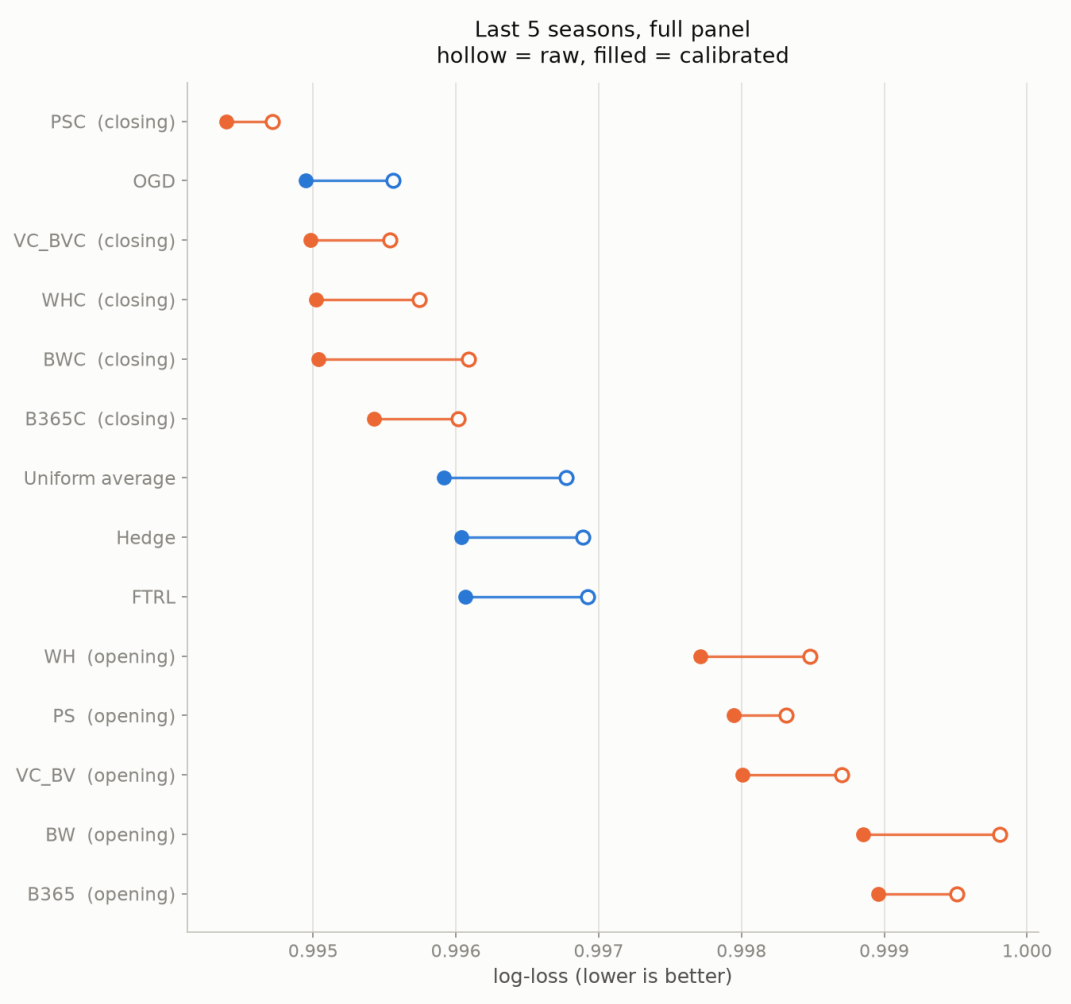


Figure 5.7: Log-loss over the last five seasons, full panel. Horizontal axis log-loss, lower better; one row per series, hollow end raw and filled end calibrated, orange for a bookmaker and blue for an algorithm. Five closing bookmakers clear the coverage threshold inside this window against three over the decade, which makes the comparison more demanding.

5.3.2 Deployment test

The test with a real date cut: fitting on 65,009 rounds up to 31/12/2024, measurement on 11,574 rounds of 2025–26, of which 8,151 are common. Table 5.9 gives the ranking and Figure 5.8 plots it.

#	Series	phase	log-loss
1	BFEC (Betfair Exchange)	closing	1.00040
2	**OGD closing-only, online**	—	**1.00110**
3	OGD full panel, online	—	1.00129
4	BWC	closing	1.00146
5	OGD closing-only, frozen	—	1.00151
6	B365C	closing	1.00165
7	OGD full panel, frozen	—	1.00183
8–10	B365, BW, BFE	opening	1.00371 – 1.00758

Table 5.9: Trained up to 31/12/2024, measured on 2025–26: calibrated log-loss over the 8,151 matches of the test window that every listed series quoted, lower better. *online* means the mixture keeps learning through the test window as results arrive, *frozen* that its weights are locked at the cut; the last row collects the three opening prices.

- **Every opening price is significantly beaten**, in every variant: -0.0019 to -0.0065, all $p \leq 0.019$.
- **The closing prices are not beaten**. The best outcome is a tie with all three, and only in the closing-only / online variant: B365C $p = 0.193$, BWC $p = 0.361$, BFEC $p = 0.052$. In the full-panel variants **BFEC significantly beats us** ($+0.00089$, $p = 0.013$ online; $+0.00143$, $p = 0.002$ frozen), so the tie with every closing price holds only for the closing-only variant.
- The online variant consistently beats the frozen one (1.00110 against 1.00151), so the precise statement is not "trained up to 2024" but "runs continuously".

Pinnacle in this window. Its coverage drops to 59% after 2025 and it stops quoting on 14/01/2026, so it does not pass the threshold. Forcing it into the common table drops the common matches from 8,151 to 2,860 and no result remains significant. It is therefore compared one to one, over its own rounds: we significantly beat PS (-0.0022 to -0.0023, p from 0.007 to 0.019, over 6,857 matches), and we tie with PSC (p from 0.532 to 0.796, over 6,892 matches).

The best forecaster in this window is not a bookmaker but **Betfair Exchange**: an exchange where players bet against one another, that is a market price without an operator's margin.

5.4 Value betting and the learned stake size

A Kelly simulation with calibrated probabilities and a panel matched to the target's phase. The stake size is not pinned to a fixed fraction but learned online, with the multiplier of §3.5.2; the fixed coefficient $\frac{1}{4}$ is retained as a reference point.

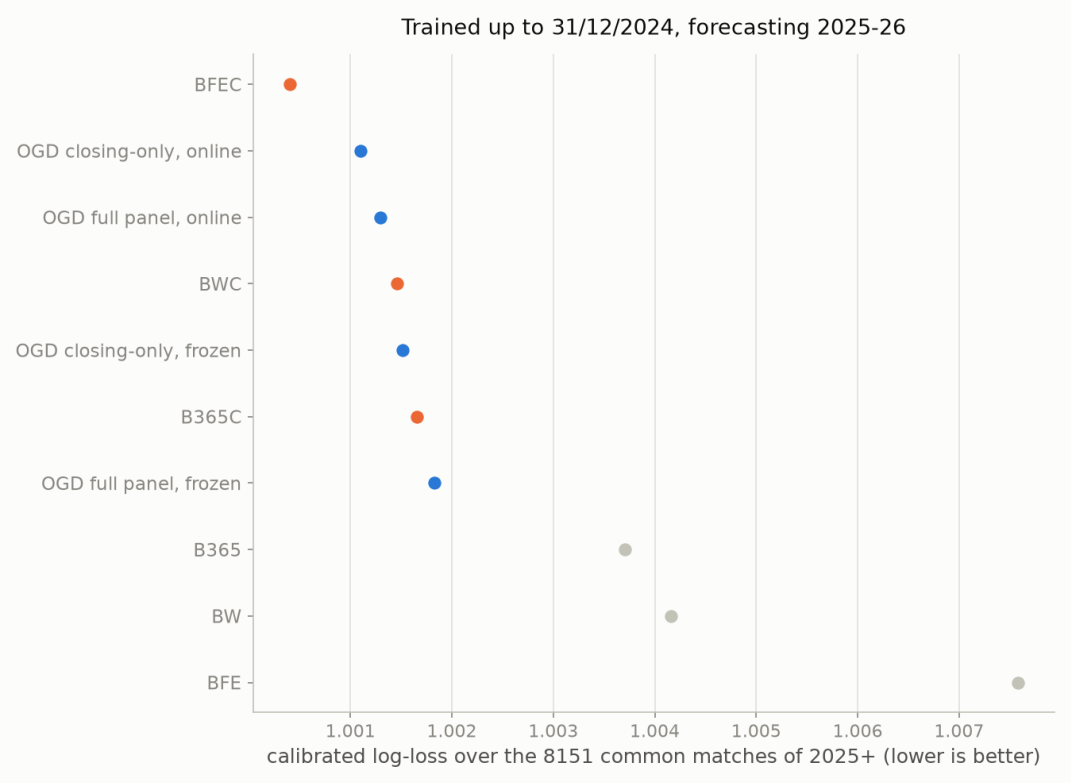


Figure 5.8: Deployment test. Horizontal axis the calibrated log-loss over the 8,151 common matches of 2025–26, lower better; one dot per series, best at the top. Blue dots are the mixture in its four variants (closing-only or full panel, online or frozen), orange dots the closing prices still quoted in the test window and grey dots the opening ones. The opening prices sit clearly further down, while the closing prices and the variants of the mixture cluster together.

Correctness check and selection. A constant $\lambda = 0.25$ reproduces the implemented simulation *exactly*, at every round of deployment and over the same set of bets, so the learned multiplier is the only difference between the columns that follow. The rate, the initial value and the upper bound of λ are selected on a chronological 75% prefix of the *bets*, jointly for all targets, and the remaining tail stays outside the selection. Neither of the two preceding stages is re-tuned here.

The selection gives $\lambda_0 = 0.5$ and $\lambda_{\max} = 1.5$ for Online Newton Step, with a tail of $+0.000106$ in log growth rate per bet, and $\lambda_0 = 0.25$, $\lambda_{\max} = 1$ for OGD, with a tail of $+0.000101$. One point is worth noting that contradicts every other tuning in the thesis: **Online Newton Step selects a rate multiplier equal to** 1, that is the constant the theory supplies needs no correction at all, whereas OGD needs $\times 10$ and in one setting the grid runs off the edge without stabilising. Where the structure of the loss is exploited correctly, the a-priori bound turns out to be correctly calibrated as well. Table 5.10 gives the closing targets and Figure 5.9 the bankroll paths behind them.

Against	Bets	Kelly $\frac{1}{4}$	OCO λ	Full Kelly	mean λ
WHC	7,253	$84.2\times$	**6,494**$\times$	$29{,}342\times$	0.79
BWC	6,920	$3.09\times$	**7.08**$\times$	$13.77\times$	0.62
VC_BVC	4,561	$1.58\times$	$1.54\times$	$1.74\times$	0.57
B365C	7,643	$1.05\times$	$0.67\times$	$0.24\times$	0.39
PSC	19,497	$0.54\times$	$0.25\times$	$0.0035\times$	0.37

Table 5.10: Final bankroll starting from 1 against each closing target, calibrated probabilities, proportional normalisation. **Bets** is the number of bets placed; the next three columns are the final bankroll with the fixed quarter of the Kelly fraction, with the multiplier λ learned online by Online Newton Step, and with the full Kelly fraction. Every column plays the *same* bets, since the choice of outcome does not depend on the size. The last column is the mean multiplier learned by Online Newton Step.

The result has two parts that do not say the same thing. On the closing targets, in the per-bet paired test the learned multiplier **ties** with the fixed $\frac{1}{4}$: over twenty cells (two normalisations $\times$ two stages $\times$ five targets) there are 17 ties, two wins and one loss. In final bankroll it is ahead on two of the five targets, and on WHC by seventy-seven times. The two reconcile numerically: the per-bet difference is $+0.00060$, minute, but it compounds over $7{,}253$ bets, while the uncertainty remains wide enough that the test does not certify it ($p = 0.099$). It is the same pattern as the "sharp" anchor-bookmaker result: systematic superiority in the point estimate, without statistical certification.

What the multiplier learns. Against the closing targets it sits almost everywhere above $\frac{1}{4}$, and its value tracks how profitable the target is: 0.37 against the loss-making PSC, 0.79 against the profitable WHC. It therefore raises the risk where there is an edge and pulls it back where there is none, behaviour no constant can express, and which emerges without its being given any information about who the opponent is.

The formulation was the problem, not the idea. The same question had been posed earlier with the learner learning the *fraction* of capital directly. That version lost system-

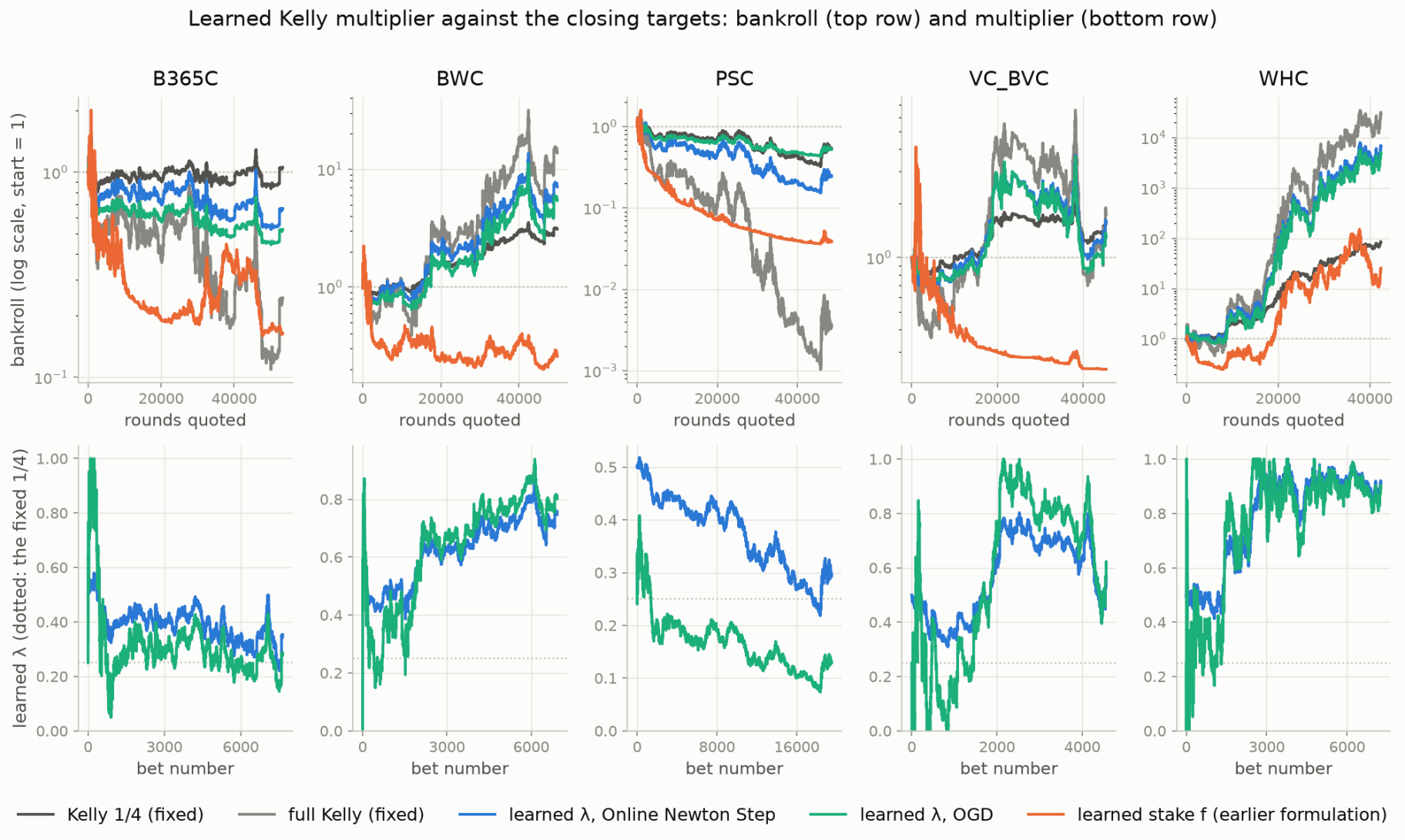


Figure 5.9: The learned Kelly multiplier against the five closing targets, one column per target; colours give the staking rule, as in the legend. Top row: the bankroll, on a logarithmic vertical axis, against the number of rounds on which the target quoted a price, with a dotted line at the starting bankroll of 1. The bankroll stays flat on rounds without a bet, which is why the two rows have different horizontal lengths. Bottom row: the learned multiplier λ against the bet number, with a dotted line at the fixed $\frac{1}{4}$.



Figure 5.10: As Figure 5.9, for the six opening targets.

atically: 9 losses out of 20 cells, no wins. The cause is exactly the one §3.5.2 predicts; the comparator "best constant stake" does not contain the closed form, and the learner throws away the $\hat{p}_t$ that the two preceding stages have just produced. Changing only what is learned, with the same data and the same test, the result goes from systematic loss to a tie with a point-estimate advantage.

A side finding: $\frac{1}{4}$ is excessively conservative. *Full* Kelly ties in 16 of the 20 cells, wins in 2 and loses in only 2. Its risk is concentrated almost entirely on PSC, where it wipes out the bankroll (0.0035). On the profitable targets, that is, the fixed $\frac{1}{4}$ leaves a substantial part of the growth unused. The learned multiplier is the middle road: it keeps most of the upside and avoids the catastrophe.

Where there is an edge. An *edge* here means an advantage over a bookmaker's price large enough to be detected statistically, that is a positive expected growth of capital when betting against it. Which bookmakers are beatable at all in this sense is mapped over eleven targets with the fixed $\frac{1}{4}$ (Table 5.11), so that the rows stay comparable with one another.

Against	Phase	Bets	Bankroll	p	
WHC	closing	6,565	$38.52\times$	< 0.001	profit
IW	opening	7,443	$7.89\times$	0.008	profit
BWC	closing	6,607	$1.81\times$	0.247	—
WH	opening	5,931	$1.38\times$	0.545	—
BW	opening	7,841	$1.09\times$	0.820	—
VC_BVC	closing	4,653	$0.75\times$	0.455	—
VC_BV	opening	8,083	$0.60\times$	0.245	—
B365C	closing	7,970	$0.58\times$	0.328	—
B365	opening	10,829	$0.41\times$	0.101	—
PS	opening	26,675	$0.26\times$	0.055	—
PSC	closing	39,997	$0.0002\times$	< 0.001	loss

Table 5.11: Value betting with the fixed $\frac{1}{4}$ of the Kelly fraction against eleven targets. **Bets** is the number of bets placed and **Bankroll** the final bankroll starting from 1. The p column tests whether the mean log growth rate per bet differs from zero (moving block bootstrap), and the last column gives the verdict where it does. The final bankrolls are not comparable across rows, since the number of bets differs by an order of magnitude.

Over eleven bookmakers there are *two* significant profits (both against "soft", recreational-facing houses) and one significant loss, against PSC, which is the best forecaster in the panel and therefore exactly the opponent an inferior model ought to lose to. The test thus works in both directions. Phase matching is essential: with the full panel instead of the closing-only one, the result gets worse on *every* closing target.

The extension to opening targets. The learned multiplier is applied to all six opening targets as well, with the corresponding opening-only panel. The causality requirement is stricter here: removing the target is not enough, the *whole* closing side must go, since no closing price exists at the moment the bet is placed (Table 5.12 and Figure 5.10). The

hyperparameters are selected separately per phase, on that phase's own prefix of bets, so that the opening result does not depend on closing bets it is supposed to be independent of.

Against	Bets	Kelly $\frac{1}{4}$	OCO λ	Full Kelly	mean λ
IW	7,443	$7.89\times$	$\mathbf{64.4}\times$	$109.6\times$	0.67
WH	5,931	$1.38\times$	$1.39\times$	$0.78\times$	0.28
BW	7,841	$1.09\times$	$1.10\times$	$0.50\times$	0.45
VC_BV	8,083	$0.60\times$	$0.67\times$	$0.03\times$	0.17
B365	10,829	$0.41\times$	$0.61\times$	$0.007\times$	0.12
PS	26,675	$0.26\times$	$\mathbf{0.68}\times$	$0.0003\times$	0.05

Table 5.12: Opening targets: final bankroll starting from 1, calibrated probabilities, proportional normalisation, with the columns as in Table 5.10. The OCO λ column gives the better of the two variants (Online Newton Step or OGD) and the mean λ corresponds to it.

The finding is that the learned multiplier pays off *more* on opening prices than on closing ones: 8 wins and 2 losses over 48 paired tests, against 4 wins and 3 losses on the closing side. Significant wins arise against PS ($p < 0.001$), IW ($p = 0.027$) and VC_BV ($p = 0.009$), while the two losses are minute (-0.000028 and -0.000080).

More interesting than the result is the *mechanism*, which becomes visible here. The mean λ ranks the targets almost perfectly by how beatable they are, without the algorithm being given any information about who the opponent is: against IW, the only profitable opening target, it raises λ to 0.67, that is nearly full Kelly; against PS, the hardest, it lowers it to 0.05 and essentially stops betting, thereby more than doubling the surviving capital, from $0.26\times$ to $0.68\times$. The opening λ values (0.05–0.75) are systematically lower than the closing ones (0.17–0.79): the opening market is harder and the algorithm discovers this by itself.

The same appears from the opposite side. *Full* Kelly, which tied on the closing targets (2 wins, 2 losses), collapses on the opening ones: **11 significant losses against 2 wins**. The opening side genuinely does need shrinkage, and the question is not whether to impose it but how much, which is precisely the question a fixed coefficient of $\frac{1}{4}$ cannot answer and a learned one can.

The conclusion is that improved log-loss does not translate into profitability. The quality difference gained is smaller than the market's margin, and whatever edge exists concerns *specific* bookmakers, not a strategy that works generally. Stake size is orthogonal to this: it changes how fast the bankroll grows where an edge exists, and how fast it is lost where none does, without creating an edge from nothing.

5.5 How much information does learning need?

5.5.1 The bandit experiment

Everything above presupposes full information: after each match all 26 bookmakers are scored. The question of §3.3.8 is what remains when the algorithm sees only *one* number per round, the loss of its own prediction. The two gradient estimators, one-point and two-point, are implemented, with ξ and the step selected on the same chronological train prefix

as every other hyperparameter, and with ten different random seeds, since the algorithm is stochastic.

Two correctness checks come first. With full information and a linearised loss the code reproduces the implemented OGD *exactly*, and the two-point estimator is identified with the true tangential gradient with relative error 0.6% and cosine 0.99998 over $200{,}000$ samples. Table 5.13 compares the two estimators with full information and Figure 5.11 shows the sweep on which their selection rests.

Series	Cal. log-loss	Raw log-loss	Difference
OGD, full info., exact gradient	0.996928	0.997455	$+0.000002$
OGD, full info., linearised	0.996926	0.997454	—
Two-point BGD ($\xi = 0.0003$)	0.997368	0.998004	$+0.000443$
No learning ($\eta = 0$)	0.997632	0.998296	$+0.000707$
One-point BGD ($\xi = 0.02$)	0.997631	0.998297	$+0.000706$
Uniform average	0.998132	0.998862	$+0.001206$

Table 5.13: Bandit against full information, the same 76,584 rounds, the same calibration; lower log-loss is better. BGD is bandit gradient descent with the one- or two-point gradient estimator and shrinkage ξ, and *No learning* is the same algorithm with its step set to zero. The last column is the paired difference from the implemented OGD in calibrated log-loss; positive means worse than it. The first row is a tie ($p = 0.934$), the rest are significant at $p < 0.001$.

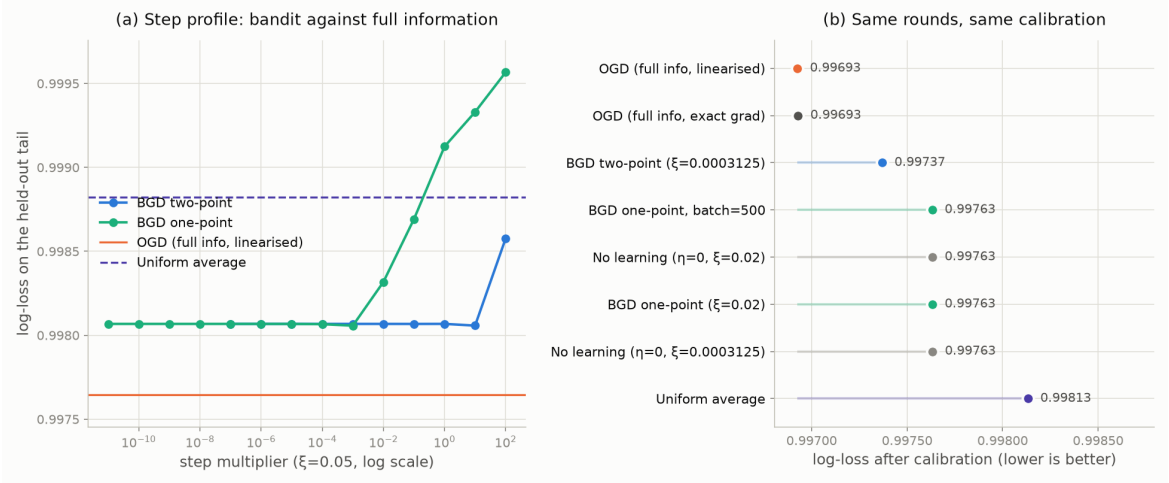


Figure 5.11: (a) The step profile on which the selection is made: horizontal axis the step multiplier on a logarithmic scale, vertical axis the log-loss on the held-out tail, one curve per gradient estimator, with horizontal lines for full-information OGD and for the uniform average. (b) The final comparison: horizontal axis the calibrated log-loss, lower better, one row per method. The flatness of panel (a) at small steps is itself the result, since a profile with no interior minimum means there is nothing for the estimator to select.

The control that makes the numbers interpretable. Write BGD for *bandit gradient descent*, the estimator of §3.3.8 run on the loss of the chosen mixture alone. The obvious reading ("BGD beats the uniform average, so the bandit feedback taught it something") is

wrong. Under the Sleeping Experts reduction the weight vector moves even with a *zero* step: every time the awake set changes, the persistent state is reprojected onto a simplex of different dimension, and that churn alone produces non-uniform weights. Measured directly: at $\eta = 0$ the weights are as far as 0.69 from uniform in max norm. The correct benchmark is therefore not the uniform average but *the same algorithm with the step set to zero*: same shrinkage, same perturbations, same projection dynamics, no learning. Whatever lies between that control and BGD is what the bandit signal actually bought, and Table 5.14 splits the margin in exactly this way.

	Projection without learning	**+ bandit learning**
Two-point BGD	$+0.000500$ (41.4%)	$+0.000264$ (21.9%)
One-point BGD	$+0.000501$ (41.5%)	$-\mathbf{0.000000}$ (0.0%)

Table 5.14: Decomposition of the full-information margin (uniform average 0.998132 against OGD 0.996926, i.e. 0.001206) into what the projection gives with no learning at all and what the bandit feedback adds on top. Each cell is the fall in calibrated log-loss below the uniform average that the step contributes and, in parentheses, its share of the full-information margin.

The one-point estimator learns nothing here. Selection on train drives its step to $1.4 \cdot 10^{-14}$, and the whole region from 10^{-14} to $9 \cdot 10^{-11}$ ties at the same train score 0.998375: this is the plateau of not learning, and the selector simply takes its endpoint. Any positive amount of learning hurts. The cause is quantified by the alignment of a single round's estimate with the true gradient: for the two-point estimator the mean cosine is $+0.27$, whereas for the one-point estimator it is $+0.002$, that is zero. The signal is 0.2% of the size of the estimate; summing over T rounds reduces the noise by $\sqrt{T} \approx 277$, which leaves an effective ratio of about 0.55, still below one. **The horizon is not long enough** for a one-point estimator on a 25-dimensional simplex.

The one legitimate fix was tried and changes nothing. The variance argument dictates a specific intervention: accumulating the estimate for m rounds before stepping, which raises the effective ratio by $\sqrt{m}$ and keeps the point in the interior of the simplex, so that the projection does not asymmetrically clip the noise. m was swept over $\{1, 20, 100, 500\}$ across five orders of magnitude of the step. *No* combination produces an interior optimum: every cell is either on the plateau or worse, and batching does exactly what the theory predicts (it tolerates a larger step without damage) while never producing a gain. The signal is not there to be amplified.

A caveat about the two-point estimator. Its shrinkage parameter was driven by the selection to the lower end of the admissible interval. This is not an accident but what is to be expected: as $\delta \to 0$ the finite difference converges to the exact directional derivative, so the two-point estimator *ceases to be bandit*. The 21.9% of Table 5.14 must therefore be read as an upper bound on what the two-query approach achieves, not as performance in a genuine bandit environment. The one-point estimator is the honest experiment, and its answer is zero.

What remains from the experiment. Two things, and neither is what it was looking for. First, that one-point feedback is insufficient at this horizon, with a quantitative explanation of why and with the obvious fix tried and rejected. Second, and more useful for the rest of the thesis, that 41% **of the mixture's margin over the uniform average does not come from learning at all**, but from the reprojection forced by the changing composition of the panel. This is an independent finding about Sleeping Experts dynamics and it agrees with the measurement of §5.3.2, where weight movement in the late stages turned out to be dominated by the projection and not by the gradient step.

A side finding in favour of linearisation. The full-information control with the *exact* gradient of the non-linear loss ties with the implemented, linearised OGD ($+0.000002$, $p = 0.934$). The simplification of Lemma 3.8, that is, costs nothing measurable on these data: a confirmation of a design choice that until now rested only on the theoretical argument that the linear problem is harder.

5.5.2 Markovian switching of market regimes

The Markov chain on the *weight vector* has already been tried and rejected: every transition kernel selects $\alpha = 0$ on train, i.e. collapses to plain OGD. Here the chain runs over a different object, over *market states*. We maintain K latent regimes, one weight vector per regime under Sleeping Experts, and a belief updated by the forward recursion of a hidden Markov model; each regime steps in proportion to its responsibility. $K = 1$, a single regime, reproduces the plain Sleeping-Experts OGD of §3.3.5 exactly.

The motivation is measured and not hypothetical: §5.2.2 and the information arrival analysis show that the closing advantage varies 26-fold depending on how much the line moved. There is therefore a genuine latent state in the data. The regime is deliberately left *latent*, so that it can be checked afterwards whether the chain discovers it on its own.

The first version failed in a diagnosable way. The mean belief stayed at exactly 0.500 for both regimes. The regimes *never differentiated*: they receive the same loss vector, so from a symmetric starting point they follow identical trajectories and their likelihoods stay equal. The result was simply OGD with the step divided by K, and it lost significantly ($+0.000128$, $p < 0.001$). The symmetric solution is a fixed point of the dynamics, and an explicit differentiation mechanism is required.

After the fix: a tie. A belief sharpness β, a forgetting factor γ and a per-regime clock were added. Selection on train gives $K = 3$, $\beta = 30$, $\gamma = 0$, $\tau = 0$, with calibrated log-loss 0.996909 against OGD's 0.996926: a difference of -0.000016, $p = 0.211$, **a tie**. The $\beta = 30$ is a genuine interior optimum; the grid was extended to 1000 and the values 100, 300, 1000 are all worse on train.

Two findings that nullify any claim that "the chain works". First, the tail is *flat*: from $\beta = 3$ to $\beta = 1000$ it moves between 0.997647 and 0.997649, a range of two millionths over three orders of magnitude of the parameter, while the train differences driving the selection are ten times larger. The selection is fitting noise, exactly as in the joint sweep of step and calibration rate (§5.6). Second, and more decisive, $\tau = 0$ is selected (no transition), and the standard deviation of the belief is 0.0000: it freezes early and never moves again.

There is therefore no Markovian switching. What remains is a fixed convex combination of three OGDs with different initialisations, that is *ensembling*, and the small point-estimate gain must be attributed to that.

And the chain discovers nothing. The correlation of the belief with the measured line movement is undefined, precisely because the belief is constant. The latent regime corresponds to no measurable property of the market. The answer to the question the experiment posed (does a model with no side information discover the structure that was measured directly?) is negative and clean. Figure 5.12 shows both halves of it.

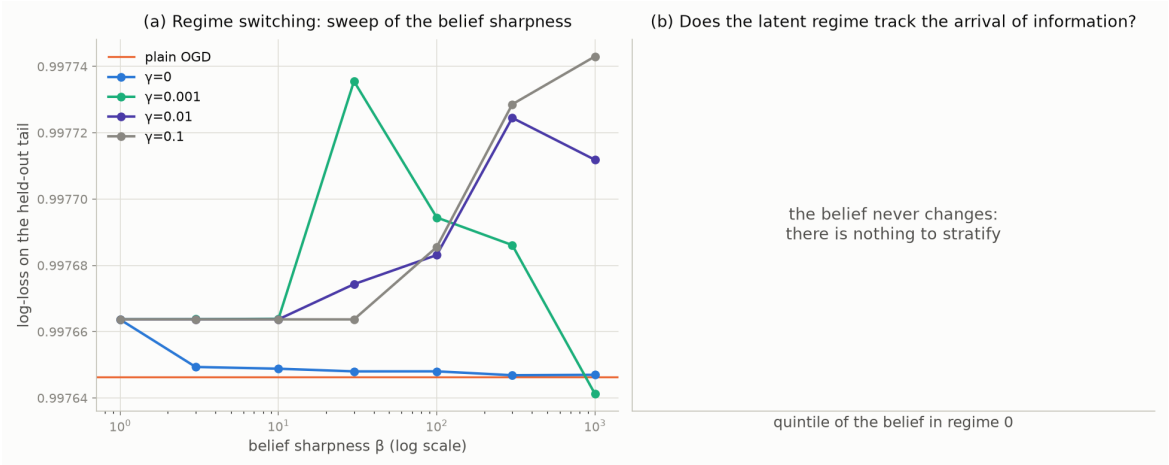


Figure 5.12: (a) The sweep of the belief sharpness β: horizontal axis β on a logarithmic scale, vertical axis the log-loss on the held-out tail, one line per forgetting factor γ and a horizontal line for plain OGD. (b) The check of whether the latent regime corresponds to the measured information arrival. It is empty because the selected model's belief never changes, so there is nothing to stratify.

This is the third independent instance of the strongest pattern in the thesis: once the step is tuned, no movement of mass over the experts offers anything measurable. Fixed-Share, transition kernels, and now regime switching. The difference here is that the failure was *diagnosed* rather than merely observed.

5.6 Negative results

A series of plausible hypotheses was tested and rejected. We report them explicitly, since each one bounds the method's scope.

Per-league specialisation does not help. A separate model per league loses to the single model in *every* one of the six leagues tested (Table A.23). Two plausible causes: each league has only 2,231–3,800 matches, far too few for the online algorithm to converge on its own, and the specialists end up trusting largely the *same* experts as the unified model, i.e. they do not discover anything local.

Fixed-Share does not help. Every $\alpha > 0$ worsens the log-loss (Table A.21: 0.99842 at $\alpha = 0$ against 0.99885 and above); the $\alpha = 0$ case reproduces plain Hedge exactly, which is what makes this a statement about Fixed-Share and not about the particular implementation of it. Sweeping α *jointly* with the step shows why: α only ever helped when the step was too large. Both control the same quantity (how fast the algorithm forgets), and the step is the more efficient of the two instruments.

A per-expert step clock significantly worsens things. Replacing the global t with the number of rounds in which each expert was awake (which is literally the quantity the Sleeping Experts guarantee speaks about) gives a result that is *better on train but* -0.00073 *on the tail.* An expert with 6.9% coverage receives a step about 14 times larger, and the mixture chases the rare bookmakers, whose apparent quality is confounded with the difficulty of the single season in which they appear. This is the clearest demonstration of overfitting in the thesis.

Joint selection of step and calibration rate offers nothing. The two parameters do not interact: the optimal calibration rate is 0.15 in 11 of 11 rows. Joint selection on train in fact picks a worse value for OGD, since the grid is nearly flat there and the selection fits noise.

Markovian transition kernels offer nothing. All three kernels of §3.3.7(a) (uniform, towards the opening/closing twin, towards the same market phase) select $\alpha = 0$ on the chronological prefix, i.e. collapse to plain OGD. The paired tests confirm this with a mean difference of 0.000000 *exactly* and $p = 1.000$, not approximately. The finding agrees with Fixed-Share: once the step itself is tuned, no post-hoc movement of mass has anything left to correct, because both control the same quantity. At *large* α, however, the structured kernels do measurably less damage than the uniform one, i.e. the panel's structure is real and detectable, merely redundant.

The Bayesian version of Kelly does not beat the fixed fraction. The formulation of §3.5.3 is theoretically more attractive, since the shrinkage emerges from the uncertainty instead of being imposed. Paired with the fixed $\frac{1}{4}$ over the same bets, for four targets and two stake levels, it gives **6 ties and 2 significant losses** (PSC at both levels, $p = 0.003$ and $p = 0.049$), with *no* significant win. The selected concentration parameter κ is moreover unstable across targets with no evident pattern, an indication that the selection is fitting noise. The most likely explanation is that the Dirichlet model does not correctly describe the mechanism generating bookmakers' disagreements: their disagreements are not independent noise around a common truth but systematic pricing differences, so the posterior is the wrong distribution and the shrinkage it produces does not correspond to the real uncertainty.

Draws are practically unpredictable as a primary prediction. Although they constitute $\sim$26% of the matches, the best algorithm selects them as the most likely outcome in 0.4% of cases (Table A.24 gives the performance of every series by outcome class). This is a known phenomenon of football prediction, confirmed here quantitatively.

5.7 Discussion

5.7.1 What the thesis showed

Summarising the findings of this chapter, three answers emerge at different levels.

At the level of **method**, the framework works, but less than a first reading suggests. An online mixture of published odds, with no data beyond the prices themselves, consistently and significantly beats both the uniform average and the overwhelming majority of individual companies. That this is achieved with no statistical assumption about the distribution of the data, and with predictions that are by construction out of sample, is the main methodological gain of the online approach over a classical batch one. The limitation emerged from the experiment of §5.5.1: with an explicit zero-step control it turns out that 41% of the margin over the uniform average is produced by the repeated projection onto the simplex, that is by the geometry of the reduction and not by learning. The weighting does add something real, but less than one would conclude by comparing it directly against the average.

At the level of **comparison with the market**, the answer is more guarded. Over the full decade and on the full panel the sharpest bookmaker remains significantly ahead. In the configuration finally selected the difference ceases to be significant, and in the deployment test the mixture beats every opening price and ties with the bookmakers' closing prices. The real limit is not a bookmaker: Betfair Exchange, which is an exchange and therefore a market price without an operator's margin, significantly beats us. The honest statement is that we *reach* the leading edge of the bookmaker market, not that we exceed it, and that the market without an intermediary remains ahead.

At the level of **economic exploitation**, the answer is negative. Improved log-loss does not translate into profitability, because the quality difference gained is smaller than the margin the market charges. This distinction (between being a better forecaster and being able to exploit it) is, in our view, the most practically useful conclusion. The third OCO stage does not overturn it, but it clarifies what stake size can and cannot do: the learned multiplier multiplies the final bankroll where an edge exists and shrinks itself where none does, without ever creating an edge. The choice of opponent remains far more important than the choice of stake.

5.7.2 When learning pays off: heterogeneity as a precondition

The recurring pattern throughout the thesis is that learned weighting pays off when, and only when, the experts differ substantially in quality. The finding appears independently at three points:

- On the full panel, which contains 13 good closing and 13 systematically worse opening experts, all three algorithms significantly beat the plain average.
- On the homogeneous panels (opening only or closing only) Hedge and FTRL *lose* to the average, in 8 tests out of 8.
- Removing Pinnacle from the *opening* panel, even OGD falls behind the average (0.99999 against 0.99990) and significantly loses to VC_BV.

The fourth case (the closing panel without Pinnacle) does **not** support the pattern any more, and is reported for precisely that reason. There OGD ranks first (0.99919) with the

average second (0.99921). The difference is 0.00002, practically zero, so the correct reading is not "weighting wins here" but "the two are indistinguishable", which still agrees with the general conclusion, since in a homogeneous panel weighting has nothing to learn.

The interpretation is that the gain comes primarily not from the algorithm but from the structure of the panel. When a real quality difference exists, the weighting finds and exploits it. When it does not, the adaptation is pure cost: the algorithm pays estimation noise with no corresponding benefit, and the average (which estimates nothing) proves more robust.

This conclusion has direct practical value for anyone who would apply the framework in another field. The criterion for a good set of experts is not their number, nor how much their numbers differ, but whether they are *produced by different processes*.

5.7.3 The character of the method

A property of the framework deserves noting that is not always made explicit. The model is not trained and then applied; there is no moment at which it "knows" and subsequently acts. At every match it states a claim, the outcome refutes it to some degree, and the claim is revised. The refutation *is* the material of learning, literally: the gradient that updates the weights equals the loss of the prediction against what happened. Without error nothing is learned, and the weights at any moment are not the answer but the current state of an unresolved disagreement.

This property was also confirmed empirically, where we did not expect it. We examined whether the mixture eventually "settles" somewhere and quietens down, given that the step decays as $1/\sqrt{t}$. It does not. The step does fall by a factor of 277 from the first round to the last, but the mean movement of the weight vector per round is 0.0353, 0.0226, 0.0227, 0.0284 over the four quarters: the last moves *more* than the second and the third, at a point where the step is two hundred and seventy times smaller. The reason is that the movement is dominated by the projection onto the simplex every time the awake set changes, not by the step: the same phenomenon that §5.5.1 isolates quantitatively, showing that the projection on its own, with no learning at all, produces 41% of the mixture's margin over the uniform average.

Relatedly, the theoretically superior alternative (the Aggregating Algorithm, which exploits the exp-concavity of the log-loss and achieves regret $\leq \ln N$ with a constant $\eta = 1$) performs in practice *worse than the uniform average*. The explanation is precisely that its bound concerns a *static* comparator, while the panel is not static: Pinnacle's coverage collapses towards the end of the period, and an algorithm that concentrates quickly on a few experts is punished exactly by an expert that disappears.

Chapter 6

Conclusions and future work

This thesis treated the betting companies that quote a football match as experts, and learned online how to combine their forecasts with three Online Convex Optimization algorithms adapted to experts that do not quote every match. Two further online stages calibrate the combined forecast and size the stake of a bet. Over 76,584 matches from 22 leagues and ten seasons, every comparison was tested with a moving block bootstrap, and a deployment test with a real date cut checked what a model trained up to the end of 2024 would have done in 2025–26. This chapter summarises the findings, interprets them, states the limits of the method and proposes future work.

6.1 Conclusions

The main conclusions of the present work are the following.

1. **The OCO framework applies successfully to betting markets**. An online mixture of published odds significantly beats the uniform average and the overwhelming majority of individual companies, with no assumption about the distribution of the data.

2. **The mixture reaches but does not exceed the market's leading edge**. In the configuration selected, the difference from the sharpest bookmaker ceases to be statistically significant. In the deployment test it significantly beats every opening price and ties with the bookmakers' closing prices. The limit is set by Betfair Exchange, which is not a bookmaker but an exchange, that is a market price without an operator's margin: that one significantly beats us.

3. **A substantial part of the apparent superiority is not learning**. Under the Sleeping Experts reduction the weight vector moves even with a zero step, because the projection is redone onto a simplex of different dimension every time the awake set changes. With an explicit zero-step control, 41% of the margin over the uniform average is attributed to this geometry and not to the algorithm. Every future comparison against the uniform average ought to carry the same control.

4. **Panel heterogeneity is a precondition, not a detail**. Learned weighting pays off only when the experts differ substantially; on a homogeneous set the plain average is at least as robust. The finding is independently confirmed at three points, while

at a fourth (the closing panel without Pinnacle) the two end up indistinguishable (§5.7.2).

5. **Worst-case theoretical steps are not optimal in practice**, and they in fact miss in opposite directions per algorithm. They protect against an adversary that betting odds are not. The only exception is the third OCO stage, where the algorithm that exploits the structure of the loss accepts the theoretical constant uncorrected.

6. **Better prediction does not mean profit**. The quality difference gained falls short of the market's margin, and whatever edge exists concerns specific companies, not a general strategy.

7. **In the third OCO stage, *what* is learned decides the outcome**. Learning the stake directly fails systematically, because the comparator "best constant stake" does not contain the closed form and because it throws away the probability the two preceding stages have just produced. Learning the *multiplier* on top of the closed form ties with the fixed coefficient on the closing targets and significantly beats it on the opening ones, by 8 wins to 2 losses, learning to stake more against some opponents and less against others without ever being told who the opponent is. It is also the only place in the thesis where theory's a-priori constant came out right without correction, precisely because the algorithm that uses it exploits the structure of the loss.

8. **Methodological discipline in comparisons is decisive**. Several intermediate "findings" turned out to be artefacts of comparing over different sets of matches, or of selection on a flat grid. The requirement of common rounds and the reporting of a held-out tail is what caught them. The sharpest instance concerns the step multipliers themselves: read off the held-out tail, two of the three appear mistuned, and the thesis carried that caveat for a time; read off the chronological prefix, which is what the protocol specifies, all three are already at their optimum. Selecting on the tail is selecting on the test set, and here the two criteria disagree.

6.2 The odds as a social relation

It is easy to read the weights as abstract parameters. Each expert here is a business, with its own costs, its own risk tolerance and its own business model, and the differences the algorithm learns are differences between those. Pinnacle is ahead not because it necessarily has a better prediction model but because it operates on a low margin and does not restrict winning players, so that its price is continuously corrected by informed bettors instead of their being shut out; in the 2025–26 window the best forecaster is not even a bookmaker but Betfair Exchange, where the price carries no operator margin; and over ten seasons the market did *not* become more accurate (no series shows a significant trend in log-loss or in the expected calibration error, ECE, the average gap between forecast probability and observed frequency; p from 0.21 to 1.00, Table A.27) but it did become more expensive, the overround rising from 4.42% to 6.00% ($p = 0.0007$).

The odds appear as a property of the match. De-vigging institutionalises the impression: you strip the margin in order to "recover" the true probability, as though beneath the commercial form there were a determinate number that the house distorts and you

restore. There is none. Remove Pinnacle from the panel and the "probability" changes, because what was being measured was not knowledge about football but position in the market. The technique itself confesses as much: Shin's parameter z (§3.2.2) **is** the estimated proportion of informed bettors, that is a social relation, and it enters the method as a *nuisance parameter* to be removed. This is what Marx [32] called the fetishism of the commodity: a definite social relation between people assumes the form of a property of a thing. Here the relation between the house, the informed and everyone else assumes the form of a number attached to a football match, and the fetishism operates *inside* the formalism rather than outside it.

It does not follow that there is nothing to be known. The uncertainty term of Murphy's decomposition is improved by no method and no amount of capital, and the closing price beats the opening price systematically. Something resists, and that something is the match, which occurs whatever anyone prices it at. The precise statement is therefore not that the odds lack objectivity but that **they are not objective about what they appear to be about**: there is no probability of the match waiting to be read off; what there is, is an objective expression of a relation, and it is measured, compared and ranked reproducibly. The fetish is not that the number is false but that we misattribute what it is a number of. If appearance and essence coincided directly, Marx [33] observed, all science would be superfluous; here the distance between them is itself a measurement.

Who pays, and why the market got dearer without getting better. The microstructure model of §3.2.2 already supplies the mechanism: the market maker charges a margin sufficient for the gains from the uninformed to cover the losses to the informed. But "informed" is not a mental state; it is a position with respect to the **material means**: data feeds, capital enough to move a line, and accounts that are not closed when you win. That same model makes the margin an increasing function of the proportion of informed traders, which appears at first to conflict with the measurement, since a market increasingly exposed to informed money should also have become more accurate. The two reconcile once it is asked *how* an operator may respond to informed money. It can absorb it, letting the price be corrected and growing more accurate, which is Pinnacle's model and the reason Pinnacle is hardest to beat. Or it can exclude it and raise the price instead, which leaves accuracy exactly where it was and the overround higher. A decade of rising margins with flat accuracy is the signature of the second response, and it is why better prediction does not convert into profit: the quality gained is measured against a toll that moves against it.

Profit and loss is the house's category, not ours. For the house no question of truth arises, only of the balance sheet, and the mechanism that secures it is not better forecasting but control of access. The thesis measures both sides of this. The significant edge appears against two targets only, WHC ($38.52\times$, $p < 0.001$) and IW ($7.89\times$, $p = 0.008$), both recreational-facing, while against the one house that does not exclude winners the simulation *loses* significantly ($0.0002\times$, $p < 0.001$). The edge exists exactly where it cannot be exercised and is absent exactly where it could be, and the reason is the same on both sides: the sharp house has the right price *because* it lets informed money correct it, the soft houses have the wrong one *because* they drive that money away. What separates the houses is not how well they predict but whom they let bet and at what price, which is Levitt's (2004) [30] observation that betting markets are organised around the bettor rather than around the price. The odds, read this way, are not a statement about the match

but **an offer, revocable against whoever accepts it too well**. Account restrictions are not measured here, since the data are prices and not customer outcomes; what is measured is the concentration of the edge at the soft houses and its absence at the sharp one. The revocation itself has been documented elsewhere: Kaunitz et al. (2017) [26] bet a profitable odds-based strategy with real money for five months and report that the response was not a correction of the prices but stake limits and account closures directed at them as clients. That is the complement of our measurement rather than a repetition of it, and the two together close the argument: the edge concentrates where access is controlled, and it is withdrawn when exercised.

No teleology, and the path is not a straight line. Every regret bound in the thesis assumes the opposite: comparison against the best **fixed** weight vector, which is the same presupposition as de-vigging, a determinate property where there is a relation in motion. The refutation is measured. The Aggregating Algorithm, holding the strongest bound in the thesis and one independent of the horizon, performs worse than the uniform average, because Pinnacle's coverage collapses and whoever concentrates quickly is punished by an expert that disappears; the schedule designed for convergence loses significantly to the non-decaying one for FTRL ($p = 0.001$); and the weights move more in the final quarter than in the middle ones, at a point where the step is orders of magnitude smaller. Nor does the thesis's own knowledge advance linearly: nine ideas were tried and rejected, findings reverse when the window changes, and a caveat about the step multipliers proved unfounded as soon as the selection criterion was corrected (§5.1). The path moves in cycles and settles into no system; the tree of knowledge, in Bitsakis's (2006) [2] image, stays green precisely because it is never finished. Characteristically, Fixed-Share, the explicit attempt to replace the fixed comparator with a sequence, does not help, because the movement **was already there** inside the step. The formalism was tracking motion from the outset while declaring a static aim: the fetish was not in the equations but in what we took them to be measuring.

That is what is at stake practically. A strictly positivist protocol would have delivered three wrong conclusions: that the entire margin over the uniform average is a product of learning, when 41% of it is not (§5.5.1); that variety among the experts suffices, when on a homogeneous panel the learned weighting loses in 8 tests out of 8 (§5.7.2); and that the market's hierarchy is a matter of technical capability, when it is a matter of margin, of access, and of who is permitted to win. All three were avoided by the same move: refusing to identify what is measured with what is real, and looking for the explanation in the relations that produce the measurement.

6.3 Limitations

The following limitations are stated explicitly.

Window choice is a form of selection. Three time windows were examined and the result improved in our favour with every narrowing. This is exactly the pattern that produces spurious findings, and for that reason all three are reported, the less favourable alongside the favourable.

Statistical power in the narrow windows. The deployment test rests on 8,151 matches against 71,818 for the full comparison. The confidence intervals are correspondingly wider, so a "tie" partly means "less data" and not only "equal quality".

The merging of identities is an inference. The VC/BetVictor merge rests on non-overlapping temporal presence, not on an official source. There may well be comparable rebrands that were not identified.

The simulation ignores liquidity. Account limits, stake restrictions and available liquidity are not modelled: precisely the mechanisms by which companies restrict winning players in practice. The simulation shows the *statistical existence* of an edge, not viable profitability.

The causality audit is not exhaustive. One leakage channel was found and fixed in the leave-one-out design, but there may be others at less obvious points.

The causal interpretations are plausible, not proved. The explanation, for example, of the margin increase as a commercial decision is consistent with the data but is not directly tested by it.

6.4 Future work

A live odds feed. Everything in this thesis is computed on a historical file, which means every result is a statement about what the method *would* have done. The framework itself does not require that: the three stages are online and causal, each round costs $O(N \log N)$, and the state is a single weight vector plus two scalars, so the whole chain runs in real time against a streaming odds feed. What such an implementation would add is not speed but the two things a file cannot supply. First, the prices would be the ones actually available for staking rather than the ones recorded afterwards, which is precisely the gap §6.3 identifies between a statistical and an exploitable edge: account limits, stake caps and liquidity become observable instead of assumed away. Second, the opening/closing distinction, which this thesis can only treat as two static snapshots, becomes the continuous object it really is: the measurement of §5.2.3, that the closing advantage is concentrated in the top quintile of line movement, suggests the interesting quantity is not the final price but the trajectory, and only a live feed records it.

A second sport. Extending to basketball or tennis would show whether the framework generalises beyond football. The cost is mainly a new data pipeline, since the algorithms do not change. A two-outcome sport, where the structure of the problem simplifies, is of particular interest.

Bayesian Model Averaging as a baseline. A full BMA baseline would give an empirical point of comparison against the online algorithms, and would further illuminate the failure of the Aggregating Algorithm.

Appendix A

Full result tables

This appendix gives in full the tables that the chapters of the main text present in abridged form. It is generated automatically from the result files, so that it cannot drift away from them.

In every table of paired tests, **Mean diff.** is the mean per-round difference in loss between the first-named series and the second, so a negative value means that the first is better; **CI low** and **CI high** bound its 95% moving block bootstrap confidence interval, n is the number of rounds the two series share and p is the p-value of the test.

A.1 Ranking and statistical significance

Table A.1 gives in full the ranking of §5.2.1, with each series' coverage, number of rounds and accuracy, and Table A.2 extends it to every series of the panel, including those below the coverage threshold. Table A.3 lists the paired tests quoted in that section.

Table A.1: Ranking over the ten seasons of the series with coverage of at least 60%, ordered by calibrated log-loss. **Coverage** is the share of the 76,584 matches for which a series quoted a price and **Rounds** their number. Log-loss and Brier are given raw and calibrated, lower better; **Acc.** is the share of matches whose most probable outcome occurred, which calibration does not change.

Series	Coverage %	Rounds	log-loss (raw)	log-loss (cal.)	Brier (raw)	Brier (cal.)	Acc.
PSC	93.8	71,818	0.99714	0.99685	0.59592	0.59577	0.50611
OGD	100.0	76,584	0.99745	0.99693	0.59612	0.59586	0.50662
Hedge	100.0	76,584	0.99842	0.99773	0.59674	0.59637	0.50649
FTRL	100.0	76,584	0.99843	0.99774	0.59674	0.59638	0.50653
Uniform average	100.0	76,584	0.99886	0.99813	0.59703	0.59665	0.50593
B365C	69.6	53,267	0.99953	0.99910	0.59732	0.59711	0.50705
BWC	65.0	49,755	0.99998	0.99919	0.59769	0.59728	0.50624
VC_BV	89.3	68,397	1.00075	1.00026	0.59833	0.59806	0.50355
WH	85.5	65,470	1.00119	1.00048	0.59860	0.59818	0.50309
PS	93.1	71,284	1.00083	1.00055	0.59838	0.59824	0.50309
IW	74.7	57,185	1.00186	1.00061	0.59899	0.59826	0.50214
BW	95.0	72,752	1.00140	1.00065	0.59873	0.59833	0.50355
B365	99.8	76,438	1.00113	1.00073	0.59852	0.59832	0.50377

Table A.2: Every series of the panel with no coverage threshold, raw scores over the matches each series quoted, ordered by log-loss. **Seasons** is the number of seasons in which a series quoted; one active in a single season is scored on that season alone, so its position mixes its quality with that season's difficulty. **Hedge weight** is the weight Hedge gave it at its own last awake round.

Series	Coverage %	Seasons	Rounds	log-loss	Brier	Acc.	Hedge weight
PSC	93.8	10	71,818	0.99714	0.59592	0.50611	0.000
BFC	10.0	1	7,680	0.99730	0.59594	0.51029	0.166
OGD	100.0	10	76,584	0.99745	0.59612	0.50662	—
BFEC	19.4	2	14,885	0.99783	0.59656	0.50890	0.045
1XBC	10.0	1	7,681	0.99786	0.59605	0.50814	0.166
LB	27.4	3	20,972	0.99797	0.59682	0.50644	0.000
LBC	7.3	1	5,587	0.99819	0.59667	0.50743	0.146
Hedge	100.0	10	76,584	0.99842	0.59674	0.50649	—
FTRL	100.0	10	76,584	0.99843	0.59674	0.50653	—
Uniform average	100.0	10	76,584	0.99886	0.59703	0.50593	—
CLC	6.9	1	5,257	0.99906	0.59732	0.50409	0.152
B365C	69.6	7	53,267	0.99953	0.59732	0.50705	0.000
BMGMC	9.8	1	7,508	0.99955	0.59776	0.50759	0.116
VC_BVC	59.3	6	45,424	0.99973	0.59757	0.50590	0.000
BWC	65.0	7	49,755	0.99998	0.59769	0.50624	0.000
WHC	55.3	6	42,366	1.00024	0.59787	0.50555	0.000
BF	10.0	1	7,673	1.00039	0.59809	0.50541	0.166
VC_BV	89.3	9	68,397	1.00075	0.59833	0.50355	0.000
PS	93.1	10	71,284	1.00083	0.59838	0.50309	0.000
BFDC	9.8	1	7,524	1.00085	0.59874	0.50704	0.115
IWC	44.4	5	34,038	1.00086	0.59820	0.50385	0.190
B365	99.8	10	76,438	1.00113	0.59852	0.50377	0.000
WH	85.5	9	65,470	1.00119	0.59860	0.50309	0.000
BW	95.0	10	72,752	1.00140	0.59873	0.50355	0.000
IW	74.7	8	57,185	1.00186	0.59899	0.50214	0.000
BMGM	9.5	1	7,309	1.00210	0.59938	0.50581	0.118
1XB	9.8	1	7,505	1.00268	0.59958	0.50393	0.169
BFD	9.9	1	7,619	1.00272	0.59991	0.50532	0.114
CL	7.0	1	5,393	1.00438	0.60099	0.50028	0.149
BFE	19.3	2	14,763	1.00716	0.60309	0.49604	0.045

Table A.3: The paired tests of §5.2.1, over the rounds the two compared series share. In a comparison of the form *raw vs calibrated*, a positive **Mean diff.** means that calibration helps.

Comparison	n	Mean diff.	CI low	CI high	p
OGD vs PSC (raw)	71,818	+0.00065	0.00044	0.00086	<0.001
OGD vs PSC (calibrated)	71,818	+0.00043	0.00020	0.00064	<0.001
OGD vs B365C (raw)	53,267	−0.00063	−0.00093	−0.00034	<0.001
OGD vs Hedge (raw)	76,584	−0.00097	−0.00118	−0.00075	<0.001
OGD vs FTRL (raw)	76,584	−0.00098	−0.00120	−0.00077	<0.001
OGD vs Uniform average (raw)	76,584	−0.00141	−0.00167	−0.00115	<0.001
Hedge vs Uniform average (raw)	76,584	−0.00044	−0.00062	−0.00026	<0.001
FTRL vs Uniform average (raw)	76,584	−0.00043	−0.00061	−0.00025	<0.001
Hedge: raw vs calibrated	76,584	+0.00069	0.00040	0.00096	<0.001

(continued)

Comparison	n	Mean diff.	CI low	CI high	p
OGD: raw vs calibrated	76,584	+0.00053	0.00027	0.00078	<0.001
FTRL: raw vs calibrated	76,584	+0.00069	0.00040	0.00096	<0.001
Uniform average: raw vs calibrated	76,584	+0.00073	0.00044	0.00101	<0.001

A.2 Panel and de-vig method

Tables A.4 and A.5 give the full grid of §5.2.2 (Table 5.4) with the Brier score and accuracy added, and Tables A.6 and A.7 compare each algorithm with the plain average of the same panel, the comparison from which that section draws its main conclusion.

Table A.4: Every algorithm on each of the three panels, with proportional normalisation: raw scores over the rounds the panel covers, lower log-loss and Brier better. **Panel** is opening-only, full or closing-only, and **Experts** is the number of bookmaker series in it.

Panel	Algorithm	Experts	Rounds	log-loss	Brier	Acc.
closing	FTRL	13	76,521	0.99761	0.59621	0.50658
full	FTRL	26	76,584	0.99843	0.59674	0.50653
opening	FTRL	13	76,557	1.00088	0.59842	0.50394
closing	Hedge	13	76,521	0.99762	0.59622	0.50659
full	Hedge	26	76,584	0.99842	0.59674	0.50649
opening	Hedge	13	76,557	1.00088	0.59842	0.50402
closing	OGD	13	76,521	0.99722	0.59598	0.50648
full	OGD	26	76,584	0.99745	0.59612	0.50662
opening	OGD	13	76,557	1.00076	0.59832	0.50391
closing	Uniform average	13	76,521	0.99748	0.59612	0.50672
full	Uniform average	26	76,584	0.99886	0.59703	0.50593
opening	Uniform average	13	76,557	1.00080	0.59836	0.50398

Table A.5: As Table A.4, with Shin's de-vig method.

Panel	Algorithm	Experts	Rounds	log-loss	Brier	Acc.
closing	FTRL	13	76,521	0.99714	0.59596	0.50659
full	FTRL	26	76,584	0.99782	0.59642	0.50646
opening	FTRL	13	76,557	1.00033	0.59812	0.50385
closing	Hedge	13	76,521	0.99714	0.59596	0.50659
full	Hedge	26	76,584	0.99781	0.59641	0.50645
opening	Hedge	13	76,557	1.00033	0.59812	0.50387
closing	OGD	13	76,521	0.99689	0.59581	0.50658
full	OGD	26	76,584	0.99709	0.59595	0.50682
opening	OGD	13	76,557	1.00024	0.59804	0.50389
closing	Uniform average	13	76,521	0.99703	0.59589	0.50674
full	Uniform average	26	76,584	0.99824	0.59670	0.50598
opening	Uniform average	13	76,557	1.00022	0.59805	0.50393

Table A.6: Each algorithm against the plain average of the same panel, proportional normalisation, for three metrics. **Mean diff.** is algorithm minus average; for log-loss and Brier a negative value favours the algorithm, for accuracy, where higher is better, a positive one does. **Verdict** gives the outcome at the 95% level.

Panel	Algorithm	Metric	n	Mean diff.	CI low	CI high	p	Verdict
opening	Hedge	log-loss	76,557	+0.00008	0.00001	0.00014	0.019	Uniform average wins
opening	Hedge	Brier	76,557	+0.00006	0.00002	0.00009	<0.001	Uniform average wins
opening	Hedge	accuracy	76,557	+0.00004	−0.00021	0.00027	0.817	tie
opening	OGD	log-loss	76,557	−0.00004	−0.00013	0.00006	0.433	tie
opening	OGD	Brier	76,557	−0.00004	−0.00010	0.00001	0.140	tie
opening	OGD	accuracy	76,557	−0.00007	−0.00043	0.00033	0.759	tie
opening	FTRL	log-loss	76,557	+0.00008	0.00002	0.00014	0.015	Uniform average wins
opening	FTRL	Brier	76,557	+0.00006	0.00002	0.00010	0.002	Uniform average wins
opening	FTRL	accuracy	76,557	−0.00004	−0.00027	0.00020	0.782	tie
closing	Hedge	log-loss	76,521	+0.00014	0.00010	0.00018	<0.001	Uniform average wins
closing	Hedge	Brier	76,521	+0.00009	0.00007	0.00012	<0.001	Uniform average wins
closing	Hedge	accuracy	76,521	−0.00013	−0.00046	0.00018	0.423	tie
closing	OGD	log-loss	76,521	−0.00026	−0.00036	−0.00016	<0.001	OGD wins
closing	OGD	Brier	76,521	−0.00014	−0.00021	−0.00008	<0.001	OGD wins
closing	OGD	accuracy	76,521	−0.00025	−0.00071	0.00022	0.297	tie
closing	FTRL	log-loss	76,521	+0.00013	0.00009	0.00018	<0.001	Uniform average wins
closing	FTRL	Brier	76,521	+0.00009	0.00006	0.00012	<0.001	Uniform average wins
closing	FTRL	accuracy	76,521	−0.00014	−0.00046	0.00017	0.367	tie
full	Hedge	log-loss	76,584	−0.00044	−0.00062	−0.00026	<0.001	Hedge wins
full	Hedge	Brier	76,584	−0.00030	−0.00042	−0.00017	<0.001	Hedge wins
full	Hedge	accuracy	76,584	+0.00056	−0.00014	0.00125	0.123	tie
full	OGD	log-loss	76,584	−0.00141	−0.00167	−0.00115	<0.001	OGD wins
full	OGD	Brier	76,584	−0.00091	−0.00108	−0.00074	<0.001	OGD wins
full	OGD	accuracy	76,584	+0.00069	−0.00025	0.00162	0.156	tie
full	FTRL	log-loss	76,584	−0.00043	−0.00061	−0.00025	<0.001	FTRL wins
full	FTRL	Brier	76,584	−0.00029	−0.00041	−0.00017	<0.001	FTRL wins
full	FTRL	accuracy	76,584	+0.00060	−0.00010	0.00131	0.108	tie

Table A.7: As Table A.6, with Shin's de-vig method.

Panel	Algorithm	Metric	n	Mean diff.	CI low	CI high	p	Verdict
opening	Hedge	log-loss	76,557	+0.00011	0.00004	0.00018	<0.001	Uniform average wins
opening	Hedge	Brier	76,557	+0.00008	0.00003	0.00012	<0.001	Uniform average wins
opening	Hedge	accuracy	76,557	−0.00005	−0.00029	0.00018	0.697	tie
opening	OGD	log-loss	76,557	+0.00002	−0.00008	0.00013	0.736	tie
opening	OGD	Brier	76,557	−0.00001	−0.00006	0.00005	0.826	tie
opening	OGD	accuracy	76,557	−0.00004	−0.00042	0.00035	0.896	tie
opening	FTRL	log-loss	76,557	+0.00011	0.00004	0.00019	<0.001	Uniform average wins
opening	FTRL	Brier	76,557	+0.00008	0.00003	0.00012	<0.001	Uniform average wins
opening	FTRL	accuracy	76,557	−0.00008	−0.00033	0.00017	0.559	tie
closing	Hedge	log-loss	76,521	+0.00011	0.00007	0.00016	<0.001	Uniform average wins
closing	Hedge	Brier	76,521	+0.00008	0.00005	0.00010	<0.001	Uniform average wins
closing	Hedge	accuracy	76,521	−0.00014	−0.00046	0.00017	0.379	tie
closing	OGD	log-loss	76,521	−0.00014	−0.00023	−0.00004	0.005	OGD wins
closing	OGD	Brier	76,521	−0.00007	−0.00013	−0.00001	0.016	OGD wins
closing	OGD	accuracy	76,521	−0.00016	−0.00060	0.00029	0.513	tie
closing	FTRL	log-loss	76,521	+0.00011	0.00006	0.00015	<0.001	Uniform average wins
closing	FTRL	Brier	76,521	+0.00007	0.00004	0.00010	<0.001	Uniform average wins
closing	FTRL	accuracy	76,521	−0.00014	−0.00046	0.00017	0.379	tie
full	Hedge	log-loss	76,584	−0.00043	−0.00062	−0.00023	<0.001	Hedge wins
full	Hedge	Brier	76,584	−0.00029	−0.00042	−0.00016	<0.001	Hedge wins
full	Hedge	accuracy	76,584	+0.00047	−0.00025	0.00116	0.219	tie
full	OGD	log-loss	76,584	−0.00115	−0.00140	−0.00090	<0.001	OGD wins
full	OGD	Brier	76,584	−0.00075	−0.00092	−0.00059	<0.001	OGD wins
full	OGD	accuracy	76,584	+0.00084	−0.00010	0.00175	0.079	tie
full	FTRL	log-loss	76,584	−0.00042	−0.00061	−0.00023	<0.001	FTRL wins
full	FTRL	Brier	76,584	−0.00028	−0.00041	−0.00015	<0.001	FTRL wins
full	FTRL	accuracy	76,584	+0.00048	−0.00024	0.00118	0.203	tie

A.3 Configurations

Table A.8 gives every configuration of §5.2.4 raw and calibrated (Table 5.6 shows the calibrated half), and Table A.9 the paired tests on which the conclusions of that section rest.

Table A.8: Log-loss of OGD in each configuration, on the same 71,818 matches: **full** over all of them, **train** over their chronological 75% prefix and **tail** over the remaining 25%, lower better; **Stage** says whether the probabilities are raw or calibrated. *PSC alone* is the benchmark bookmaker, scored on the same matches.

Configuration	Stage	full	train	tail
full panel, proportional	raw	0.99779	0.99860	0.99536
full panel, proportional	cal.	0.99728	0.99801	0.99508
closing-only, proportional	raw	0.99733	0.99801	0.99528
closing-only, proportional	cal.	0.99693	0.99756	0.99506
full panel, Shin	raw	0.99745	0.99822	0.99512
full panel, Shin	cal.	0.99731	0.99804	0.99512
closing-only + Shin	raw	0.99702	0.99767	0.99506
closing-only + Shin	cal.	0.99692	0.99754	0.99507
closing-only + Shin, no single-season	raw	0.99702	0.99767	0.99509
closing-only + Shin, no single-season	cal.	0.99692	0.99754	0.99509
PSC alone	raw	0.99714	0.99776	0.99526
PSC alone	cal.	0.99685	0.99746	0.99502

Table A.9: Each configuration against PSC alone and against the full panel with proportional normalisation, paired over the same 71,818 matches. **Mean diff.** is the configuration's log-loss minus that of the series in **Against**, so a negative value means the configuration is better.

Configuration	Stage	Against	Mean diff.	CI low	CI high	p
full panel, proportional	raw	PSC alone	+0.00065	0.00044	0.00086	<0.001
full panel, proportional	cal.	PSC alone	+0.00043	0.00020	0.00064	<0.001
closing-only, proportional	raw	PSC alone	+0.00019	0.00009	0.00029	<0.001
closing-only, proportional	raw	full panel, proportional	−0.00046	−0.00064	−0.00029	<0.001
closing-only, proportional	cal.	PSC alone	+0.00009	−0.00003	0.00020	0.143
closing-only, proportional	cal.	full panel, proportional	−0.00034	−0.00053	−0.00016	<0.001
full panel, Shin	raw	PSC alone	+0.00031	0.00008	0.00054	0.011
full panel, Shin	raw	full panel, proportional	−0.00034	−0.00046	−0.00022	<0.001
full panel, Shin	cal.	PSC alone	+0.00046	0.00021	0.00069	<0.001
full panel, Shin	cal.	full panel, proportional	+0.00004	−0.00002	0.00009	0.181
closing-only + Shin	raw	PSC alone	−0.00012	−0.00025	0.00000	0.057
closing-only + Shin	raw	full panel, proportional	−0.00077	−0.00100	−0.00056	<0.001
closing-only + Shin	cal.	PSC alone	+0.00007	−0.00004	0.00018	0.234
closing-only + Shin	cal.	full panel, proportional	−0.00036	−0.00056	−0.00016	<0.001
closing-only + Shin, no single-season	raw	PSC alone	−0.00011	−0.00024	0.00001	0.072
closing-only + Shin, no single-season	raw	full panel, proportional	−0.00077	−0.00099	−0.00055	<0.001
closing-only + Shin, no single-season	cal.	PSC alone	+0.00008	−0.00003	0.00018	0.181
closing-only + Shin, no single-season	cal.	full panel, proportional	−0.00035	−0.00055	−0.00015	<0.001

A.4 Without Pinnacle

Tables A.10 and A.11 give in full the three panels of §5.2.5, which Table 5.7 summarises.

Table A.10: Every series of each panel refitted without the two Pinnacle columns, ranked by calibrated log-loss on the matches common to the whole panel (**#** is the rank within the panel, and the number of experts left is in parentheses). Log-loss is given raw and calibrated over all common matches, and calibrated over their chronological 75% prefix (**train**) and the remaining 25% (**tail**).

Panel	#	Series	Coverage %	log-loss (raw)	log-loss (cal.)	train	tail
full (24)	1	FTRL	100.0	0.99854	0.99771	1.00112	0.98747
full (24)	2	Hedge	100.0	0.99854	0.99771	1.00112	0.98747
full (24)	3	OGD	100.0	0.99858	0.99782	1.00131	0.98735
full (24)	4	Uniform average	100.0	0.99927	0.99837	1.00162	0.98862
full (24)	5	VC_BV	89.3	1.00039	0.99987	1.00278	0.99113
full (24)	6	WH	85.5	1.00116	1.00030	1.00310	0.99191
full (24)	7	BW	95.0	1.00113	1.00038	1.00321	0.99187
full (24)	8	B365	99.8	1.00081	1.00038	1.00326	0.99173
full (24)	9	IW	74.7	1.00173	1.00050	1.00352	0.99146
closing (12)	1	OGD	69.6	0.99997	0.99919	1.00405	0.98459
closing (12)	2	Uniform average	69.6	1.00000	0.99921	1.00407	0.98462
closing (12)	3	FTRL	69.6	1.00001	0.99921	1.00407	0.98463
closing (12)	4	Hedge	69.6	1.00001	0.99921	1.00407	0.98463
closing (12)	5	VC_BVC	59.3	0.99981	0.99929	1.00430	0.98426
closing (12)	6	B365C	69.6	1.00008	0.99955	1.00443	0.98493
closing (12)	7	IWC	44.4	1.00074	0.99957	1.00423	0.98558
closing (12)	8	BWC	65.0	1.00047	0.99968	1.00469	0.98463
closing (12)	9	WHC	55.3	1.00085	1.00017	1.00496	0.98579
opening (12)	1	VC_BV	89.3	1.00039	0.99987	1.00278	0.99113
opening (12)	2	Uniform average	100.0	1.00071	0.99990	1.00277	0.99130
opening (12)	3	FTRL	100.0	1.00072	0.99991	1.00277	0.99131
opening (12)	4	Hedge	100.0	1.00072	0.99991	1.00277	0.99131
opening (12)	5	OGD	100.0	1.00069	0.99999	1.00287	0.99136
opening (12)	6	WH	85.5	1.00116	1.00030	1.00310	0.99191
opening (12)	7	BW	95.0	1.00113	1.00038	1.00321	0.99187
opening (12)	8	B365	99.8	1.00081	1.00038	1.00326	0.99173
opening (12)	9	IW	74.7	1.00173	1.00050	1.00352	0.99146

Table A.11: OGD against each bookmaker of each panel without Pinnacle, paired over the panel's common matches, calibrated log-loss. A negative **Mean diff.** means that OGD is better.

Panel	Bookmaker	n	Mean diff.	CI low	CI high	p	Verdict
full (24)	IW	54,877	−0.00315	−0.00366	−0.00262	<0.001	OGD better
full (24)	WH	54,877	−0.00259	−0.00311	−0.00208	<0.001	OGD better
full (24)	BW	54,877	−0.00255	−0.00309	−0.00205	<0.001	OGD better
full (24)	B365	54,877	−0.00223	−0.00273	−0.00176	<0.001	OGD better
full (24)	VC_BV	54,877	−0.00181	−0.00230	−0.00134	<0.001	OGD better
closing (12)	WHC	32,210	−0.00088	−0.00135	−0.00039	<0.001	OGD better
closing (12)	IWC	32,210	−0.00077	−0.00108	−0.00047	<0.001	OGD better
closing (12)	BWC	32,210	−0.00051	−0.00085	−0.00017	0.007	OGD better
closing (12)	B365C	32,210	−0.00011	−0.00040	0.00021	0.505	tie
closing (12)	VC_BVC	32,210	+0.00016	−0.00015	0.00043	0.284	tie
opening (12)	IW	54,877	−0.00105	−0.00130	−0.00076	<0.001	OGD better
opening (12)	WH	54,877	−0.00048	−0.00074	−0.00019	0.002	OGD better
opening (12)	BW	54,877	−0.00045	−0.00070	−0.00020	<0.001	OGD better
opening (12)	B365	54,877	−0.00012	−0.00033	0.00007	0.203	tie
opening (12)	VC_BV	54,877	+0.00030	0.00009	0.00050	0.004	VC_BV better

A.5 Recent window and deployment test

Tables A.12 and A.13 complete §5.3.1. Tables A.14 and A.15 add a comparison not shown in the main text: the market a bettor can actually use today, that is the series still quoting

in the last quarter of the full history. There OGD significantly beats every opening price and ties with every closing one. Tables A.16 and A.17 complete §5.3.2.

Table A.12: Ranking over the last five seasons (38,749 matches), full panel, ordered by calibrated log-loss: Table 5.8 in full. **Coverage** is computed inside the window; log-loss and Brier are given raw and calibrated, lower better, each over the matches the series quoted.

Series	Phase	Coverage %	log-loss (raw)	log-loss (cal.)	Brier (raw)	Brier (cal.)
PSC	closing	87.9	0.99471	0.99439	0.59424	0.59410
OGD	—	100.0	0.99556	0.99495	0.59482	0.59452
VC_BVC	closing	79.7	0.99554	0.99498	0.59483	0.59456
WHC	closing	72.1	0.99575	0.99502	0.59490	0.59450
BWC	closing	91.9	0.99609	0.99504	0.59513	0.59460
B365C	closing	100.0	0.99602	0.99543	0.59500	0.59471
Uniform average	—	100.0	0.99677	0.99591	0.59559	0.59516
Hedge	—	100.0	0.99689	0.99604	0.59567	0.59524
FTRL	—	100.0	0.99692	0.99607	0.59569	0.59526
WH	opening	71.9	0.99848	0.99771	0.59677	0.59634
PS	opening	87.7	0.99831	0.99794	0.59665	0.59649
VC_BV	opening	79.1	0.99870	0.99801	0.59696	0.59660
BW	opening	91.5	0.99981	0.99885	0.59765	0.59716
B365	opening	99.8	0.99951	0.99896	0.59736	0.59709

Table A.13: The paired tests of §5.3.1, over the last five seasons.

Comparison	n	Mean diff.	CI low	CI high	p
OGD vs PSC (raw)	34,065	+0.00065	0.00037	0.00097	<0.001
OGD vs PSC (calibrated)	34,065	+0.00036	0.00008	0.00067	0.014
OGD vs B365C (raw)	38,748	−0.00047	−0.00086	−0.00012	0.014
OGD vs Hedge (raw)	38,749	−0.00133	−0.00166	−0.00102	<0.001
OGD vs FTRL (raw)	38,749	−0.00136	−0.00169	−0.00104	<0.001
OGD vs Uniform average (raw)	38,749	−0.00121	−0.00151	−0.00092	<0.001
Hedge vs Uniform average (raw)	38,749	+0.00012	0.00004	0.00019	0.006
FTRL vs Uniform average (raw)	38,749	+0.00015	0.00006	0.00023	0.002
Hedge: raw vs calibrated	38,749	+0.00085	0.00036	0.00133	0.001
OGD: raw vs calibrated	38,749	+0.00061	0.00021	0.00100	0.001
FTRL: raw vs calibrated	38,749	+0.00086	0.00037	0.00134	0.001
Uniform average: raw vs calibrated	38,749	+0.00086	0.00036	0.00135	0.001

Table A.14: The market of today: the series that clear the coverage threshold in the last quarter of the full history (19,146 matches), ranked by calibrated log-loss on the 7,699 of those matches that all of them quoted. The mixtures are fitted on the full history in one causal run, and **Coverage** is computed on the last quarter alone.

#	Series	Phase	Coverage %	log-loss
1	OGD	—	100.0	1.00013
2	BWC	closing	85.1	1.00022
3	BFEC	closing	77.7	1.00026
4	PSC	closing	75.5	1.00028
5	B365C	closing	100.0	1.00070
6	Uniform average	—	100.0	1.00076

(continued)

#	Series	Phase	Coverage %	log-loss
7	Hedge	—	100.0	1.00086
8	FTRL	—	100.0	1.00089
9	PS	opening	75.3	1.00258
10	B365	opening	99.8	1.00296
11	BW	opening	84.8	1.00319
12	BFE	opening	77.1	1.00809

Table A.15: OGD against each bookmaker of Table A.14, paired over the same 7,699 matches; a negative **Mean diff.** means that OGD is better.

Bookmaker	Phase	n	Mean diff.	CI low	CI high	p	Verdict
BFE	opening	7,699	−0.00796	−0.01093	−0.00527	<0.001	OGD better
BW	opening	7,699	−0.00306	−0.00479	−0.00141	0.001	OGD better
B365	opening	7,699	−0.00283	−0.00454	−0.00118	0.001	OGD better
PS	opening	7,699	−0.00244	−0.00406	−0.00094	0.002	OGD better
B365C	closing	7,699	−0.00057	−0.00146	0.00031	0.205	tie
PSC	closing	7,699	−0.00015	−0.00056	0.00029	0.503	tie
BFEC	closing	7,699	−0.00013	−0.00092	0.00074	0.797	tie
BWC	closing	7,699	−0.00009	−0.00082	0.00064	0.815	tie

Table A.16: Deployment test: trained up to 31/12/2024 and scored by calibrated log-loss on the 8,151 matches of 2025–26 that every listed series quoted; Table 5.9 in full. **Coverage** is the share of the test window's matches a series quoted, and *online* and *frozen* are the two readings of the mixture defined in §5.3.2.

#	Series	Type	Coverage %	log-loss
1	BFEC	closing	96.2	1.00040
2	OGD closing-only, online	mixture	100.0	1.00110
3	OGD full panel, online	mixture	100.0	1.00129
4	BWC	closing	75.5	1.00146
5	OGD closing-only, frozen	mixture	100.0	1.00151
6	B365C	closing	100.0	1.00165
7	OGD full panel, frozen	mixture	100.0	1.00183
8	B365	opening	99.8	1.00371
9	BW	opening	75.1	1.00416
10	BFE	opening	95.3	1.00758

Table A.17: Each variant of the mixture against each bookmaker still quoting in 2025–26, paired, calibrated log-loss. A negative **Mean diff.** means that the mixture is better. Pinnacle (PS, PSC), whose coverage falls below the threshold in this window, is compared over its own rounds, hence its smaller n.

Variant	Bookmaker	n	Mean diff.	CI low	CI high	p	Verdict
OGD full panel, online	B365	8,151	−0.00241	−0.00382	−0.00102	0.001	mixture better
OGD full panel, online	B365C	8,151	−0.00036	−0.00121	0.00054	0.390	tie
OGD full panel, online	BFE	8,151	−0.00629	−0.00878	−0.00386	<0.001	mixture better
OGD full panel, online	BFEC	8,151	+0.00089	0.00016	0.00166	0.013	BFEC better
OGD full panel, online	BW	8,151	−0.00287	−0.00431	−0.00141	<0.001	mixture better
OGD full panel, online	BWC	8,151	−0.00017	−0.00087	0.00058	0.685	tie
OGD full panel, online	PS	6,857	−0.00228	−0.00408	−0.00058	0.007	mixture better
OGD full panel, online	PSC	6,892	−0.00013	−0.00058	0.00030	0.568	tie
OGD full panel, frozen	B365	8,151	−0.00188	−0.00313	−0.00066	0.005	mixture better
OGD full panel, frozen	B365C	8,151	+0.00018	−0.00080	0.00122	0.751	tie
OGD full panel, frozen	BFE	8,151	−0.00575	−0.00821	−0.00337	<0.001	mixture better
OGD full panel, frozen	BFEC	8,151	+0.00143	0.00046	0.00238	0.002	BFEC better
OGD full panel, frozen	BW	8,151	−0.00233	−0.00358	−0.00112	<0.001	mixture better
OGD full panel, frozen	BWC	8,151	+0.00037	−0.00043	0.00123	0.342	tie
OGD full panel, frozen	PS	6,857	−0.00219	−0.00407	−0.00042	0.015	mixture better
OGD full panel, frozen	PSC	6,892	−0.00004	−0.00038	0.00030	0.796	tie
OGD closing-only, online	B365	8,151	−0.00260	−0.00417	−0.00106	0.002	mixture better
OGD closing-only, online	B365C	8,151	−0.00055	−0.00137	0.00030	0.193	tie
OGD closing-only, online	BFE	8,151	−0.00647	−0.00904	−0.00389	<0.001	mixture better
OGD closing-only, online	BFEC	8,151	+0.00070	−0.00000	0.00138	0.052	tie
OGD closing-only, online	BW	8,151	−0.00306	−0.00466	−0.00140	0.001	mixture better
OGD closing-only, online	BWC	8,151	−0.00036	−0.00107	0.00042	0.361	tie
OGD closing-only, online	PS	6,857	−0.00223	−0.00418	−0.00036	0.019	mixture better
OGD closing-only, online	PSC	6,892	−0.00009	−0.00038	0.00021	0.532	tie
OGD closing-only, frozen	B365	8,151	−0.00220	−0.00374	−0.00068	0.007	mixture better
OGD closing-only, frozen	B365C	8,151	−0.00014	−0.00097	0.00076	0.714	tie
OGD closing-only, frozen	BFE	8,151	−0.00607	−0.00860	−0.00353	<0.001	mixture better
OGD closing-only, frozen	BFEC	8,151	+0.00111	0.00028	0.00192	0.003	BFEC better
OGD closing-only, frozen	BW	8,151	−0.00265	−0.00419	−0.00110	<0.001	mixture better
OGD closing-only, frozen	BWC	8,151	+0.00005	−0.00061	0.00076	0.850	tie
OGD closing-only, frozen	PS	6,857	−0.00223	−0.00422	−0.00035	0.019	mixture better
OGD closing-only, frozen	PSC	6,892	−0.00009	−0.00031	0.00013	0.444	tie

A.6 Value betting

Table A.18 gives the edge map of Table 5.11 with raw as well as calibrated probabilities, Table A.19 the full sweep of de-vig method, training panel and target behind the phase matching of §5.4, and Table A.20 the same exercise over the last five seasons. All three stake the fixed $\frac{1}{4}$ of the Kelly fraction. **Bets** is the number of bets placed, **Bankroll** the final bankroll starting from 1, and **log-growth** the mean logarithmic growth of capital per bet, with its 95% confidence interval and the p-value of the moving block bootstrap test that it differs from zero; here, unlike in the loss tables, higher is better.

Table A.18: Value betting against each of the eleven targets, with raw and with calibrated probabilities; the mixture is trained on the panel of the target's own phase, with the target itself removed.

Bookmaker	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
B365	raw	15,836	0.24	−0.00009	−0.00017	−0.00001	0.041
B365	cal.	10,829	0.41	−0.00008	−0.00017	0.00001	0.101
BW	raw	4,611	0.68	−0.00008	−0.00021	0.00006	0.248

(continued)

Bookmaker	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
BW	cal.	7,841	1.09	+0.00001	−0.00010	0.00011	0.820
PSC	raw	42,881	0.0000	−0.00028	−0.00038	−0.00019	<0.001
PSC	cal.	39,997	0.0002	−0.00021	−0.00033	−0.00009	<0.001
PS	raw	29,835	0.0246	−0.00012	−0.00018	−0.00007	<0.001
PS	cal.	26,675	0.26	−0.00005	−0.00010	0.00000	0.055
VC_BV	raw	9,508	0.24	−0.00015	−0.00026	−0.00005	0.007
VC_BV	cal.	8,083	0.60	−0.00006	−0.00018	0.00005	0.245
WH	raw	5,793	0.68	−0.00007	−0.00024	0.00010	0.425
WH	cal.	5,931	1.38	+0.00005	−0.00012	0.00024	0.545
IW	raw	3,101	3.70	+0.00042	0.00015	0.00067	0.004
IW	cal.	7,443	7.89	+0.00028	0.00006	0.00048	0.008
B365C	raw	9,570	0.36	−0.00011	−0.00023	0.00002	0.093
B365C	cal.	7,970	0.58	−0.00007	−0.00020	0.00008	0.328
BWC	raw	3,694	1.34	+0.00008	−0.00014	0.00032	0.520
BWC	cal.	6,607	1.81	+0.00009	−0.00006	0.00024	0.247
VC_BVC	raw	6,521	0.46	−0.00012	−0.00025	0.00003	0.106
VC_BVC	cal.	4,653	0.75	−0.00006	−0.00025	0.00012	0.455
WHC	raw	6,852	19.79	+0.00044	0.00016	0.00068	<0.001
WHC	cal.	6,565	38.52	+0.00056	0.00027	0.00083	<0.001

Table A.19: Value betting against each target, for every combination of de-vig method, training panel (**Train panel**: opening-only, closing-only or full) and stage. The rows that bet on an opening price with the full panel use closing prices that do not yet exist when the bet is placed, and are shown only for completeness.

De-vig	Train panel	Bookmaker	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
basic	closing	PSC	raw	20,580	0.0812	−0.00012	−0.00019	−0.00005	0.001
basic	closing	PSC	cal.	19,497	0.54	−0.00003	−0.00010	0.00004	0.368
basic	closing	B365C	raw	7,941	1.02	+0.00000	−0.00012	0.00012	0.991
basic	closing	B365C	cal.	7,643	1.05	+0.00001	−0.00013	0.00014	0.968
basic	closing	BWC	raw	3,134	2.02	+0.00022	−0.00004	0.00049	0.091
basic	closing	BWC	cal.	6,920	3.09	+0.00016	0.00001	0.00033	0.042
basic	closing	VC_BVC	raw	5,142	1.01	+0.00000	−0.00015	0.00018	0.936
basic	closing	VC_BVC	cal.	4,561	1.58	+0.00010	−0.00010	0.00030	0.339
basic	closing	WHC	raw	6,633	38.45	+0.00055	0.00026	0.00082	<0.001
basic	closing	WHC	cal.	7,253	84.24	+0.00061	0.00028	0.00088	<0.001
basic	full	B365	raw	34,022	725,029,747	+0.00060	0.00046	0.00072	<0.001
basic	full	B365	cal.	35,894	147,950,203,510	+0.00072	0.00057	0.00086	<0.001
basic	full	BW	raw	24,313	65,881,997	+0.00074	0.00060	0.00090	<0.001
basic	full	BW	cal.	29,845	40,182,375,505	+0.00082	0.00066	0.00098	<0.001
basic	full	PSC	raw	42,881	0.0000	−0.00028	−0.00038	−0.00019	<0.001
basic	full	PSC	cal.	39,997	0.0002	−0.00021	−0.00033	−0.00009	<0.001
basic	full	PS	raw	45,926	43,477,338,975	+0.00053	0.00042	0.00064	<0.001
basic	full	PS	cal.	48,919	282,147,608,799,934	+0.00068	0.00056	0.00080	<0.001
basic	full	VC_BV	raw	25,631	5,855,729	+0.00061	0.00046	0.00075	<0.001
basic	full	VC_BV	cal.	29,112	1,175,733,570	+0.00072	0.00057	0.00087	<0.001
basic	full	WH	raw	22,036	6,476,491	+0.00071	0.00055	0.00088	<0.001
basic	full	WH	cal.	27,493	1,397,287,497	+0.00077	0.00061	0.00094	<0.001
basic	full	IW	raw	16,562	2,296,631	+0.00088	0.00070	0.00108	<0.001
basic	full	IW	cal.	21,808	1,277,831,748	+0.00096	0.00076	0.00117	<0.001
basic	full	B365C	raw	9,570	0.36	−0.00011	−0.00023	0.00002	0.093
basic	full	B365C	cal.	7,970	0.58	−0.00007	−0.00020	0.00008	0.328
basic	full	BWC	raw	3,694	1.34	+0.00008	−0.00014	0.00032	0.520
basic	full	BWC	cal.	6,607	1.81	+0.00009	−0.00006	0.00024	0.247
basic	full	VC_BVC	raw	6,521	0.46	−0.00012	−0.00025	0.00003	0.106
basic	full	VC_BVC	cal.	4,653	0.75	−0.00006	−0.00025	0.00012	0.455
basic	full	WHC	raw	6,852	19.79	+0.00044	0.00016	0.00068	<0.001
basic	full	WHC	cal.	6,565	38.52	+0.00056	0.00027	0.00083	<0.001

(continued)

De-vig	Train panel	Bookmaker	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
basic	opening	B365	raw	15,836	0.24	−0.00009	−0.00017	−0.00001	0.041
basic	opening	B365	cal.	10,829	0.41	−0.00008	−0.00017	0.00001	0.101
basic	opening	BW	raw	4,611	0.68	−0.00008	−0.00021	0.00006	0.248
basic	opening	BW	cal.	7,841	1.09	+0.00001	−0.00010	0.00011	0.820
basic	opening	PSC	raw	57,078	0.0000	−0.00049	−0.00061	−0.00036	<0.001
basic	opening	PSC	cal.	56,645	0.0000	−0.00044	−0.00057	−0.00031	<0.001
basic	opening	PS	raw	29,835	0.0246	−0.00012	−0.00018	−0.00007	<0.001
basic	opening	PS	cal.	26,675	0.26	−0.00005	−0.00010	0.00000	0.055
basic	opening	VC_BV	raw	9,508	0.24	−0.00015	−0.00026	−0.00005	0.007
basic	opening	VC_BV	cal.	8,083	0.60	−0.00006	−0.00018	0.00005	0.245
basic	opening	WH	raw	5,793	0.68	−0.00007	−0.00024	0.00010	0.425
basic	opening	WH	cal.	5,931	1.38	+0.00005	−0.00012	0.00024	0.545
basic	opening	IW	raw	3,101	3.70	+0.00042	0.00015	0.00067	0.004
basic	opening	IW	cal.	7,443	7.89	+0.00028	0.00006	0.00048	0.008
basic	opening	B365C	raw	27,138	0.0000	−0.00066	−0.00084	−0.00049	<0.001
basic	opening	B365C	cal.	27,203	0.0000	−0.00061	−0.00079	−0.00042	<0.001
basic	opening	BWC	raw	19,475	0.0000	−0.00075	−0.00094	−0.00058	<0.001
basic	opening	BWC	cal.	20,931	0.0000	−0.00071	−0.00092	−0.00051	<0.001
basic	opening	VC_BVC	raw	22,741	0.0000	−0.00066	−0.00085	−0.00046	<0.001
basic	opening	VC_BVC	cal.	22,708	0.0000	−0.00061	−0.00082	−0.00041	<0.001
basic	opening	WHC	raw	16,306	0.0000	−0.00067	−0.00087	−0.00045	<0.001
basic	opening	WHC	cal.	16,588	0.0000	−0.00065	−0.00086	−0.00042	<0.001
Shin	closing	PSC	raw	13,136	1.03	+0.00000	−0.00008	0.00008	0.947
Shin	closing	PSC	cal.	18,398	0.98	−0.00000	−0.00008	0.00007	0.974
Shin	closing	B365C	raw	5,198	1.40	+0.00007	−0.00008	0.00022	0.420
Shin	closing	B365C	cal.	7,374	1.25	+0.00003	−0.00010	0.00016	0.687
Shin	closing	BWC	raw	2,535	2.69	+0.00039	0.00008	0.00074	0.014
Shin	closing	BWC	cal.	6,511	3.32	+0.00018	0.00002	0.00036	0.029
Shin	closing	VC_BVC	raw	2,900	1.61	+0.00016	−0.00009	0.00042	0.204
Shin	closing	VC_BVC	cal.	4,097	1.72	+0.00013	−0.00008	0.00036	0.245
Shin	closing	WHC	raw	4,980	60.98	+0.00083	0.00042	0.00117	0.001
Shin	closing	WHC	cal.	6,675	75.41	+0.00065	0.00032	0.00095	<0.001
Shin	full	B365	raw	31,621	1,887,794,279	+0.00068	0.00055	0.00081	<0.001
Shin	full	B365	cal.	34,959	35,721,495,032	+0.00070	0.00056	0.00084	<0.001
Shin	full	BW	raw	23,893	435,973,320	+0.00083	0.00068	0.00099	<0.001
Shin	full	BW	cal.	28,919	24,022,128,668	+0.00083	0.00067	0.00099	<0.001
Shin	full	PSC	raw	35,750	0.0002	−0.00024	−0.00035	−0.00013	<0.001
Shin	full	PSC	cal.	39,367	0.0003	−0.00021	−0.00032	−0.00009	<0.001
Shin	full	PS	raw	44,003	675,151,510,774	+0.00062	0.00051	0.00073	<0.001
Shin	full	PS	cal.	48,190	71,263,276,547,990	+0.00066	0.00054	0.00078	<0.001
Shin	full	VC_BV	raw	23,908	35,276,900	+0.00073	0.00057	0.00088	<0.001
Shin	full	VC_BV	cal.	28,218	578,810,361	+0.00072	0.00056	0.00086	<0.001
Shin	full	WH	raw	21,430	29,379,522	+0.00080	0.00063	0.00098	<0.001
Shin	full	WH	cal.	26,564	483,169,194	+0.00075	0.00059	0.00093	<0.001
Shin	full	IW	raw	17,465	13,207,065	+0.00094	0.00074	0.00112	<0.001
Shin	full	IW	cal.	21,369	745,207,659	+0.00096	0.00076	0.00114	<0.001
Shin	full	B365C	raw	6,413	0.58	−0.00008	−0.00023	0.00006	0.253
Shin	full	B365C	cal.	7,786	0.66	−0.00005	−0.00019	0.00009	0.467
Shin	full	BWC	raw	2,740	1.56	+0.00016	−0.00012	0.00046	0.281
Shin	full	BWC	cal.	6,132	2.26	+0.00013	−0.00003	0.00031	0.120
Shin	full	VC_BVC	raw	3,871	0.69	−0.00010	−0.00029	0.00010	0.300
Shin	full	VC_BVC	cal.	4,216	0.86	−0.00003	−0.00023	0.00016	0.668
Shin	full	WHC	raw	4,774	32.03	+0.00073	0.00037	0.00105	0.001
Shin	full	WHC	cal.	5,914	32.86	+0.00059	0.00028	0.00089	<0.001
Shin	opening	B365	raw	8,796	0.50	−0.00008	−0.00017	0.00003	0.188
Shin	opening	B365	cal.	10,578	0.33	−0.00010	−0.00019	−0.00000	0.049
Shin	opening	BW	raw	2,315	1.16	+0.00006	−0.00011	0.00029	0.400
Shin	opening	BW	cal.	7,441	1.33	+0.00004	−0.00007	0.00015	0.442
Shin	opening	PSC	raw	55,765	0.0000	−0.00046	−0.00058	−0.00033	<0.001
Shin	opening	PSC	cal.	56,441	0.0000	−0.00044	−0.00057	−0.00030	<0.001
Shin	opening	PS	raw	18,357	0.35	−0.00006	−0.00011	−0.00000	0.038
Shin	opening	PS	cal.	24,953	0.25	−0.00006	−0.00011	0.00000	0.054
Shin	opening	VC_BV	raw	4,268	0.93	−0.00002	−0.00015	0.00012	0.722
Shin	opening	VC_BV	cal.	7,333	0.64	−0.00006	−0.00018	0.00005	0.265
Shin	opening	WH	raw	2,996	1.85	+0.00020	−0.00007	0.00051	0.155
Shin	opening	WH	cal.	6,418	1.16	+0.00002	−0.00014	0.00021	0.737
Shin	opening	IW	raw	3,579	5.60	+0.00048	0.00020	0.00074	<0.001
Shin	opening	IW	cal.	7,179	9.84	+0.00032	0.00011	0.00052	0.001
Shin	opening	B365C	raw	26,159	0.0000	−0.00063	−0.00081	−0.00045	<0.001

(continued)

De-vig	Train panel	Bookmaker	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
Shin	opening	B365C	cal.	27,097	0.0000	−0.00061	−0.00081	−0.00042	<0.001
Shin	opening	BWC	raw	19,224	0.0000	−0.00074	−0.00093	−0.00055	<0.001
Shin	opening	BWC	cal.	20,970	0.0000	−0.00071	−0.00091	−0.00051	<0.001
Shin	opening	VC_BVC	raw	21,857	0.0000	−0.00064	−0.00084	−0.00043	<0.001
Shin	opening	VC_BVC	cal.	22,573	0.0000	−0.00061	−0.00081	−0.00041	<0.001
Shin	opening	WHC	raw	15,400	0.0000	−0.00066	−0.00088	−0.00044	<0.001
Shin	opening	WHC	cal.	16,333	0.0000	−0.00064	−0.00086	−0.00041	<0.001

Table A.20: Value betting over the last five seasons, with the mixture trained either on the full panel or on the panel matched to the target's phase (**Train panel**).

Bookmaker	Train panel	Stage	Bets	Bankroll	log-growth	CI low	CI high	p
B365	matched	raw	5,183	0.68	−0.00007	−0.00023	0.00005	0.216
B365	matched	cal.	4,493	0.62	−0.00011	−0.00028	0.00005	0.178
B365C	matched	raw	4,621	0.87	−0.00003	−0.00018	0.00015	0.733
B365C	matched	cal.	5,537	0.78	−0.00005	−0.00021	0.00014	0.650
B365C	full	raw	6,305	0.22	−0.00024	−0.00040	−0.00006	0.015
B365C	full	cal.	7,362	0.33	−0.00015	−0.00032	0.00002	0.088
BW	matched	raw	1,108	0.85	−0.00015	−0.00041	0.00020	0.440
BW	matched	cal.	4,546	1.20	+0.00004	−0.00011	0.00019	0.645
BWC	matched	raw	1,377	2.05	+0.00052	0.00006	0.00099	0.019
BWC	matched	cal.	5,751	2.65	+0.00017	−0.00000	0.00036	0.051
BWC	full	raw	2,307	1.02	+0.00001	−0.00030	0.00037	0.931
BWC	full	cal.	6,674	1.23	+0.00003	−0.00013	0.00020	0.698
IW	matched	raw	562	1.09	+0.00015	−0.00028	0.00071	0.522
IW	matched	cal.	4,645	1.50	+0.00009	−0.00010	0.00027	0.335
IWC	matched	raw	1,332	2.67	+0.00074	0.00025	0.00122	0.002
IWC	matched	cal.	5,092	8.51	+0.00042	0.00016	0.00068	<0.001
IWC	full	raw	1,981	1.38	+0.00016	−0.00020	0.00056	0.421
IWC	full	cal.	5,947	5.78	+0.00029	0.00006	0.00054	0.010
PS	matched	raw	13,103	0.0955	−0.00018	−0.00026	−0.00009	<0.001
PS	matched	cal.	12,643	0.21	−0.00012	−0.00022	−0.00003	0.011
PSC	matched	raw	14,512	0.15	−0.00013	−0.00022	−0.00004	0.004
PSC	matched	cal.	15,306	0.61	−0.00003	−0.00012	0.00006	0.499
PSC	full	raw	17,953	0.0499	−0.00017	−0.00026	−0.00007	<0.001
PSC	full	cal.	18,526	0.21	−0.00008	−0.00019	0.00002	0.134
VC_BV	matched	raw	2,408	0.47	−0.00032	−0.00049	−0.00014	<0.001
VC_BV	matched	cal.	1,503	0.84	−0.00012	−0.00041	0.00017	0.418
VC_BVC	matched	raw	3,094	1.05	+0.00002	−0.00019	0.00023	0.901
VC_BVC	matched	cal.	3,283	1.08	+0.00002	−0.00022	0.00026	0.905
VC_BVC	full	raw	4,523	0.23	−0.00033	−0.00052	−0.00012	0.001
VC_BVC	full	cal.	4,722	0.35	−0.00023	−0.00042	−0.00003	0.020
WH	matched	raw	2,023	1.41	+0.00017	−0.00020	0.00056	0.359
WH	matched	cal.	2,817	1.65	+0.00018	−0.00012	0.00052	0.256
WHC	matched	raw	4,327	17.07	+0.00066	0.00026	0.00101	0.001
WHC	matched	cal.	5,517	33.68	+0.00064	0.00029	0.00096	<0.001
WHC	full	raw	4,352	4.77	+0.00036	0.00004	0.00064	0.029
WHC	full	cal.	5,471	8.56	+0.00039	0.00008	0.00070	0.018

A.7 Extensions and negative results

The remaining tables document the results of §5.6 and two analyses that the main text mentions only in passing. Fixed-Share is in Tables A.21 and A.22, the per-league specialists in Table A.23 and the performance by outcome class in Table A.24. Tables A.25 and A.26 test two variants of the calibration stage: restarted every season, it still beats the raw probabilities for 11 of the 13 series but never beats the single global calibration, and starting each season from the previous season's temperature does not change that. Table A.27 gives

the trend tests of §6.2, Table A.28 the full step sweep of §5.1, Table A.29 the rankings over the two time windows and Table A.30 the paired tests of §5.2.3.

Table A.21: Fixed-Share (§3.3.6) by leak rate α, with the Hedge step in force: raw scores over all 76,584 rounds, lower log-loss and Brier better. $\alpha = 0$ is plain Hedge.

α	log-loss	Brier	Acc.
0	0.99842	0.59674	0.50649
0.005	0.99885	0.59702	0.50597
0.01	0.99885	0.59702	0.50595
0.02	0.99885	0.59703	0.50595
0.05	0.99886	0.59703	0.50594

Table A.22: Hedge weight of the eleven series that quote only in the last seasons, read at each one's own last awake round, by α. Without a leak ($\alpha = 0$) each holds 0.11 to 0.17, more than an equal share among the experts awake with it; any $\alpha > 0$ brings it back to almost exactly that equal share.

Bookmaker	$\alpha = 0$	$\alpha = 0.005$	$\alpha = 0.01$	$\alpha = 0.02$	$\alpha = 0.05$
1XB	0.169	0.100	0.100	0.100	0.100
1XBC	0.166	0.100	0.100	0.100	0.100
BF	0.166	0.100	0.100	0.100	0.100
BFC	0.166	0.100	0.100	0.100	0.100
BFD	0.114	0.062	0.062	0.062	0.062
BFDC	0.115	0.062	0.062	0.062	0.062
BMGM	0.118	0.062	0.062	0.062	0.062
BMGMC	0.116	0.062	0.062	0.062	0.062
CL	0.149	0.063	0.063	0.063	0.063
CLC	0.152	0.063	0.063	0.063	0.063
LBC	0.146	0.062	0.062	0.062	0.062

Table A.23: Per-league specialists: OGD trained on one league alone (**Specialist**) against the single OGD trained on all of them (**Unified**), both scored by log-loss on that league's matches, lower better. **Spec. wins** says whether the specialist is better, and **Most trusted** lists the three series to which the specialist gave the largest weight.

League	Matches	Unified	Specialist	Spec. wins	Most trusted
Scotland Premiership	2,231	0.93590	0.93635	no	CLC, BFEC, BFD
Italy Serie A	3,800	0.94636	0.94692	no	CL, BF, LBC
England Premier League	3,800	0.94928	0.94978	no	BFEC, WHC, CLC
Spain La Liga	3,800	0.96476	0.96534	no	LBC, BFC, PSC
Germany Bundesliga	3,060	0.97734	0.97823	no	BMGMC, LBC, 1XBC
France Ligue 1	3,477	0.98084	0.98132	no	LBC, 1XBC, BFC

Table A.24: Performance by outcome class: for each series and each actual outcome, n is the number of matches with that outcome, **Acc.** the share of them in which the series' most probable outcome was the right one, and log-loss and Brier are computed on those matches alone.

Series	Outcome	n	Acc.	log-loss	Brier
Hedge	1 (Home win)	33,280	0.82058	0.76764	0.43084
Hedge	X (Draw)	20,080	0.00249	1.31494	0.83571
Hedge	2 (Away win)	23,224	0.49216	1.05547	0.62784
OGD	1 (Home win)	33,280	0.81830	0.76463	0.42884
OGD	X (Draw)	20,080	0.00418	1.31341	0.83548
OGD	2 (Away win)	23,224	0.49440	1.05791	0.62889
FTRL	1 (Home win)	33,280	0.82061	0.76769	0.43088
FTRL	X (Draw)	20,080	0.00249	1.31497	0.83570
FTRL	2 (Away win)	23,224	0.49225	1.05541	0.62781
Uniform average	1 (Home win)	33,280	0.82263	0.76722	0.43061
Uniform average	X (Draw)	20,080	0.00159	1.31593	0.83603
Uniform average	2 (Away win)	23,224	0.48816	1.05666	0.62887
B365	1 (Home win)	33,216	0.82557	0.76485	0.42878
B365	X (Draw)	20,050	0.00165	1.32349	0.84164
B365	2 (Away win)	23,172	0.47695	1.06090	0.63147
B365C	1 (Home win)	22,894	0.81729	0.76566	0.42918
B365C	X (Draw)	13,984	0.00279	1.32752	0.84407
B365C	2 (Away win)	16,389	0.50394	1.04637	0.62166
BW	1 (Home win)	31,644	0.82755	0.76947	0.43235
BW	X (Draw)	19,102	0.00162	1.32083	0.83778
BW	2 (Away win)	22,006	0.47333	1.05764	0.63048
BWC	1 (Home win)	21,384	0.81673	0.77242	0.43422
BWC	X (Draw)	13,106	0.00351	1.31919	0.83683
BWC	2 (Away win)	15,265	0.50292	1.04470	0.62136
IW	1 (Home win)	24,912	0.82671	0.78010	0.43964
IW	X (Draw)	14,973	0.00147	1.30499	0.82684
IW	2 (Away win)	17,300	0.46809	1.05885	0.63125
PS	1 (Home win)	30,943	0.81944	0.76503	0.42925
PS	X (Draw)	18,694	0.00332	1.31817	0.83832
PS	2 (Away win)	21,647	0.48247	1.06384	0.63295
PSC	1 (Home win)	31,186	0.81668	0.76231	0.42734
PSC	X (Draw)	18,837	0.00695	1.31270	0.83592
PSC	2 (Away win)	21,795	0.49314	1.06041	0.62971
VC_BV	1 (Home win)	29,698	0.82480	0.76751	0.43096
VC_BV	X (Draw)	17,933	0.00112	1.31780	0.83746
VC_BV	2 (Away win)	20,766	0.47799	1.06051	0.63120
WH	1 (Home win)	28,469	0.82528	0.77363	0.43519
WH	X (Draw)	17,150	0.00146	1.31082	0.83259
WH	2 (Away win)	19,851	0.47438	1.06002	0.63078

Table A.25: Calibration restarted at the start of every season, against the raw probabilities and against the single global calibration of §3.4.1, paired over each series' rounds; a negative **Mean diff.** means that the per-season variant is better.

Series	Comparison	n	Mean diff.	CI low	CI high	p
Hedge	per-season vs raw	76,584	−0.00056	−0.00081	−0.00030	<0.001
Hedge	per-season vs global	76,584	+0.00013	−0.00003	0.00031	0.108
OGD	per-season vs raw	76,584	−0.00041	−0.00064	−0.00017	0.001
OGD	per-season vs global	76,584	+0.00012	−0.00005	0.00030	0.158
FTRL	per-season vs raw	76,584	−0.00056	−0.00082	−0.00030	<0.001
FTRL	per-season vs global	76,584	+0.00013	−0.00003	0.00031	0.109
Uniform average	per-season vs raw	76,584	−0.00059	−0.00087	−0.00032	<0.001
Uniform average	per-season vs global	76,584	+0.00014	−0.00004	0.00032	0.137
B365	per-season vs raw	76,438	−0.00031	−0.00054	−0.00007	0.007
B365	per-season vs global	76,438	+0.00009	−0.00008	0.00026	0.283
B365C	per-season vs raw	53,267	−0.00033	−0.00059	−0.00007	0.012
B365C	per-season vs global	53,267	+0.00010	−0.00004	0.00025	0.154
BW	per-season vs raw	72,752	−0.00061	−0.00091	−0.00031	<0.001
BW	per-season vs global	72,752	+0.00014	−0.00005	0.00032	0.149
BWC	per-season vs raw	49,755	−0.00065	−0.00106	−0.00022	0.001
BWC	per-season vs global	49,755	+0.00014	−0.00008	0.00036	0.205
IW	per-season vs raw	57,185	−0.00110	−0.00150	−0.00069	<0.001
IW	per-season vs global	57,185	+0.00016	−0.00006	0.00039	0.146
PS	per-season vs raw	71,284	−0.00017	−0.00039	0.00005	0.140
PS	per-season vs global	71,284	+0.00011	−0.00009	0.00030	0.246
PSC	per-season vs raw	71,818	−0.00016	−0.00033	0.00001	0.082
PSC	per-season vs global	71,818	+0.00013	−0.00003	0.00028	0.108
VC_BV	per-season vs raw	68,397	−0.00039	−0.00066	−0.00013	<0.001
VC_BV	per-season vs global	68,397	+0.00009	−0.00009	0.00028	0.305
WH	per-season vs raw	65,470	−0.00061	−0.00095	−0.00026	<0.001
WH	per-season vs global	65,470	+0.00010	−0.00011	0.00033	0.373

Table A.26: Per-season calibration that starts each season from the previous season's final temperature (warm start), against the raw probabilities, the global calibration and the per-season variant restarted from scratch; a negative **Mean diff.** means that the warm start is better.

Series	Comparison	n	Mean diff.	CI low	CI high	p
Hedge	warm-start vs raw	76,584	−0.00066	−0.00093	−0.00037	<0.001
Hedge	warm-start vs global	76,584	+0.00003	−0.00005	0.00012	0.476
Hedge	warm-start vs per-season	76,584	−0.00010	−0.00021	0.00001	0.085
OGD	warm-start vs raw	76,584	−0.00048	−0.00070	−0.00026	<0.001
OGD	warm-start vs global	76,584	+0.00005	−0.00001	0.00011	0.136
OGD	warm-start vs per-season	76,584	−0.00008	−0.00023	0.00007	0.313
FTRL	warm-start vs raw	76,584	−0.00066	−0.00094	−0.00038	<0.001
FTRL	warm-start vs global	76,584	+0.00003	−0.00005	0.00012	0.479
FTRL	warm-start vs per-season	76,584	−0.00010	−0.00021	0.00001	0.086
Uniform average	warm-start vs raw	76,584	−0.00069	−0.00097	−0.00040	<0.001
Uniform average	warm-start vs global	76,584	+0.00004	−0.00005	0.00013	0.418
Uniform average	warm-start vs per-season	76,584	−0.00010	−0.00024	0.00004	0.157
B365	warm-start vs raw	76,438	−0.00036	−0.00061	−0.00011	0.005
B365	warm-start vs global	76,438	+0.00004	−0.00009	0.00016	0.548
B365	warm-start vs per-season	76,438	−0.00005	−0.00015	0.00004	0.270
B365C	warm-start vs raw	53,267	−0.00040	−0.00068	−0.00011	0.005
B365C	warm-start vs global	53,267	+0.00004	−0.00004	0.00012	0.373
B365C	warm-start vs per-season	53,267	−0.00007	−0.00018	0.00005	0.276
BW	warm-start vs raw	72,752	−0.00071	−0.00106	−0.00039	<0.001
BW	warm-start vs global	72,752	+0.00004	−0.00011	0.00018	0.601
BW	warm-start vs per-season	72,752	−0.00010	−0.00020	0.00001	0.064
BWC	warm-start vs raw	49,755	−0.00074	−0.00116	−0.00030	0.001
BWC	warm-start vs global	49,755	+0.00005	−0.00014	0.00025	0.595
BWC	warm-start vs per-season	49,755	−0.00009	−0.00019	0.00002	0.100
IW	warm-start vs raw	57,185	−0.00122	−0.00161	−0.00082	<0.001
IW	warm-start vs global	57,185	+0.00004	−0.00004	0.00012	0.388

(continued)

Series	Comparison	n	Mean diff.	CI low	CI high	p
IW	warm-start vs per-season	57,185	−0.00012	−0.00031	0.00005	0.179
PS	warm-start vs raw	71,284	−0.00022	−0.00044	0.00000	0.056
PS	warm-start vs global	71,284	+0.00006	−0.00009	0.00021	0.389
PS	warm-start vs per-season	71,284	−0.00005	−0.00015	0.00004	0.322
PSC	warm-start vs raw	71,818	−0.00022	−0.00038	−0.00005	0.009
PSC	warm-start vs global	71,818	+0.00007	−0.00001	0.00014	0.074
PSC	warm-start vs per-season	71,818	−0.00006	−0.00016	0.00004	0.254
VC_BV	warm-start vs raw	68,397	−0.00043	−0.00066	−0.00020	<0.001
VC_BV	warm-start vs global	68,397	+0.00006	−0.00001	0.00013	0.107
VC_BV	warm-start vs per-season	68,397	−0.00004	−0.00021	0.00013	0.638
WH	warm-start vs raw	65,470	−0.00067	−0.00100	−0.00035	<0.001
WH	warm-start vs global	65,470	+0.00003	−0.00009	0.00016	0.571
WH	warm-start vs per-season	65,470	−0.00006	−0.00023	0.00009	0.444

Table A.27: Mann–Kendall tests for a monotone trend over the ten seasons, per series and metric. S is the Mann–Kendall statistic, Z its normal approximation and p the two-sided p-value; **Trend** gives the sign of S, which means something only where p is small. ECE is the expected calibration error and *overround* the bookmaker's margin.

Series	Metric	S	Z	p	Trend
Algorithm average	ECE	−9	−0.72	0.474	decreasing
B365	ECE	−3	−0.18	0.858	decreasing
BW	ECE	5	0.36	0.721	increasing
Bookmaker average	ECE	7	0.54	0.592	increasing
FTRL	ECE	−11	−0.89	0.371	decreasing
Hedge	ECE	−11	−0.89	0.371	decreasing
OGD	ECE	−15	−1.25	0.210	decreasing
PS	ECE	3	0.18	0.858	increasing
PSC	ECE	−3	−0.18	0.858	decreasing
Uniform average	ECE	−11	−0.89	0.371	decreasing
Algorithm average	log-loss	1	0.00	1.000	increasing
B365	log-loss	9	0.72	0.474	increasing
BW	log-loss	7	0.54	0.592	increasing
Bookmaker average	log-loss	7	0.54	0.592	increasing
FTRL	log-loss	1	0.00	1.000	increasing
Hedge	log-loss	1	0.00	1.000	increasing
OGD	log-loss	3	0.18	0.858	increasing
PS	log-loss	3	0.18	0.858	increasing
PSC	log-loss	1	0.00	1.000	increasing
Uniform average	log-loss	1	0.00	1.000	increasing
B365	overround	31	2.68	0.007	increasing
BW	overround	23	1.97	0.049	increasing
Bookmaker average	overround	39	3.40	0.001	increasing
PS	overround	31	2.68	0.007	increasing
PSC	overround	33	2.86	0.004	increasing

Table A.28: The full step sweep of §5.1: for each family of step schedules, every setting tried, with the log-loss on the full sample (**full**), on the chronological 75% prefix on which the selection is made (**train**) and on the held-out 25% tail (**tail**). *Multiplier* rows scale the textbook decaying schedule, *constant* η rows replace it with a constant step, and the remaining rows are adaptive (AdaGrad) or loss-bound M variants of OGD, and the settings in force.

Family	Setting	full	train	tail
Hedge, multiplier	0.001	0.99885	0.99886	0.99881
Hedge, multiplier	0.005	0.99879	0.99880	0.99877
Hedge, multiplier	0.01	0.99873	0.99873	0.99873
Hedge, multiplier	0.02	0.99862	0.99860	0.99867
Hedge, multiplier	0.05	0.99845	0.99838	0.99866
Hedge, multiplier	0.1	0.99840	0.99829	0.99873
Hedge, multiplier	0.25	0.99842	0.99828	0.99883
Hedge, multiplier	0.5	0.99844	0.99828	0.99891
Hedge, multiplier	0.75	0.99845	0.99828	0.99894
Hedge, multiplier	1	0.99845	0.99828	0.99896
Hedge, multiplier	1.5	0.99846	0.99829	0.99899
Hedge, multiplier	2	0.99847	0.99829	0.99901
Hedge, multiplier	5	0.99854	0.99833	0.99916
Hedge, multiplier	10	0.99864	0.99841	0.99934
Hedge, multiplier	25	0.99881	0.99852	0.99969
Hedge, multiplier	50	0.99896	0.99855	1.00018
Hedge, multiplier	100	0.99907	0.99859	1.00051
Hedge, multiplier	200	0.99914	0.99863	1.00066
Hedge, multiplier	500	0.99919	0.99867	1.00077
Hedge, multiplier	1000	0.99922	0.99868	1.00085
OGD, multiplier	1	0.99826	0.99833	0.99803
OGD, multiplier	2	0.99822	0.99829	0.99800
OGD, multiplier	5	0.99814	0.99820	0.99794
OGD, multiplier	10	0.99804	0.99808	0.99789
OGD, multiplier	25	0.99784	0.99785	0.99780
OGD, multiplier	50	0.99761	0.99757	0.99770
OGD, multiplier	75	0.99749	0.99744	0.99766
OGD, multiplier	90	0.99747	0.99741	0.99765
OGD, multiplier	100	0.99745	0.99739	0.99765
OGD, multiplier	110	0.99745	0.99738	0.99765
OGD, multiplier	125	0.99745	0.99738	0.99766
OGD, multiplier	150	0.99748	0.99741	0.99769
OGD, multiplier	200	0.99758	0.99751	0.99778
OGD, multiplier	500	0.99822	0.99809	0.99858
OGD, multiplier	1000	0.99852	0.99838	0.99895
Hedge, constant η	10^{-5}	0.99878	0.99879	0.99878
Hedge, constant η	$3 \cdot 10^{-5}$	0.99866	0.99863	0.99875
Hedge, constant η	10^{-4}	0.99847	0.99835	0.99882
Hedge, constant η	$3 \cdot 10^{-4}$	0.99845	0.99829	0.99892
Hedge, constant η	0.001	0.99848	0.99830	0.99901
Hedge, constant η	0.003	0.99856	0.99835	0.99920

(continued)

Family	Setting	full	train	tail
Hedge, constant η	0.01	0.99874	0.99846	0.99957
Hedge, constant η	0.03	0.99898	0.99855	1.00030
Hedge, constant η	0.1	0.99913	0.99862	1.00066
Hedge, constant η	0.3	0.99922	0.99870	1.00078
Hedge, constant η	0.5	0.99927	0.99875	1.00084
Hedge, constant η	1	0.99932	0.99877	1.00097
Hedge, constant η	2	0.99934	0.99876	1.00107
OGD, constant η	10^{-4}	0.99829	0.99837	0.99806
OGD, constant η	$3 \cdot 10^{-4}$	0.99828	0.99836	0.99804
OGD, constant η	0.001	0.99824	0.99833	0.99799
OGD, constant η	0.003	0.99816	0.99825	0.99792
OGD, constant η	0.01	0.99800	0.99806	0.99781
OGD, constant η	0.03	0.99771	0.99772	0.99767
OGD, constant η	0.1	0.99760	0.99748	0.99795
OGD, constant η	0.3	0.99827	0.99808	0.99882
OGD, constant η	1	0.99861	0.99843	0.99916
OGD, constant η	3	0.99871	0.99858	0.99911
OGD, constant η	10	0.99881	0.99867	0.99920
OGD, constant η	30	0.99891	0.99871	0.99951
OGD, constant η	100	0.99904	0.99886	0.99959
FTRL, constant η	10^{-9}	0.99886	0.99888	0.99882
FTRL, constant η	$3 \cdot 10^{-9}$	0.99886	0.99887	0.99882
FTRL, constant η	10^{-8}	0.99886	0.99887	0.99882
FTRL, constant η	$3 \cdot 10^{-8}$	0.99886	0.99887	0.99882
FTRL, constant η	10^{-7}	0.99885	0.99886	0.99881
FTRL, constant η	$3 \cdot 10^{-7}$	0.99883	0.99884	0.99880
FTRL, constant η	10^{-6}	0.99877	0.99877	0.99875
FTRL, constant η	$3 \cdot 10^{-6}$	0.99863	0.99859	0.99876
FTRL, constant η	10^{-5}	0.99841	0.99828	0.99879
FTRL, constant η	$3 \cdot 10^{-5}$	0.99845	0.99828	0.99894
FTRL, constant η	10^{-4}	0.99846	0.99829	0.99896
FTRL, constant η	$3 \cdot 10^{-4}$	0.99849	0.99830	0.99903
FTRL, constant η	0.001	0.99858	0.99835	0.99926
FTRL, constant η	0.003	0.99869	0.99838	0.99963
FTRL, constant η	0.01	0.99906	0.99858	1.00047
FTRL, constant η	0.1	0.99926	0.99875	1.00078
FTRL, constant η	1	0.99934	0.99876	1.00110
FTRL, multiplier	0.001	0.99881	0.99882	0.99879
FTRL, multiplier	0.005	0.99865	0.99862	0.99872
FTRL, multiplier	0.01	0.99852	0.99844	0.99879
FTRL, multiplier	0.02	0.99841	0.99828	0.99878
FTRL, multiplier	0.05	0.99843	0.99828	0.99889
FTRL, multiplier	0.1	0.99845	0.99828	0.99895
FTRL, multiplier	0.25	0.99845	0.99828	0.99896
FTRL, multiplier	0.5	0.99846	0.99829	0.99899
FTRL, multiplier	0.75	0.99848	0.99830	0.99903
FTRL, multiplier	1	0.99850	0.99830	0.99907

(continued)

Family	Setting	full	train	tail
FTRL, multiplier	1.5	0.99853	0.99832	0.99915
FTRL, multiplier	2	0.99854	0.99833	0.99920
FTRL, multiplier	5	0.99867	0.99841	0.99946
FTRL, multiplier	10	0.99889	0.99861	0.99976
FTRL, multiplier	25	0.99913	0.99869	1.00046
FTRL, multiplier	50	0.99920	0.99873	1.00060
FTRL, multiplier	100	0.99923	0.99874	1.00069
FTRL, multiplier	200	0.99926	0.99876	1.00076
FTRL, multiplier	500	0.99930	0.99876	1.00092
FTRL, multiplier	1000	0.99932	0.99875	1.00102
OGD, loss bound M	4.60517	0.99819	0.99826	0.99797
OGD, loss bound M	3.763	0.99817	0.99824	0.99796
OGD, AdaGrad	AdaGrad per-expert	0.99781	0.99761	0.99842
OGD, AdaGrad	AdaGrad scalar	0.99815	0.99821	0.99795
in force	Hedge (in force)	0.99842	0.99828	0.99883
in force	OGD (in force)	0.99745	0.99739	0.99765
in force	FTRL (in force)	0.99843	0.99828	0.99889
in force	Uniform average	0.99886	0.99888	0.99882

Table A.29: Ranking over the full ten seasons and over the last five, for the series that clear the coverage threshold in each window; log-loss raw and calibrated, lower better, each over the matches the series quoted.

Window	Series	Coverage %	Rounds	log-loss (raw)	log-loss (cal.)
5 seasons	PSC	87.9	34,065	0.99471	0.99439
5 seasons	OGD	100.0	38,749	0.99556	0.99495
5 seasons	VC_BVC	79.7	30,882	0.99554	0.99498
5 seasons	WHC	72.1	27,952	0.99575	0.99502
5 seasons	BWC	91.9	35,624	0.99609	0.99504
5 seasons	B365C	100.0	38,748	0.99602	0.99543
5 seasons	Uniform average	100.0	38,749	0.99677	0.99591
5 seasons	Hedge	100.0	38,749	0.99689	0.99604
5 seasons	FTRL	100.0	38,749	0.99692	0.99607
5 seasons	WH	71.9	27,864	0.99848	0.99771
5 seasons	PS	87.7	33,981	0.99831	0.99794
5 seasons	VC_BV	79.1	30,644	0.99870	0.99801
5 seasons	BW	91.5	35,468	0.99981	0.99885
5 seasons	B365	99.8	38,687	0.99951	0.99896
10 seasons	PSC	93.8	71,818	0.99714	0.99685
10 seasons	OGD	100.0	76,584	0.99745	0.99693
10 seasons	Hedge	100.0	76,584	0.99842	0.99773
10 seasons	FTRL	100.0	76,584	0.99843	0.99774
10 seasons	Uniform average	100.0	76,584	0.99886	0.99813
10 seasons	B365C	69.6	53,267	0.99953	0.99910
10 seasons	BWC	65.0	49,755	0.99998	0.99919
10 seasons	VC_BV	89.3	68,397	1.00075	1.00026
10 seasons	WH	85.5	65,470	1.00119	1.00048
10 seasons	PS	93.1	71,284	1.00083	1.00055

(continued)

Window	Series	Coverage %	Rounds	log-loss (raw)	log-loss (cal.)
10 seasons	IW	74.7	57,185	1.00186	1.00061
10 seasons	BW	95.0	72,752	1.00140	1.00065
10 seasons	B365	99.8	76,438	1.00113	1.00073

Table A.30: The paired tests behind Table 5.5 and Figure 5.4: in each quintile of each stratification, OGD fitted on the closing-only panel against OGD fitted on the opening-only panel, raw log-loss. A negative **Mean diff.** means that the closing panel is better.

Stratified by	Quintile	n	Mean diff.	CI low	CI high	p
line movement	Q1	10,650	−0.00044	−0.00084	0.00000	0.052
line movement	Q2	10,649	−0.00072	−0.00140	−0.00000	0.050
line movement	Q3	10,649	−0.00200	−0.00296	−0.00095	<0.001
line movement	Q4	10,649	−0.00452	−0.00596	−0.00301	<0.001
line movement	Q5	10,649	−0.01158	−0.01403	−0.00909	<0.001
Shin z anomaly	Q1	15,315	−0.00295	−0.00397	−0.00187	<0.001
Shin z anomaly	Q2	15,311	−0.00371	−0.00486	−0.00250	<0.001
Shin z anomaly	Q3	15,313	−0.00281	−0.00401	−0.00169	<0.001
Shin z anomaly	Q4	15,303	−0.00440	−0.00562	−0.00314	<0.001
Shin z anomaly	Q5	15,252	−0.00498	−0.00641	−0.00353	<0.001

References

[1] Agarwal, A., Dekel, O., & Xiao, L. (2010). Optimal algorithms for online convex optimization with multi-point bandit feedback. *COLT*, 28–40.

[2] Bitsakis, E. (2006). Το αειθαλές δέντρο της γνώσεως [*The evergreen tree of knowledge*]. Athens: Agra.

[3] Blum, A., & Mansour, Y. (2007). From external to internal regret. *Journal of Machine Learning Research*, 8, 1307–1324.

[4] Bousquet, O., & Warmuth, M. K. (2002). Tracking a small set of experts by mixing past posteriors. *Journal of Machine Learning Research*, 3, 363–396.

[5] Breiman, L. (1961). Optimal gambling systems for favorable games. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1, 65–78.

[6] Browne, S., & Whitt, W. (1996). Portfolio choice and the Bayesian Kelly criterion. *Advances in Applied Probability*, 28(4), 1145–1176.

[7] Cesa-Bianchi, N., & Lugosi, G. (2006). *Prediction, Learning, and Games*. Cambridge University Press.

[8] Cover, T. M. (1991). Universal portfolios. *Mathematical Finance*, 1(1), 1–29.

[9] Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.

[10] Duchi, J., Shalev-Shwartz, S., Singer, Y., & Chandra, T. (2008). Efficient projections onto the ℓ_1-ball for learning in high dimensions. *ICML*, 272–279.

[11] Duchi, J., Hazan, E., & Singer, Y. (2011). Adaptive subgradient methods for online learning and stochastic optimization. *JMLR*, 12, 2121–2159.

[12] Freund, Y., & Schapire, R. E. (1997). A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1), 119–139.

[13] Freund, Y., Schapire, R. E., Singer, Y., & Warmuth, M. K. (1997). Using and combining predictors that specialize. *STOC*, 334–343.

[14] Flaxman, A. D., Kalai, A. T., & McMahan, H. B. (2005). Online convex optimization in the bandit setting: gradient descent without a gradient. *SODA*, 385–394.

[15] Football-Data.co.uk. Historical football results and betting odds data. `https://www.football-data.co.uk/data.php`. Accessed 2026.

[16] Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100.

[17] Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.

[18] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *ICML*, PMLR 70, 1321–1330.

[19] Hall, P., Horowitz, J. L., & Jing, B.-Y. (1995). On blocking rules for the bootstrap with dependent data. *Biometrika*, 82(3), 561–574.

[20] Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362.

[21] Hastings, W. K. (1970). Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1), 97–109.

[22] Hazan, E., Agarwal, A., & Kale, S. (2007). Logarithmic regret algorithms for online convex optimization. *Machine Learning*, 69(2–3), 169–192.

[23] Hazan, E. (2016). Introduction to online convex optimization. *Foundations and Trends in Optimization*, 2(3–4), 157–325.

[24] Herbster, M., & Warmuth, M. K. (1998). Tracking the best expert. *Machine Learning*, 32(2), 151–178.

[25] Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301), 13–30.

[26] Kaunitz, L., Zhong, S., & Kreiner, J. (2017). Beating the bookies with their own numbers – and how the online sports betting market is rigged. *arXiv:1710.02824*.

[27] Kelly, J. L. (1956). A new interpretation of information rate. *Bell System Technical Journal*, 35(4), 917–926.

[28] Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241.

[29] Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.

[30] Levitt, S. D. (2004). Why are gambling markets organised so differently from financial markets? *The Economic Journal*, 114(495), 223–246.

[31] Littlestone, N., & Warmuth, M. K. (1994). The weighted majority algorithm. *Information and Computation*, 108(2), 212–261.

[32] Marx, K. (1867). *Das Kapital*, Band I, Kapitel 1, Abschnitt 4: Der Fetischcharakter der Ware und sein Geheimnis.

[33] Marx, K. (1894). *Das Kapital*, Band III, Kapitel 48: Die trinitarische Formel.

[34] Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4), 595–600.

[35] Pfitzner, L., Wintenberger, O., & Mestre, O. (2025). Contribution of expert aggregation to temperature prediction part II: Second order bounds with sleeping experts. *arXiv:2506.15216*.

[36] Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286.

[37] Shalev-Shwartz, S. (2012). Online learning and online convex optimization. *Foundations and Trends in Machine Learning*, 4(2), 107–194.

[38] Shin, H. S. (1992). Prices of state contingent claims with insider traders, and the favourite-longshot bias. *The Economic Journal*, 102(411), 426–435.

[39] Shin, H. S. (1993). Measuring the incidence of insider trading in a market for state-contingent claims. *The Economic Journal*, 103(420), 1141–1153.

[40] Sun, T., Sun, Y., & Yin, W. (2018). On Markov chain gradient descent. *Advances in Neural Information Processing Systems (NeurIPS)*, 31, 9896–9905.

[41] Štrumbelj, E. (2014). On determining probability forecasts from betting odds. *International Journal of Forecasting*, 30(4), 934–943.

[42] Thaler, R. H., & Ziemba, W. T. (1988). Anomalies: parimutuel betting markets: racetracks and lotteries. *Journal of Economic Perspectives*, 2(2), 161–174.

[43] Vovk, V. G. (1990). Aggregating strategies. *COLT*, 371–383.

[44] Zhu, H., Soen, A., Cheung, Y. K., & Xie, L. (2024). Online learning in betting markets: profit versus prediction. *ICML*.

[45] Zinkevich, M. (2003). Online convex programming and generalized infinitesimal gradient ascent. *ICML*, 928–936.